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--- Summary ---
Mathematical Literacy/P1/ Wiskundige Geletterdhei d/V1 3 NW/June 2024 /Junie 2024
Grade 11 – Marking Guidelines/ Nasienriglyne

Copyright reserved/Kopiereg voorbehou Please turn over/Blaai om asseblief QUESTION 1 /VRAAG 1 [14 MARKS/ PUNTE ] AO – Full marks
Q/V Solution /Oplossing Explanation /Verduideliking T/L
1.1.1 C ✓✓A 2A Correct option
(2) DH
L1
1.1.2 A ✓✓A 2A Correct option
(2) F
L1
1.1.3 B ✓✓A 2A Correct option
(2) DH
L1

1.1.4 C ✓✓A 2A Correct option
(2) DH
L1
1.2.1 Gross income / Bruto inkomste
= R14 000,00 + R11 500,00 ✓MA
= R25 500,00 ✓A
1MA Adding income values
1A Answer gross income
(2) F
L1
1.2.2 B ✓✓A 2A Correct answer
(2) F
L1
1.2.3 Net income / Netto inkomste ✓✓A 2A Answer
(2) F
L1
[14]

QUESTION 2/VRAAG 2 [22 MARKS/ PUNTE ]
Q/V Solution/ Oplossing Explanation/ Verduideliking T/L
2.1.1 Cell phone make = Samsung ✓✓A
2A Correct answer
(2) F
L1
2.1.2 ✓RT
Subtotal = R2 607,83 × 15% ✓MA
= R391, 17

1RT Use correct value
1MA Multiply with 15%
(2) F
L1
2.1.3 ENG:
Two thousand, Nine hundred and
ninety nine Rand ✓✓A

AFR:
Tweeduisend, Negehonderd Nege -en-
negentig Rand ✓✓A

2A Correct answer

(2) F
L1
2.1.4
a) Quote valid from
15 March + 14 days ✓MA
28 March ✓A

1MA add 14 days to 15 March
1A Answer correct date
(2) F
L2
2.1.4
b) Quotes are only valid for a certain
time period because prices of goods

and services go up after a while, and
the supplier don’t want to lose money /
✓✓O
Kwotasies is slegs geldig vir ‘n sekere
tydperk want pryse van goedere en
dienste gaan op na ‘n tydperk, en die
verskaffer wil nie geld verloor nie
✓✓O

2O Explanation

(2) F
L4
2.2.1 ✓✓A
Number of minutes / Aantal minute
2A Answer
(2) F
L1
2.2.2 Fixed cost / Vaste koste
= R200 ✓✓A

2A Answer
(2) F
L1
2.2.3 Prepaid option / Vooruitbetaalde opsie
= R40 0 ÷ 200 minutes ✓MA
= R2 per minute

A = Prepaid price plan /
Vooruitbetaalde prysplan
✓A
Total cost = R2,00 × number of
minutes ✓A

1MA Dividing with 20 minutes

1A Correct value R2,00
1A Multiply with number of minutes
(3) F
L3
2.2.4 Break even pint / Gelykbreekpunt
= 130 minutes ✓✓RT

2RT Correct answer
(2) F
L2

2.2.5
Contract option / Kontrak opsie
= R344 – R200✓MA
= R144 for 200 minutes

* 80 free minutes
= 200 minutes – 80 minutes ✓MA
= 120 minutes

Cost per minute / Koste per minuut
= R144 ÷ 120 minutes
= R1,20 per minute ✓

1MA Subtract fixed cost

1MA Subtract free minutes

1A Answer cost per minute - contract
(3)
Any value s from table can be used
to calculate the cost per minute F
L3
[22]

QUESTION 3/ VRAAG 3 [20 MARKS/ PUNTE ]
Q/V Solution/ Oplossing Explanation/ Verduideliking T/L
3.1.1 Other colo ur cars / Ander kleur motors
= 2 + 3 + 5 + 2 + 4 + 2 ✓MA
= 18 ✓A
1MA Adding correct values
1A Answer coloured cars
AO (2) DH
L2
3.1.2 Average (mean) / Gemiddeld
= 7 + 9 + 12 + 8 + 13 + 11
6 ✓
= 60 ÷ 6 months ✓
= 10 ✓

1M Adding white cars
1M Dividing with 6 months
1CA Average white cars
(3) DH
L2
3.1.3 7; 8; 9; 11; 12; 13 ✓MA
Median = 9+ 11
2 ✓MA
Median = 20 ÷ 2
Median = 10 ✓A
1MA Arrange values

1MA Median concept

1A Median white cars
(3) DH
L2
3.1.4 Range = biggest – smallest/
Omvang = Grootste – kleinste
8 = Big – 2 ✓SF
Big = 8 + 2 ✓M
Big = 10 ✓CA

1SF Substitution
1M Change subject of formula
1CA Answer
(3) DH
L3
3.1.5 Probability ( March white car )/
Waarskynlikheid (Maart wit motor)
= 12 ✓A
60 ✓A
= 0,2 ✓CA

1A Numerator
1A Denominator
1CA Decimal number
(3) P
L3
3.2 ANSWER SHEET /
ANTWOORDBLAD 1A Correct bar s – White cars Feb
1A Correct bar – White cars May
1A Correct bar – Other cars Jan
1A Correct bar – Other cars May
(4) DH
L2

3.3 Fuel consumption / Brandstof
verbruik ✓✓ O
OR
Number of kilometres on the clock /
Aantal kilometre op die klok ✓✓ O

OR
Price of the car/ Prys van die kar ✓✓ O

2O Reason

(2) DH
L4
[20]

QUESTION 4/VRAAG 4 [19 MARKS/PUNTE]
Q/V Solution/ Oplossing Explanation/ Verduideliking T/L
4.1 % cleaning water of total water
= 2,5 ■■
5,5+2,5+4,8+1,3+5,9 ✓■ ×100
= 2,5 ■■
20 ■■ ×100 ✓M
= 12,5% ✓CA

1M Adding total water consumption

1M Multiply fraction with 100

1CA Answer %
(3) F
L2
4.2 P (bath water) /(badwater )
= 2,8 ■ ✓A
4,8 ✓A
= 7
12 ✓CA

1A Numerator
1A Denominator
1CA Simplified fraction
(3) P
L2
4.3 Block 1
✓MA
0-6 = 6 k■ × 0,00c = 0,00c
7-15 = 9 k■ × 1095c = 9855c ✓MA
16-20 = 5 k■ × 1248c = 6240c

Total cost in cent
= 0,00c + 9855c + 6240 c
= 16095c ✓CA

Total cost in Rand
= 16095c ÷ 100
= R160,95 ✓C

Total cost in Rand, VAT added
= R160,95 × 1,15 ✓M
= R185,09 ✓CA

His claim is NOT valid ✓O
Sy bewering is NIE geldig NIE

1MA Dividing 20 kl into b locks
1MA Multiply with correct cost per block

1CA Total cost in cent

1C Convert to rand

1M Multiply with VAT
1CA Total cost with VAT

1O Opinion
(7) F
L4
4.4 Continuous /Kontinu ✓A
It is values that can be measured ./
Dit is waardes wat gemeet kan
word .✓✓O
1A Answer - Continuous

2O Reason
(3) DH

L4
4.5 Pie chart /Sirkeldiagram ✓A
It is better to compare percentages
than only indicating different values.
QUESTION 1/VRAAG [19 MARKS/ PUNTE ] Answer only AO – full marks

Q/V Solution /Oplossing Explanation/ Verduideliking T/L
1.1.1
(a) Thermometer/ Termometer ■■A 2A answer/ antwoord
(2) M
L1
E
(b)
The degree of hotness or coldness of a substance/ Die
grade van hoe warm of koud ‘n voorwerp is ■■A 2A answer/ antwoord
(2) M
L1
E
(c)
43 oC■■A 2A answer/ antwoord
(2) M
L1
E
1.1.2
400 ml■RT
0,4 litres/ liters■A 1RT correct reading/ korrek
gelees
1CA conversion/ herleiding
(2) M
L1
E
1.1.3 Distance/ Afstand = 9,7 cm – 2,9 cm■RT
= 6,8 cm■CA
= 68 mm■C
1RT reading from the diagram
/lees vanaf die diagram
1CA answer/ antwoord
1C conversion/ herleiding M
L1
M

OR/OF
Distance /Afstand = 97 mm – 29 mm■RT■C
= 68 mm■CA OR/OF
1RT reading values/ lees
waardes
1C conversion/ herleiding
1CA answer// antwoord
(3)
1.2.1 Strip Chart/map/ Strookkaart ■■A 2A answer/ antwoord
(2) MP
L1
E
1.2.2 Distance /Afstand = 117 km ■■RT

OR/OF
Distance/ Afstand = 557 km – 440 km■RT
= 117 km ■A 2RT reading from the map/ lees
vanaf kaart
OR/OF

1RT read and subtraction/ lees
en aftrek
1A answer/ antwoord
(2) MP
L1
M
1.2.3 Escourt■■RT 2RT reading from the map/ lees
Vanaf kaart
(2) MP
L1
E
1.2.4 5 towns/ dorpe■■RT
Accept 6 towns /aanvaar 6 dorpe 2RT reading from the map/ lees
Vanaf kaart
(2) MP
L1
E
[19]

QUESTION/ VRAAG 2 [24 MARKS/ PUNTE ]

2.1.1 Food/Kitchen /Kos/kombuis ■■RT 2RT answer /antwoord

(2) MP
L1
E
2.1.2 South/ Suid (S) ■A and/en East/ Oos (E)■A
OR/OF
Southern/ Suidelik and/en Eastern/ Oostelik
1A South/ Suid
1A East/ Oos

(2) MP
L2
E
2.1.3
(a) 37■RT and/en 38■RT 1RT room/ kamer 37
1RT room/ kamer 38
(2) MP
L2
E
(b) The rooms are close to the corner ■A of Mimosa ■RT
and Flower Street ■RT/
Die klaskamers is naby die hoek ■A van Mimosa
■RT en Flower Straat ■RT
OR/OF
Adjacent/next to corner of Mimosa and Flower Street
Op/langs die hoek van Mimosa en Flower str aat
1A corner/adjacent/next to
/hoek van/langs aan
1RT Mimosa
1RT Flower

(3) MP
L3
M

2.1.4
Classroom/ Klaskamer 14■■RT 2RT number/ nommer 14

(2) MP

L2
M
2.1.5 Traffic congestion can be avoided when learners are picked up or dropped off/ Verkeersopeenhoping kan verhoed word as leerders op gelaai of afgelaai word.■■O
OR
Learners might be crossing the road, and a two -way traffic can cause accidents/ Leerders mag dalk die pad oorsteek, en tweerigting -verkeer kan ongelukke veroorsaak 2O reason/ rede

(2) MP
L4
M
2.1.6 P = 4■A
38■A ×100

= 10,52631579% ■CA

= 10,526%■R 1A numerator/ teller
1A denominator/ noemer
1CA simplification/
vireenvoudiging
1R rounding/ afronding
(4) P
L3
D
2.2.1 12 km = average speed × 60 min ■SF
12 000 m ■C = average speed × 60 min
Average speed/ gemiddelde spoed
= 12 000 m ÷ 60 min ■M
= 200 metres/ minute■CA

OR/OF

Average speed = distance ÷ time ■M
12 km ÷ 60 min ■SF
= 12 000 m ÷ 60 min ■C
= 200 m/min ■CA
1SF substitution/ vervanging
1C conversion/ herleiding
1M changing the subject of the
formula/ verander die
onderwerp van die formule
1CA simplification/
vereenvoudiging
OR/OF

1M changing the subject of the
formula/ verander die
onderwerp van di e formule
1SF substitution/ vervan gding
1C conversion/ herleiding
1CA simplification/
vereenvoudiging
(4) MP

L3
M
2.2.2 200 m/min is too fast for walking and too slow for travelling by car or taxi./ 200 m/min is te vining om te loop en te stadig om met ‘n motor of minibus te ry.■■O
Thus, the learner was cycling/running/ Dus, moet die leerder fiets ry of hardloop ■O
OR/OF
Any other sensible answer/ Enige sinvole antwoord
2O reason/ rede

1O opinion/ opinie

(3) MP
L4
D
[24]

3.2.2

Number of pallets/ hoeveelheid palette
= 8 386 ÷ 594 ■MCA
= 14,11784512 ■CA
= 15 pallets/ palette■R
OR/OF
Number of pallets/ hoeveelheid palette
= 8 385,048 ÷ 594 ■MCA
= 14,11624242 ■CA
= 15 pallets/ palette ■R

CA number of bricks from /
hoeveelheid stene van 3.2.1
1MCA dividing by/ deël met
594
1CA number of pallets/
hoeveelheid palette
1R rounding up/ rond op

(3)

M
L2
M

QUESTION /VRAAG 3 [ 26 MARKS/ PUNTE ]

3.1.1 Area of 1 bicycle/ Oppv.
Number of bricks/ Hoeveelheid stene
= 48 × 166,37 ■MA
= 7 985,76■A
Increase by 5% / verhoog met 5%
= 7 985,76 × 1,05 ■MCA OR/OF 105%
= 8 385,048■CA
= 8 386■R 1MA multiplying/
vermenigvuldig
1A number of bricks/
hoeveelheid stene
1MCA increase percentage/
verhoogde per sentasie
1CA simplification/
vereenvoudiging
1R rounding/ afronding

(5)
M

L3
D 3.2.3 Mass/ gewig = 8 386 × 2,230 kg ■MCA
= 18 700,78 kg ■CA
= 18 700,78 ÷ 1 000
= 18,70078 ton ■C
= 18,7 tons ■R CA number of bricks from
/hoeveelheid stene van 3.2.1
1MCA multiplying /
vermenigvuldig
1CA simplification /
vereenvoudiging
1C conversion /herleiding
1R rounding /afronding
(4)

M
L2
M
3.3.1 Indirect/inverse proportion/ Indirekte/Omgekeerde eweredigheid ■A
As one quantity increases, the other one decreases/ As een eenheid verhoog, neem die ander een af ■■O
1A correct proportion / korrekte
proporsie
2O explanation/ verduidelikking

(3) M
L1
E
3.3.2 Number of days = 30■RT ÷ number of people ■A
/ Hoeveelheid dae = 30 ÷ hoeveelheid mense

OR/OF
Number of days × number of people ■ = 30■
Hoeveelheid dae × hoeveelhed mense = 30
1RT 30
1A divided by number of
people/ deël met
hoeveelheid mense
OR/OF
1RT 30
1A number of days multiplied
by number of people/
hoeveelheid dae maal met
hoeveelhed mense
(2) M
L2
E

3.3.3
Number of days/ hoeveelheid dae = 30 ÷ 8■MCA

= 3,75 days/ dae■CA
Accept 4 days CA formula fro m/vanaf 3.3.2
1MCA substituting/ trek af 8

1CA answer/ antwoord
(2) M
L1
E
[26]
QUESTION /VRAAG 4 [31 MARKS/ PUNTE ]
Q/V Solution /Oplossing Explanation/ Verduideliking T/L

4.1.1 Anti clockwise/ Anti-kloksgewys
OR/OF
Left/ Links■■A 2A direction/ rigting

(2) MP
L1
E
4.1.2 Diagram 3 ■■A 2A correct/ korrekte diagram

(2) MP
L2
E
4.1.3 1,55 m = diagram height/ hoogte x 25
Diagram height/ hoogte = 1,55 m ÷ 25 ■MA
= 0,062 m ■A
= 6,2 cm■C
OR/OF
1,55 m = 155 cm ■C
155 cm ÷ 25■MA
= 6,2 cm■A
1MA divide by scale /deël met
skaal
1A length/ lengte in m
1C conversion/ herleiding
OR/OF
1C conversion/ herleiding
1MA dividing by scale/ deël
met skaal
1A length/ lengte (3) MP
L2
M

4.2.1 Space occupied by a 3D object/ Spasie wat deur 'n 3D
voorwerp opgeneem word ■■A 2A answer/ antwoord
(2) M
L1
E
4.2.2 Volume = 340 mm × 325 mm × 180 mm ■SF
= 0,34 m × 0,325 m × 0,18 m ■C
= 0,01989 m3■CA
OR/OF

Volume of the box/ volume van die boks
= 340 mm × 325 mm × 180 mm ■SF
= 19 890 000 mm3 ÷ 1 000 000 000■CA
= 0,01989 m3■C 1C conversion/ herleiding
1SF substitution/ vervanging
1CA simplification/
vereenvoudiging

NPR (3) M
L2
M
4.2.3 Number of boxes/ Hoeveelheid bokse :
Length wise/ Lengte = 1 700 mm ÷ 340 mm ■MA
= 5■A
Width wise/ Breedte = 1 490 mm ÷ 325 mm

= 4,584615385
= 4■A
Height wise/ Hoogte = 1 200 mm ÷ 180 mm
= 6,6666667
= 6■A
Total/ Totaal = 5 × 4 × 6 ■MCA
= 120 boxes/ bokse■CA
OR/OF
Length wise/ lengte : 1,7 m ÷ 0,34 m ■MA = 5■A
Width wise/ breedte : 1,49 m ÷ 0,325 m = 4,584615…
= 4■A
Height wise/ hoogte : 1,2 m ÷ 0,18 m = 6,666…
= 6■A
Total/ totaal = 5 × 4 × 6 ■MCA
= 120 boxes ■CA 1MA dividing/ deë/
1A answer/ antwoord
1A number width wise/
hoeveelheid in breedte
1A number height wise/
hoeveelheid in hoogte
1MCA multiplying/
vermenigvuldig
1CA number of boxes/
hoeveelheid bokse

(6) MP
L3
D
4.2.4 30 June from 14:50 to 24:00 = 9 hours 10 min ■A
30 Junie van 14:50 tot 24:00 = 9 uur 10 min
1 July = 24 hours
1 Julie = 24 uur
2 July from 00:00 to 8:15 = 8 h 15 min ■A
2 Julie van 00:00 tot 8:15 = 8 h 15 min
Total elapsed time = 9h10 + 24h00 + 8h15
= 41 hours 25 min ■CA
Totale tyd verloop = 41 uur 25 min
This is within the 48 -hour service ■O
Dit is binne die 48 -uur diens
OR/OF
1A time 30 June/ tyd 30 Junie

1A time 1 and 2 July/ tyd 1 en
2 Julie
1CA adding time/ tel tyd
bymekaar
1O opinion/ opinie

OR/OF

M
L4
D

30 June from 14:50
To 1 July 14:50 = 24 hours/ first day ■ A
/30 Junie van 14:50
Tot 1 Julie 14:50 = 24 uur/ dag 1
To 2 July 14:50 = 48 hours/second day ■ A
Tot 2 Julie 14:50 = 48 ure / dag 2
But 2 July   8:15 is before 48 hours ■ A
Maar 2 Julie 8:15 is voor 48 uur
It is within 48 hours ■O
Dit is binne 48 uur  1A 1st  day/dag 1

1A 2nd day/dag 2

1A conclusion/ gevolgtrekking

1O conclusion /gevolgtrekking
(4)
4.3.1  Body Mass Index/ Liggaamsmassa -indeks■■A 2A answer/ antwoord
(2) M
L1
E
4.3.2  Underweight/ Ondergewig ■■RT 2RT answer/ antwoord
(2) M
L2
E
4.3.3  18,2 kg/m2 = mass/ gewig  ÷ (1,56 m)2 ■SF
Mass/ gewig  = 18,2 × (1,56)2 ■M
= 44,29 kg ■A 1SF substitution/ vervanging
1M changing the subject of
the formula/ verander die
onderwerp van die f ormule
1A simplification/
vereenvoudiging
NPR (3) M
L2
M
4.3.4  Eat a balanced diet/ Eet ‘n gebalenseerde dieët
OR/OF eat food with lots of fibre/fats/ eet kos met baie
vessel/ vette  OR/OF eat food that will assist her to gain
weight/ eet kos wat sal help om gewig op te tel ■■O
2O opinion/ opinie

(2) M
L4
E
[31]
TOTAL/ TOTAAL : 100

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NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER  2024

MATHEMATICAL LITERACY P1
MARKING GUIDELINE

MARKS:  100

Symbol  Explanation
M Method
MA Method with accuracy
CA Consistent accuracy
A Accuracy
C Conversion
S Simplification
RT Reading from a table/a graph/document/diagram
SF Correct substitution in a formula
O Opinion/Explanation
P Penalty, e.g.
■■A
OR
The rate at which price increase s over time if there is a decline  in the
purchasing value of  money  ■■A
2A Explanation

(2) F
L1

2.4.2
Price of brown bread in 2017
= (1 +  6,59 %) × R9,99
= R10,65
1M Multiply
correct values
1A Cost (2) F
L1

[29]

QUESTION 3  [17]
Ques  Solution  Explanation  T&L

3.1.1
12 miles
Distance = 12  ×1,609
= 19,308 km  ■A
1RT Correct value
1A Answer in km
NPR (2) M
L1

3.1.2
Humidity of Cape Town =
10068
=
2517

1RT Correct value

1A Simplified
fraction (2) M
L1

3.1.3
Time of sunset in Cape Town = 17:27
12-hour format = 05:27 pm

2A Correct time (2) M
L1

3.1.4
■=(■ ×1,8) + 32
= (17 × 1,8 ) + 32
= 62,6
= 63

1SF Correct value
1S Simplification
1R Rounding (3) M
L2

3.2.1
Volume = Length × Width × Height
= 30 in × 12 in × 7,1 in
= 2 556 in3
1SF Substitution
1S Simplification
1A Correct unit (3) M
L2

■ RT ■ A ■ M
■ SF ■ A
■■ A
■ S
■ ■ R

■ SF
■S ■A ■ RT

3.2.2
Volume of the tank = 2 556 in3 × 85%
= 2 172,6 in3

Increased volume after stones added = 2 556 × 97%
= 2479,32 in3
Volume of stones = 2479,32 – 2 172,6
= 306,72 in3 ■CA

OR
Volume of stone s = 97%  - 85% ■M ■M
= 12% × 2556 ■M ■M
= 306,72  in3 ■CA CA from 3.2.1
1M Multiply by
85%
1CA Volume

1CA Volume
1M subtraction
1CA Volume

1M Using correct
values
1M Subtraction
2M Multiplication
by 12% and 2556
1CA  Volume of
stones (5) M
L3

[17]

QUESTION 4  [13]
Ques  Solutions  Explanation  T&L

4.1
Hartford  ■■RM
2RM Correct  city
(2) M&P
L1

4.2
Distance on the map = 7,5 cm
2,5 cm = 100 miles  ■M

5,25,7 = 3 cm ■S
3 × 100 = 300 miles  ■CA

1M Scale measure
(use the scale as
from actual map)
1S Division

1CA Multiply by
100 (3) M&P
L2

4.3
84■and 87 ■ RG
OR
81,■88 and 90 ■RG
2RG Combination
of roads
(2) M&P
L1

4.4
North East ■■A

2A Correct direction
(2) M&P
L1

4.5
Road 80

2RG Correct road
(2) M&P
L1

4.6
Probability = ■
■■
1A Numerator
1A Denominator (2) P
L2

[13]
■M
■CA
■M ■CA

■■ RG
■ RG ■ RG

Copyright reserved Please turn over QUESTION 5 [20]
Ques Solutions Explanation T&L

5.1
Poults not hatched in December = 28 795 – 25 422 ■M
= 3 373 ■CA 1M Subt racting
correct
values
1CA Not hatched (2) D
L1

5.2
Mean
= 28 927 + 28 409 + 27 179 + 28 795 + 29 961 + 29 906 + 30 030 +

28 597 + 28 825 + 29 441 + 29 271 + 29 725
= 349 066
12 ■M
= 29 088,93 ■CA

1M Adding
1M Dividing by 12
1CA Mean
NPR (3) D
L2

5.3
No modal value ■■A
2A Modal value (2) D
L1

5.4
Range = highest value – lowest value
= 25 719 – 22 782 ■MA
= 2 937 ■CA
1MA Subtracting
correct values
1CA Range (2) D
L2

5.5
25 719, 25 422, 25 332, 25 075, 24 786, 24 616, 24 067, 23 645,
23 598, 23 572, 23 179, 22 782 ■A

Median =
2067 24 616 24■ ■MA
= 24 341 ■ CA
1M Arranged
1MA Median concept
with correct values
1CA Median (3) D
L2

5.6 Total hatched during 201 6-2107= 291 7893 ■M
Hatched in March= 25 719 ■ RT
Ratio.
2 A
Verduideliking
L1(2)
1.2.2 9,50
500 × 100■ = 0,019 × 100■ = 1,9%■ 1M deel deur 500
1M
1CA wanneer die
waarde wat gebruik
word verskillend van
die staat is L1(3)
1.2.3 R110,00 + R55,00+R1,10 × 2■
R165,00 + R2,20 ■

= R167,20 ■ 1M
1S
1CA L2(3)
1.2.4 Agtienduisend Vyfhonderd en Twee en tagtig
Rand ■en sewe sent ■ 2A Uitbreiding
L1(2)

VRAAG 2 [19]
Vr Oplossing Verduideliking Punt
2.1.1 Lengte = 30 vt■
Breedte = 25 vt■ 2 RD
L1(2)
2.1.2 30 vt × 25 vt ■
= 750 ft2 ■ OF
=750 ■■
10,764■
= 69,68 m2 ■
(Aanvaar 69,677 ■2 ) 30 vt = 9,1435
25 vt = 7,6196
■A = 9,1435 × 7,6196
= 69,6698 m2
= 69,67 m2 1M
1S
1 M Deel deur 10,764
1CA
L2(4)
2.1.3 Aantal kunsmis =20 × 15■
100 ■
= 300 vt2
100 vt2 ■
= 3 × 2 ■ pond
= 6 pond ■
OF
2 pond × 3■■ = 100 vt2 × 3■
6 pond ■ = 300 vt2■ 1M teller
1M noemer
1S
1M Vermenigvuldig met
2
1CA
L2(5)
2.1.4 0,15 × 2 ■ = 0,3 ■
Ongeveer = 1
3 koppie ■
OF
1
0,15■= 6,66666667
Dus 2
0,666666667 ■
= 0,3 koppies = ■ koppie ■ 1M Vermenigvuldig met
2
1 A
1A
L1(3)

2.2.1 Oggend + Aand
(10■■+15■■+10■■ ) ■ × 2■
=35■■ × 2
= 70 ■■■
2 M

1CA L1(3)
2.2.2 10 ■■+10 ■■■ =20 ■■■ 1M
1A L1(2)

VRAAG 3 [19]
Vr Oplossing Verduideliking Punt
3.1.1 4■■ 2 RT L1(2)
3.1.2 Water moniteringstasie ■■ 2 RT L1 (2)
3.1.3 MA-1 ■■ 2 RT L1(2)
3.1.4 103 mm ■■ (10,3 cm) 2 RT L1(2)
3.1.5 10,3 × 250 000■
100 000■
= 25,75 km ■ 1C
1 Deling
1CA L3(3)

3.2.1 Tafel B■■(Aanvaar H/l) 2A RP L1(2)
3.2.2 Suidoos ■■ 2A RP L1(2)
3.2.3 Onmoontlik/Zero/0% ■■ 2A L1(2)
3.2.4 Tafel C ■■ 2RP L1(2)

VRAAG 4 [27]
Vr Oplossing Verduideliking Punt
4.1 32,49 ; 29,63 ; 23,62 ; 17,89 ; 14,59 ; 12,03 ; 10,31 ;
9,89; 9,57■■
2 M L1(2)
4.2 Mediaan =R14,59 ■■ 2 M L2(2)
4.3 Omvang = R33,73 – R9,68■■ =R24,05 ■ 2RT L2(3)
4.4 R-3,32 ■■ 2M L1(3)
4.5 Margarien 500g■■ 2 RT L1(2)
4.6 Geen modus ■■ 2RT L2 (2)
4.7 Gebied Margarien Rys Olie Tee Witsuiker
Stedelik 21,68 23,45 17,25 9,68 26,31

1punt vir elke voedselprys korrek afgesteek × 5 = 5 punte + 1 = 6 punte
1punt vir korrekte grafiek L2(6)
4.8 Mense in landelike gebiede betaal meer of minder vir
sekere items ■■ 2A Verduideliking
L3(2)
4.9 0,56 + 0,72 + 1,11 + 1,24 + 3,79 + (-0,17) + 2,66 +
(-0,21) + (-3,32) ■
R6,38
9■ =■0,71■ 1M
1deel deur
1CA
L2(3)
4.10
2■
9■ × 100 = 22,2% =22% ■ 1teller
1noemer
1CA L2(3)

VRAAG 5 [12]
Vr Oplossing Verduideliking Punt
5.1 R36,99 ■■ 2RT L1(2)
5.2 R200,00 ■■ 2 RT L1(2)
5.3 ■278 ,78
13 ■ = R21,44 ■ = R20,00 ■ 1M deel deur 13
1S
1A L1(3)
5.4 6 × R17,99 OF

= ■107 ,94 ■
1,14 ■ = R94,68 ■  17,99
1,14 ■ = 15,78 ■ × 6
= R94,68 ■ 1A vir 6 1M ÷ 1,14
1CA L2(3)
5.5  R99 ,98
2■ = R49,99 ■ 1M
1A L1(2)
5.6 Totaal = 22,49 + 29,26 + 25,59 + 99,98 + 22,00 +
17,99 + 9,99 + 13,49 + 1,00 ■ = R241,79 ■
OF Totaal = 478,78 – 200 – 278,78 = R241,79
1M
1CA L2(2)

TOTAAL : 100

051015202530
Margarien Rys Olie Tee Wit SuikerPryse in Rand
Geselekteerde voedselpryseStedelike voedselpryse
Stedelike voedselpryse
■ ■
■
■ ■
■

NATIONAL
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GRADE 11

NOVEMBER  2016

MATHEMATICAL LITERACY P1
MEMORANDUM

MARKS:  100

This memorandum  consists of 4 pages .
1A two years
1CA a dding i  to
outstanding amount
(4) F
L1(2)
L2(2)

2.3
Amount = 2419,3067R
= R304,4245.....
≈ R304,42
1M dividing by 24
1A answer
(2) F
L2

2.4
Height = 12
× 2,54 cm = 30,48 cm

1M multiplying by 2,54 1A height

(2) M
L1

2.5.1
Amount = 12 × R300
= R3 600
1M multiplying by 12
1A answer
(2) F
L1

2.5.2
Amount = R300 – (R120 + R150)
= R30
1M subtracting from R300 1M adding costs
1A answer
(3) F
L1(2)
L2(1)
[15]

■M
■CA
■A
■A ■A
■CA
■M
■A
■A
■M
■A
■M ■M
■A ■M

QUESTION 3 [ 20 marks ]
Ques Solution Explanation Topic

3.1.1
Height of tin = 430 mm = 43 cm
1A length
1C conversion
(2) M
L1

3.1.2
Distance = 4 570 mm – 1 780 m m
= 2 790 m m

1M subtracting cor rect
values
1A correct answer
(2) M
L1

3.1.3
Length = 4 260 – 2 610
= 1 650 mm
Area = 1 600 × 1 650

= 2 640 000 m m2
1 A value of length
1SF substitution
1CA answer
(3) M
L1(1)
L2(2)

3.1.4
A = 4, 830 m (2 × 9,75mm + 6,4 m ) – 6,4 m × 0,43 m
= 4, 83 m (25,9 m ) – 2,752 m2
= 122,345 m2
1C conversion
1SF value of k
1SF value of b
1SF value of t and p.
1S simplification
1CA area
(6) M
L1(2)
L2(2)
L3(2)

3.2.1
Amount of paint = ■8345,122
= 15,293.. ■
≈ 16 ■
1M dividing by 8
1CA amount of paint
1R rounding up
(3) M
L1(2)
L2(1)

3.2.2
Number of 5 ■ = ■■
516
= 3 ,2 = 4
Cost = 4
× R215,85
= R 863,40
1M dividing by 5 ■
1 CA number of containers
1M multiplying by cost
1CA cost of paint
(4) M/F
L1(2)
L2(2)
[20]
■M
■A
■SF ■A
■M ■SF ■SF
■CA ■CA
■S ■CA
■M
■CA ■M
■CA
■R ■SF ■C

## QUESTION 4 [14 marks]
Ques Solution Explanation Topic

4.1.1
R63
2A correct regional road
(2) M
L1

4.1.2
Pearston; Somerset E ast; Cookhouse; Bedford
3A any three correct
(3) M
L1

4.1.3
25 km + 49 km + 10 km
= 84 km
1A correct reading from map
1M adding 1CA total distance
(3) M
L1

4.1.4
North -easterly OR NE

2A correct direction
(2) M/P
L1

4.2
3 cm : 15 km
= 3 cm : 1 500 000 cm
= 1 : 500 000
1A correct measurement
1M writing as a ratio
1C converting to cm
1CA simplified ratio
(4) S/M/P
L2
[14]
■■A
■A ■A ■A
■M
■CA
■■A
■M ■A
■C
■CA ■A

## QUESTION 5 [ 32 marks ]
Ques Solution Explanation Topic

5.1.1
Total = 20 + 5 + 5 + 30 + 40 + 15 + 5 + 10
= 130
1M adding 1A sum
(2) D
L1
5.1.2 Percentage preferring casual wear

= %10013040×
= 30,77%
1A correct values
1M calculating %
1CA solution
(3) D
L1

5.1.3
Fancy dress
2A correct answer
(2) D
L2

5.1.4 DRESS CODE FOR THE DANCE
051015202530354045
FORMAL
TRADITIONAL
DRESS
FANCY DRESS
CASUAL
WEAR
Dress CodeNumber of Learners

7A each bar

(7) D
L1

5.1.5 (a)
P= 13060
1A numerator
1A denominator
(2) P
L2

5.1.5 (b)
P = 13010 130–

= 130120
1A numerator

1A solution
(2) P
L3
■A
■A ■A ■A
■A ■A
■A ■A ■M
■M ■A
■CA
■■A
■A
■A ■A
■A

Ques Solution Explanation Topic

5.2.1
39 49 56 58 59 67 75 75 75 79 89 98 99

1A solution
(2) D
L1

5.2.2
75
2A solution
(2) D
L1

5.2.3
Mean = 14976
= 69,7…
≈70
1M adding 1A dividing by 14
1A simplification

(3) D
L1
L2

5.2.4
Median = 268 66+
= 67
1M concept
1A solution (2) D
L2

5.2.5
Range = 99 – 39
= 60
1M subtracting correct values
1A solution (2) D
L1

5.2.6
P= 133
1A numerator
1A denominator
(2) P
L2

5.2.7
P = 145
1A numerator
1A denominator
(2) P
L2
[32]

TOTAL: 100 ■M
■A
■A
■A
■A
■A ■A ■M
■A ■M
■A ■A ■A ■A

MARKS:

SYMBO L
A
CA

C
J
M
MA
P
R
RT/RG
S
SF
O

L
MAT H
100

EXPLA
Accura c
Consist e
Conver s
Justific a
Method
Method
Penalty
Roundi n
Readin g
Simplifi c
Correct
Own o p
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OR CE R

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--- Full Combined Text ---

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GRADE 11

NOVEMBER 2025

MATHEMATICAL LITERACY P2
MARKING GUIDELINES

MARKS: 100

| Symbol | Explanation |
|---|---|
| M | Method |
| MA | Method with accuracy |
| CA | Consistent accuracy |
| A | Accuracy |
| C | Conversion |
| S | Simplification |
| RT | Reading from a table/graph/ diagram |
| SF | Correct substitution in a formula |
| O | Opinion/Explanation/Reasoning |
| P | Penalty, e.g. for no units, incorrect rounding off , etc. |
| R | Rounding Off/Reason |
| NPR | No penalty for correct rounding minimum two decimal places |
| AO | Answer only |
| MCA | Method with consistent accuracy |
| RCA | Rounding with consistent accuracy |

This marking guidelines consist of 10 pages .

## MARKING GUIDELINES

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled) version.
- Consistent Accuracy (CA) applies in ALL aspects of the marking guidelines; however, it stops at the second calculation error.
- If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalize for every extra incorrect item presented.

KEY TO TOPIC SYMBOL:
F = Finance; M = Measurement; MP = Maps, plans and other representations; P = Probability

QUESTION 1 [2 4 MARKS]                                    ANSWER
ONLY FULL MARKS

Ques.  Solution  Explanation  Level
1.1.1  Total time = 1,5 minutes + 3 minutes + 15 minutes
= 19,5 minutes  1M adding time
1A total time
(2) M
L1

1.1.2  Cream cheese = 10 ounces
35,274
= 0,28349…
≈ 0,28 kg  1C dividing with
35,274
1A cream cheese in kg
NPR
(2) M
L1

1.1.3  Salt = 1½ × 5 ml
= 7,5 ml  1M multiplying
1A correct answer
(2) M
L1

1.1.4  ■ = 5
9 × (■ - 32)
= 5
9 × (356 ■ - 32)
= 180 ■
1SF substitution
1A temperature in ■
(2) M
L1

1.2.1 Capacity refers to the maximum amount of liquid a
container can hold.  2A correct definition
(2)  M
L1

1.2.2 Capacity = 500 ml   2RT correct capacity
(2) M
L1

1.3.1 Circumference of a circle = $\pi$ × diameter
= 3,142 × 13
= 40,846 m 1SF substitution
1A circumference
NPR
(2) M
L1

1.3.2 Z : X
16 : 8
2 : 1 1A correct ratio
1S simplification
(2) M
L1

1.4.1 Oribi Animal Clinic 2RT correct clinic
(2) MP
L1

1.4.2 Southeast OR SE 2A correct direction
(2) MP
L1 ■M
■A
■C
■A
■M
■A
■SF
■A
■✓RT ■✓A
■✓A ■✓A ■SF
■A
■A
■S If ratio swopped = 1 mark
■✓RT

1.4.3 Travelling distance = 21,1 km × 1 000
= 21 100 m 1C multiply by 1 000
1A distance in m
(2) MP
L1

1.4.4 Arrival time = 10:15
+ :26
= 10:41 1M adding time
1A correct time
(2) MP
L1
[24]

■A
■M
■A ■C

QUESTION 2 [ 27 MARKS]

Ques.  Solution  Explanation  Level
2.1.1  Rober's Grave  2RT correct attraction
(2) MP
L2

2.1.2  Probability = 0  OR None  2A correct answer
(2) P
L2

2.1.3  Distance = 6,4 cm
(Accept 6,2 cm – 6,6 cm) 2A correct distance
(2) MP
L1

2.1.4

Scale line = 2 ,6cm (Accept  2,5 – 2,7 cm)

Scale = 2,6 cm
2,6 = 600  m
2,6
= 1 cm : 230,7692308  m

Actual distance  = 6,4 cm × 230,7692308
= 1 476 ,923077  m

1 000
= 1,48 km

OR

Scale line = 2 ,6 cm

Scale = 2 ,6 cm = 600 m
= 2,6
2,6 : 60 000
2,6
= 1 : 23 076,92308

Actual distance = 6,4 × 23 076,92308
= 147 692,3077 cm
= 147 692 ,3077
100 000
= 1,48 km CA from 2.1.3
1A scale line in cm

1M dividing correct
values
1CA scale

1MCA multiplication

1CA distance in km

OR

1A scale line in cm

1M dividing correct
values

1CA scale

1MCA multiplication

1CA distance in km
NPR
(5) MP
L3

2.1.5
No of people = 1 721
25,40
= 67,7559…
≈ 68 people
(Accept 67 people) 1M dividing correct
values
1R number of people

(2) MP
L1

2.2.1
Mozambique OR Swaziland 2RT any correct
country
(2) MP
L1

■✓RT
■✓A ■✓A
■✓A
■A
■M
■CA
■MCA
■CA
■M
■R
■✓RT ■✓RT ■A
■CA ■M
■CA
■MCA

2.2.2 6 national roads 2RT no of national
roads
(2) MP
L2

2.2.3 Scale = 8,3 cm : 276 km
= 276 km × 100 000
= 27 600 000 cm

Scale = 8,3
8,3 : 27 600 000
8,3
= 1 : 3 325 301,205
1C multiplying by
100 000

1M dividing distances
1CA scale
NPR
(3) MP
L2

2.2.4 Speed = Distance
Time

80 km/h = 276 km
Time

Time = 276 ■■
80 ■■/■
= 3,45 hours
= 0,45 × 60 = 27 minutes
Travel time = 3h27min
Arrival time = 12:45
+ 03:27
= 16:12
■ Siphokhazi’s claim is not valid

1SF substitution

1M changing subject
of formula
1CA time
1C converting hrs to
min

1M adding time

1CA arrival time
1O opinion
(7) MP
L4
[27]

■✓RT
■C
■M
■CA
■SF
■M
■CA
■C
■M
■CA
■O

QUESTION 3 [ 23 MARKS]
Ques. Solution Explanation Level
3.1.1 Radius = 50
10 = 5 cm

Volume = 3,142 × 52 × 12
= 942,6 cm3 1C radius in cm

1SF substitution
1CA volume in cm3
(3) M
L2

3.1.2 Width = 0,08 × 1 000
= 80 mm

Height = 14,5 × 10
= 145 mm

TSA = 2 (150 × 80) + 2 (150 × 145) + 2 (145 × 80)
= 24 000 mm2 + 43 500 mm2 + 23 200 mm2

= 90 700 mm2
≈ 91 000 mm2 1C multiply by 1  000
1A width in mm
1C height in mm
1SF substitution

1S simplification
1R rounding

(6) M
L3

3.1.3
Amount of plastic needed = 91 000
1 000  000

= 0,091 m2 × 50
= 4,55 m2
≈ 5 m2

Cost of plastic = 5 m2 × R27,55
= R137,75
CA from 3.1.2
1C divide by
1 000 000

1M multiply by 50

1M multiply area with
cost
1CA total cost
(4) M
L3

3.1.4
Height = 7
100 × 12 cm =
0,84 cm
■ 12cm – 0,84 cm
= 11,16
≈ 11,2 cm
OR

Height = 100% - 7%
= 93%  ■ CA

■ 93
100 × 12 cm
= 11,16 cm
≈ 11,2 cm  1M calculating 7% of
12
1CA answer
1M subtracting 7%
from container height

OR

1M subtracting 7%
1CA beans height in
%
1M calculating 93%

of 12

(3) M
L2

■C
■SF
■CA
■C
■M
■M
■CA
■M
■CA
■M
■M
■M ■C
■A
■C
■SF
■S
■R

3.1.5 Volume of cylindrical -shaped container = 942,6 cm3
Rectangular shaped container: Length = 15 cm
Width = 8 cm
Height = 14,5 cm
Volume = L × W × H
= 15 cm × 8 cm × 14,5 cm
= 1 740 cm3
Her statement is incorrect / not valid

1C length in cm

1C width in cm

1SF substitution

1CA volume in cm3

1O opinion
(5) M
L4

3.1.6 To keep the coffee beans fresh
OR
To protect container against breakage
(Accept any other relevant explanation) 2O reason

(2) M
L4
[23]

■C
■C
■SF
■CA
■O
■✓O
■✓O

QUESTION 4 [26 MARKS]

Ques. Solution Explanation Level

4.1.1 Probability = 7
11 × 100%

= 63,64% 1A correct fraction
1M multiply by 100%
1CA probability as a
%
(3) P
L2

4.1.2 Travelling time = 10:15
- 08:30
= 01:45
■ 0,45 ÷ 60 = 0,75 h
Time = 1 h + 0,75 h = 1,75 hours

Distance = Speed × Time
= 1,4 miles/h × 1,75 h
= 2,45 miles
■ 2,45 miles × 1,60934
= 3,942883 km
Her statement is invalid

OR

Travelling time = 10:15
- 08:30
= 01:45
■ 0,45 ÷ 60 = 0,75 h
Time = 1 h + 0,75 h = 1,75 hours

Speed = 1,4 miles/h × 1,60934
= 2,253076km/h
Distance = Speed × Time
= 2,253076km/h × 1,75 h
= 3,942883 km
Her statement is invalid
1M subtracting time

1C minutes to hours
1CA traveling time in
hours

1SF substitution
1C multiply with
conversion factor
1CA distance in km
1O opinion

OR
1M subtracting time

1C minutes to hours
1CA traveling time in
hours

1C convert speed to
km/h
1SF substitution
1CA distance in km
1O opinion
NPR
(7) MP
L4

4.2.1 Area of square = side × side
= 45 cm × 45 cm
= 2 025 cm2
≈ 2 030 cm2 1SF substitution
1A area of box

1R rounding
(3) M
L2

4.2.2 Value of D:
2D = 45 cm – 42 cm
= 3 cm
D = 3 ■■
2
= 1,5 cm 1M subtraction
1M divide by 2
1CA value of D

(3) M
L2

■A ■M

■CA
■M
■C
■CA
■SF
■C
■CA
■O
■SF
■A
■R
■M
■CA ■M
■C
■CA
■SF ■C
■CA
■O
■M

4.2.3 Area of pizza (circle) = 3,142 × 212
= 1 385,622 cm2
Area of one slice = 1 385 ,622
8
= 173,20 cm2 1A radius
1SF substitution
1M dividing area by
8

1CA area of one
slice
(4) M
L3

4.3.1 Diameter = 160 cm + 35 cm + 35 cm
= 230 cm
Radius = 230
2
= 115 cm 1M adding correct
values

1M divide diameter
by 2
1CA radius
(3) M
L2

4.3.2 Height of support pole = 78 cm ■M – (2,8 cm + 0,7 cm) ■M
= 74,5 cm 1M subtraction
1M adding base and
thickness
1CA height in cm
(3) M
L2
[26]

TOTAL : 100

■A ■SF

■M
■CA
■M
■M
■CA
■CA

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 202 5

MATHEMATICAL LITERACY P1
MARKING GUIDELINE

MARKS: 100

This marking guideline consists of 8 pages.

| Symbol | Explanation |
|---|---|
| M | Method |
| MA | Method with accuracy |
| CA | Consistent accuracy |
| A | Accuracy |
| C | Conversion |
| S | Simplification |
| RT | Reading from a table/a graph/document/diagram |
| SF | Correct substitution in a formula |
| O | Opinion/Explanation |
| P | Penalty, e.g. for no units, incorrect rounding off, etc. |
| R | Rounding off |
| NPR | No penalty for correct rounding minimum two decimal places |
| AO | Answer only |
| MCA | Method with constant accuracy |

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
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- Consistent accuracy applies in ALL aspects of the marking guideline. Stop marking at the second calculation error.
- NOTE : Consistent accuracy (CA) does NOT apply in cases of a breakdown.
- If the candidate presents any extra solution when reading from a graph and table , then penalise for every extra item presented.
- As a general marking principle, if a candidate has incurred one mistake and there is evidence of sound Mathematics thereafter, then that candidate should lose ONE mark only.

Topics: F – Finance, DH – Data Handling, P – Probability

QUESTION 1 [22 Marks]

No. Solution Explanation Topic and
Level
1.1 1.1.1 2 457 164 834 ✓✓A 2A correct answer
(2) F
L1
1.1.2 2 Cheques/Two cheques ✓✓A 2A correct number
of cheques
(2) F
L1
1.1.3 Amount of cash = R276,70 ✓✓A 2A correct amount
(2) F
L1
1.1.4 Total Amount = R260,00 + R16,70 + R340,00 +
R480,00 ✓MA
= R1 097,50 ✓A 1MA Adding correct
amounts
1A correct answer

(2) F
L1
1.2 1.2.1 Numerical ✓✓A 2A correct answer
(2) D
L1
1.2.2 73 ✓✓A 2A correct points
(2) D
L1
1.2.3 TSA and AMA ✓✓A 2A correct teams
(2) D
L1
1.3 1.3.1 C ✓✓ A 2A correct definition
(2) D
L1
1.3.2 E ✓✓A 2A correct definition
(2) F
L1
1.3.3 A ✓✓A 2A correct definition
(2) P
L1
1.3.4 F ✓✓A 2A correct definition
(2) F
L1
[22]

AO – FULL MARKS

QUESTION 2 [27 Marks]

No. Solution Explanation Topic and
Level
2.1 2.1.1 Deposit saved =16
100×■499 900 ,00 ✓MA

=■79 984 ,00 ✓A

1MA multiplying

correct amount by
16%
1A correct answer

(2) F
L2

2.1.2 Balance = R499 900,00 – R79 984,00 ✓MA
= R419 916,00 ✓A

Repayment =■419 916 ,00+(9,5
100×
■419 916 ,00)×3 ✓MA
= R419 916,00 + (R39 892,02 x 3)
= R419 916,00 + R119 676,06 ✓MA
= R539 592,06 ✓CA

Loan amount can also be calculated as :

■■
■■■×■■■■ ■■■ =■■■■ ■■■ ,■■ CA from 2.1.1
1MA subtracting
correct deposit
1A correct balance
1MA multiplying
by 9,5% and 3
1MA adding
correct balance
1CA answer

(5) F
L4

2.1.3 Monthly instalment = 539 592 ,06
36 ✓
= R14 988 ,67 ✓ CA from 2.1.2
1M division by 36
1A answer rounded
to 2 DP

(2) F
L2

2.1.4 P(Silver Car) =3✓A
14✓A=0,21 ✓R 1A numerator
1A denominator
1R rounding

(3) P
L2

2.2 2.2.1 Number of builders x number of days = 36 ✓✓A
OR
Number of Builders = 36
■■■■■■ ■■ ■■■■ ✓✓A 2A correct formula

(2) F
L2

2.2.2
No. of builders 3 4 6 9 12
No. of days 12 9 ✓ 6 ✓ 4 ✓ 3
1A 9 days
1A 6 days
1A 4 days

(3) F
L2

2.2.3 Number of builders vs No. of days

Number of builders

✓✓✓ All
points plotted
✓ joining the
points with a
smooth
curve

(4)

2.2.4 Direct Proportion ✓✓A 2A direct
(2) F
L1

2.3 Inflation rate = Price Difference x 100
Original Amount
✓M
= ■20 600 – ■20 000
■20 000×100 ✓M ✓M

= 3% ✓CA

1M
subtracting
correct values
1M
multiplying by
100
1M division
by R20 000
1CA answer
(4) F
L2
[27]

Number of days

QUESTION 3 [23 Marks]

No. Solution Explanation Topic
and
Level

3.1 3.1.1 % = 100 – (12 + 26 + 8 + 20 + 27 + 7) ✓M
= 100 – 93 ✓M
= 7% ✓CA 1M adding
correct values
1M subtracting
1CA % (3) D
L1

3.1.2 Number of learners in Buffalo City = 26

100×20 910 ✓M

= 5 436,6 ✓S
= 5 437 learners ✓R 1M correct
value
1S simplification
1R whole
number (3) D
L2
3.1.3 7% 8% 12% 20% 26% 27% ✓M

Median = 12% + 20% ✓M
2
= 16% ✓CA 1M arranging in
order
1M concept of
median
1CA answer in
percentages
(3) D
L3
3.1.4 Sarah Baartman ✓✓RT 2RT correct
district (2) D
L2
3.1.5 Mean =7+8+12+20+26+27
6 ✓M ✓M
=16,67% ✓CA
1M adding
correct values
1M concept of a
mean
1CA correct mean
(3) D
L2

3.1.6 P(District less than 10%) =2
6 ✓A✓A
=1
3 ✓CA 1A numerator
1A denominator
1CA simplified
fraction
(3) D
L2
3.2
3A 1 mark for
every 3 points
plotted correctly
1A for title
2A for x - and y -
axis

(6) D
L4
[23]

0102030405060
0 10 20 30 40 50 60No of learners contracted TB
No of AntibioticsThe results of learners who contracted TBCA from 3.1.1
CA from 3.1.1
CA from 3.1.1

QUESTION 4 [28 Marks]

No. Solutions Explanations Topic
and
Level
4.1 4.1.1 Cost = R4 950 + R4 750 ✓M
= R9 700 ✓A
= R9 700 + R1 455 ✓M
= R11 155 ✓CA 1M adding correct
amounts
1A answer
1M adding VAT
amount
1CA answer (4) F
L3
4.1.2 Profit = Income – Expenses
= R135 – R95 ✓RT ✓M
= R40 ✓A
= R40 × 25 ✓M
= R1 000 ✓CA
INVALID ✓O
OR
Cost = 25 x 95
= R2 375
Income = 135 x 25
= 3 375
Profit = 3 375 – 2 375
= 1 000
INVALID ✓O 1RT correct values
1M subtracting
correct values
1M multiplying by
25
1CA answer
1O conclusion
OR
1RT correct values
1M subtracting
correct values
1M multiplying by
25
1CA answer
1O conclusion (6) F
L4

4.1.3 (a) Values = W(5; 675) ✓A
The income is equal to expenses ✓A 1A correct points
1A correct meaning
(2) FL1
(b) Cost = R2 00,00 + R9 5,00 x number of liters ✓✓MA 2MA correct
formula (2) F
L2

4.2 R1 = € 0,049669
? = € 45 000
€45 000

€0,049669×■1 ✓RT ✓M

= R905 997,70 ✓CA 1RT correct amount
1M dividing by the
correct rate
1CA simplification

(3) F
L4

4.3 Range of marks = 75 – 15 ✓RT ✓MA
= 60 ✓CA
Range as a %: 60
75×100 =80% ✓M
VALID ✓O 1MA concept of
range
1RT correct values
1CA range of the
data
1M multiply by 100
1O opinion (5) D
L3

4.4 ✓MA ✓M
Block 1 = 350 kWh x R2,216 = R775,60
Block 2 = 24 kWh x R2,711 = R65,064 ✓M
= R840,664 x 1,15 ✓MA
= R966,7636 ✓CA
= R966,76 ✓R 1MA correct kWh
1M cost of 350 kWh
1M cost for 24 kWh
1MA multiplying
by 1,15
1CA answer
1R rounding
(6) F
L4
[28]

TOTAL: 100

MARKS /PUNTE : 75

| Symbol/ Kode | Explanation/ Verduideliking |
|---|---|
| M | Method/ Metode |
| MA | Method with accuracy/ Metode met akkuraatheid |
| CA | Consistent accuracy/ Volgehoue akkuraatheid |
| A | Accuracy/ Akkuraatheid |
| C | Conversion/ Herleiding |
| S | Simplification/ Vereenvoudiging |
| RT | Reading from table/a graph/document/diagram/ Lees vanaf tabel/grafiek/ document/ diagram |
| SF | Correct substitution in a formula /Korrekte vervanging in formule |
| O | Opinion/Explanation/Reasonig/ Opinie/Verduideliking/Redenasie |
| P | Penalty, e.g. for no units, incorrect rounding off, etc./ Penalisasie, bv. Vir geen eenhede/verkeerde afronding, ens. |
| R | Rounding/ Afronding |
| NPR | No penalty for correct rounding/ Geen penalisasie vir korrekte afronding nie |
| AO | Answer only/ Slegs antwoord |
| MCA | Method with constant accuracy/ Metode met volgehoue akkuraatheid |
| RCA | Rounding consistent with accuracy/ Afronding met volgehoue akkuraatheid |

These marking guidelines consist of 9 pages. /
Hierdie nasienriglyne bestaan uit 9 bladsye.

PROVINCIAL ASSESSMENT /
PROVINSIALE ASSESSERING
GRADE 11 /
GRAAD 11

MATHEMATICAL LITERACY P1/
WISKUNDIGE GELETTERDHEID V1
JUNE 2024 / JUNIE 2024
MARKING GUIDELINES/ NASIENRIGLYNE
`

NOTE:

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- As a general principle, if a candidate has incurred one mistake and there is evidence of sound mathematics thereafter, then that candidate should lose one mark only.
- Rounding is an independent mark.
- In order to award the verification/conclusion mark the candidate must have scored at least one mark in the calculation preceding the final conclusion.

LET WEL:

- As ‘n kandidaat ‘n vraag TWEE KEER beantwoord, sien slegs die EERSTE poging na.
- As ‘n kandidaat ‘n antwoord van ‘n vraag doodtrek (kanselleer) en nie oordoen nie, sien die doodgetrekte (gekanselleerde) poging na.
- Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyne toegepas, dit hou op by die tweede berekeningsfout.
- LET WEL : volgehoue akkuraathe id (CA) geld nie  in die geval van ‘n afbreuk nie.
- Wanneer ‘n kandidaat aflesings vanaf ‘n grafiek, tabel, uitlegplan en kaart geneem en ekstra antwoorde gee, penaliseer vir elke ekstra item.
- ‘n Algemene nasienbeginsel is dat indien ‘n kandidaat een fout maak en daarna voortgaan met korrekte wiskunde, dat die kandidaat slegs een punt verloor.
- Afronding tel as ‘n afsonderlike punt.
- Ten einde die verifikasie/gevolgtrekking punt toe te ken moes die kandidaat ten minste een punt gekry het in die berekeninge wat lei tot die finale gevolgtrekking.

QUESTION 1
/VRAAG 1   [14 MARKS/ PUNTE ] AO – Full marks
Q/V Solution /Oplossing  Explanation /Verduideliking  T/L
1.1.1  C ✓✓A 2A Correct option
(2) DH
L1
1.1.2  A ✓✓A 2A Correct option
(2) F
L1
1.1.3  B ✓✓A 2A Correct option
(2) DH
L1
1.1.4  C ✓✓A 2A Correct option
(2) DH
L1
1.2.1  Gross income / Bruto inkomste
= R14 000,00 + R11 500,00  ✓MA
= R25 500,00  ✓A
1MA Adding income values
1A Answer gross income
(2) F
L1
1.2.2  B ✓✓A 2A Correct answer
(2) F
L1
1.2.3 Net income / Netto inkomste  ✓✓A 2A Answer
(2) F
L1
[14]

QUESTION
2/VRAAG 2  [22 MARKS/ PUNTE ]
Q/V Solution/ Oplossing  Explanation/ Verduideliking  T/L

2.1.1 Cell phone make = Samsung ✓✓A
2A Correct answer
(2) F
L1
2.1.2 ✓RT
Subtotal = R2 607,83 × 15% ✓MA
= R391, 17

1RT Use c orrect value
1MA Multiply with 15%
(2) F
L1
2.1.3 ENG:
Two thousand, Nine hundred and
ninety nine Rand ✓✓A

AFR:
Tweeduisend, Negehonderd Nege -en-
negentig Rand ✓✓A

2A Correct answer

(2) F
L1
2.1.4
a) Quote valid from
15 March + 14 days ✓MA
28 March ✓A

1MA add 14 days to 15 March
1A Answer correct date
(2) F
L2
2.1.4
b) Quotes are only valid for a certain time period because prices of good s and services go up after a while, and the supplier don't want to lose money /
✓✓O
Kwotasies is slegs geldig vir 'n sekere tydperk want pryse van goedere en dienste gaan op na 'n tydperk, en die verskaffer wil nie geld verloor nie
✓✓O

2O Explanation

(2) F
L4
2.2.1 ✓✓A
Number of minutes / Aantal minute
2A Answer
(2) F
L1
2.2.2 Fixed cost / Vaste koste
= R200 ✓✓A

2A Answer
(2) F
L1
2.2.3 Prepaid option / Vooruitbetaalde opsie

= R40 0 ÷ 200 minutes ✓MA
= R2 per minute

A = Prepaid price plan /
Vooruitbetaalde prysplan
✓A
Total cost = R2,00 × number of
minutes ✓A

1MA Dividing with 20 minutes

1A Correct value R2,00
1A Multiply with number of minutes
(3) F
L3
2.2.4 Break even pint / Gelykbreekpunt
= 130 minutes ✓✓RT

2RT Correct answer
(2) F
L2

2.2.5
Contract option / Kontrak opsie
= R344 – R200✓MA
= R144 for 200 minutes

* 80 free minutes
= 200 minutes – 80 minutes ✓MA
= 120 minutes

Cost per minute / Koste per minuut
= R144 ÷ 120 minutes
= R1,20 per minute ✓

1MA Subtract fixed cost

1MA Subtract free minutes

1A Answer cost per minute - contract
(3)
Any value s from table can be used
to calculate the cost per minute F
L3
[22]

QUESTION

3/ VRAAG 3   [20 MARKS/ PUNTE ]
Q/V Solution/ Oplossing  Explanation/ Verduideliking  T/L
3.1.1  Other colour cars / Ander kleur motors
= 2 + 3 + 5 + 2 + 4 + 2  ✓MA
= 18 ✓A
1MA Adding correct values
1A Answer coloured cars
AO (2) DH
L2
3.1.2  Average (mean) / Gemiddeld
= 7 + 9 + 12 + 8 + 13 + 11
6  ✓
= 60 ÷ 6 months ✓
= 10 ✓

1M Adding white cars
1M Dividing with 6 months
1CA Average white cars
(3) DH
L2
3.1.3  7; 8; 9; 11; 12; 13 ✓MA
Median = 9+ 11
2 ✓MA
Median = 20 ÷ 2
Median = 10 ✓A
1MA Arrange values

1MA Median concept

1A Median white cars
(3) DH
L2
3.1.4 Range = biggest – smallest/
Omvang = Grootste – kleinste
8 = Big – 2 ✓SF
Big = 8 + 2  ✓M
Big = 10  ✓CA

1SF Substitution
1M Change subject of formula
1CA Answer
(3) DH
L3
3.1.5 Probability ( March white car )/
Waarskynlikheid (Maart wit motor)
= 12  ✓A
60  ✓A
= 0,2 ✓CA

1A Numerator
1A Denominator
1CA Decimal number
(3) P
L3
3.2 ANSWER SHEET /
ANTWOORDBLAD  1A Correct bars – White cars Feb
1A Correct bar  – White cars May
1A Correct bar  – Other cars Jan
1A Correct bar – Other cars May
(4) DH
L2

3.3 Fuel consumption / Brandstof
verbruik ✓✓ O
OR
Number of kilometres on the clock /
Aantal kilometre op die klok ✓✓ O

OR
Price of the car/ Prys van die kar ✓✓ O

2O Reason

(2) DH
L4
[20]

QUESTION 4/VRAAG 4 [19 MARKS/PUNTE]
Q/V Solution/ Oplossing Explanation/ Verduideliking T/L
4.1 % cleaning water of total water
= 2,5 ■■
5,5+2,5+4,8+1,3+5,9 ✓■ ×100
= 2,5 ■■
20 ■■ ×100 ✓M
= 12,5% ✓CA

1M Adding total water consumption

1M Multiply fraction with 100

1CA Answer %
(3) F
L2
4.2 P (bath water) /(badwater )
= 2,8 ■ ✓A
4,8 ✓A
= 7
12 ✓CA

1A Numerator
1A Denominator
1CA Simplified fraction
(3) P
L2
4.3 Block 1
✓MA
0-6 = 6 k■ × 0,00c = 0,00c
7-15 = 9 k■ × 1095c = 9855c ✓MA
16-20 = 5 k■ × 1248c = 6240c

Total cost in cent
= 0,00c + 9855c + 6240 c
= 16095c ✓CA

Total cost in Rand

= 16095c ÷ 100
= R160,95 ✓C

Total cost in Rand, VAT added
= R160,95 × 1,15 ✓M
= R185,09 ✓CA

His claim is NOT valid ✓O
Sy bewering is NIE geldig NIE

1MA Dividing 20 kl into blocks
1MA Multiply with correct cost per block

1CA Total cost in cent

1C Convert to rand

1M Multiply with VAT
1CA Total cost with VAT

1O Opinion
(7) F
L4

4.4 Continuous /Kontinu ✓A
It is values that can be measured ./ Dit is waardes wat gemeet kan word .✓✓O
1A Answer - Continuous

2O Reason
(3) DH
L4

4.5 Pie chart /Sirkeldiagram ✓A
It is better to compare percentages than only indicating different values. / Dit is beter om persentasies te vergelyk as om net die verskillende waardes aan te dui . ✓✓O
1A Type of graph

2O Percentage’s reason

(3) DH
L3
[19]
TOTAL /TOTAAL : 75

02468101214
January February March April May JuneNumber of cars
MonthsNumber of cars sold by Mr. Botha from January -June

White cars
Other colour cars
✓A
✓A ✓A
✓A

MARKS : 75

| Symbol | Explanation |
|---|---|
| M | Method |
| MA | Method with accuracy |
| MCA | Method with consistent accuracy |
| CA | Consistent accuracy |
| A | Accuracy |
| C | Conversion |
| S | Simplification |
| RT | Reading from a table/a graph/document/ diagram |
| SF | Correct substitution in a formula |
| O | Opinion/Explanation/Reasoning |
| P | Penalty, e.g. for no units, incorrect rounding off, etc. |
| R | Rounding off |
| NPR | No penalty for correct rounding |
| AO | Answer only |

These marking guidelines consist of 5 pages.

# MATHEMATICAL LITERACY P2
JUNE 2024
MARKING GUIDELINES PROVINCIAL ASSESSMENT
GRADE 11

NOTE:
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- NOTE: consistent accuracy (CA) does not apply in cases of a breakdown.
- If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra item presented.
- General principle of marking, if a candidate has incurred one mistake and there is evidence of sound mathematics thereafter, then that candidate should lose one mark only.

- Rounding is an independent mark.
- In opinion type questions marks will only be awarded if relevant calculations are shown.

QUESTION 1 [14 MARKS ] ANSWER ONLY – FULL MARKS
Q Solution Explanation T/L

1.1.1
4 people✓✓A
2A correct answer
(2) M
L1

1.1.2
10 + 40 min ✓MA
= 50 min✓A
1MA adding
1A answer
(2)
M
L1

1.1.3
✓MA
1
4 cup : 60 ml OR 60 ml ÷ 1
4 ✓MA OR 60 ml × 4
= 1 ÷ 1
4 × 60 ml✓MA = 60 ml ÷ 0,25 = 240 ml ✓A
= 240 ml✓ A = 240 ml ✓A

1MA multiplying and /or
dividing
1A answer
(2) M
L1

1.1.4
18 : 37✓✓A
2A correct format
(2) M
L1

1.2.1
1 unit on the map represents 50 units on the actual
ground/in reality ✓✓A
2A correct explanation
(2) MP
L1

1.2.2
Bar/ Linear / Graphic scale ✓✓A
2A correct answer
(2) MP
L1

1.3
Results of an experiment/trial ✓✓A
2A correct definition
(2) P
L1

[14]

Copyright reserved    Please turn over  QUESTION 2     [20 MARKS]
QUES  SOLUTION  EXPLANATION  T&L

2.1.1
4 provinces ✓✓RT
2RT reading from the map
(2) MP
L1

2.1.2
Zeerust, Swartruggens, Rustenburg,
Hartebeespoort ✓✓ RT (Accept Kroondal)
2RT ANY two towns
(2) MP
L1

2.1.3
(a)

104  882
1 220  813✓A ×100✓MA

= 8,59%✓R
1A correct fraction
1MA finding a %
1A rounded answer
(3) MP
L2

(b)

Botswana is not in South Africa . ✓✓O
OR
North West is in another country (South Africa)  ✓✓O
OR
People need passports to travel from one country to
another.  ✓✓O
2O opinion
(2) MP
L4

2.2.1
Top/ Aerial/Bird's eye view ✓✓A
2A correct answer
(2) MP
L2

2.2.2
Distance = speed ×  time
460 km   = speed ×  4,5 hours✓SF
Speed = 460 km ÷ 4,5 hours ✓M
= 102,2 km/h ✓A

1SF substitution
1M changing the subject
1A answer
(3) MP
L2

2.2.3
Amount of petrol = 6,42 ×460
100✓MA
= 29,532 litres ✓A

Return trip = 29,532 ×2 ✓MCA
= 59,064 litres ✓CA
OR
Return trip = 460 km × 2✓MA = 920 km ✓A
Amount of petrol = 6,42 × 920
100✓MCA
= 59,064 litres ✓CA
1MA multiplying and
dividing
1A answer
1MCA multiplying by 2
1CA simplification

(4) MP
L3

2.2.4
Less likely ✓✓A

2A correct answer
(2) P
L2
[20]

QUESTION 3 [24 MARKS]

3.1.1 Total height = (15 + 17 + 19 + 21) cm ✓M
= 72 cm✓CA
= 720 mm ✓C

OR
21 cm = 210 mm: 19 cm = 190 mm; 17 cm = 170 mm
and 15 cm = 150 mm ✓C

Total height = (210 + 190 + 170 + 150) mm ✓MA
= 720 mm ✓CA 1MA adding all values
1CA answer
1C conversion

OR

1C Conversion
1MA adding
1CA answer
(3) M
L2

3.1.2
Pounds → grams : 3,5 × 453,592 ✓C
= 1 587,572 g ✓A
g → kg : 1 587,572 ÷ 1 000
= 1,59 kg ✓C
1C conversion
1A answer
1C conversion
NPR (3) M
L3

3.1.3
(a)
Diameter = 14 cm × 2 ✓MA
= 28 cm✓A 1MA multiplying
1A answer
AO (2) M
L1

(b)
Volume = 3,142 × (14 cm)2 ✓SF × 15 cm✓SF
= 9 237,48 cm3
The volume is CORRECT ✓O
1SF radius squared
1SF substitution
1O opinion
(3) M
L4

3.1.4
■–32° = 9
5 ×5✓SF
■ = 9 + 32✓M
= 41✓CA

1SF substitution

1M changing the subject
1CA simplification
(3) M
L2

3.2.1
15
60✓MA
= 0,25✓A
1MA dividing
1A answer
AO (2) M
L1

3.2.2
(a)
True✓✓A
2A correct answer
(2) M
L2

(b)
False✓A
The distance from Tom's school to Neo's school is
5 km✓O
1A correct choice

1O reason
(2) M
L2

3.2.3
Time taken = 06:47 + 50 min ✓A + 20 min✓MCA
= 07 : 57 a.m. ✓CA
before 8:00 Statement is correct ✓O
OR
Time taken: 50 min ✓A + 20 min = 1hr + 10 min
06:47 + 1 hr 10 min ✓MCA = 07:5 7✓CA
before 8:00 Statement is correct ✓O
1A identifying 50 min
1MCA adding
1CA answer
1O opinion
(4) M
L4
[24]

QUESTION 4 [17 MARKS ]
Q Solution Explanation T/L

4.1.1
Circumference = 3,142 × 144 cm ✓SF
= 452,448 cm ✓A
1SF substitution
1A answer
NPR (2) M
L2

4.1.2
Area of a circle = 3,142 × (144
2)2✓SF
= 16 288,128 cm2✓A
Area of a square = 2302✓SF
= 52 900 cm2✓A
Shaded area = 52 900 cm2 - 16 288,128 cm2✓MCA
= 36 611,872 cm2✓CA
1SF substitution
1A answer
1SF substitution
1A answer
1MCA subtraction
1CA answer
(6) M
L3

4.1.3
36 611,872 ÷ 10 000/1002✓MA
= 3,6611872 m2✓CA CA area from 4.1.2
1MA dividing
1CA simplification
NPR (2) M
L2

4.2.1
4 offices✓✓RT
2RT reading from the
diagram

(2) MP
L1

4.2.2
Actual length = 2,2 cm × 300 ✓MA
= 660 cm ✓A
= 6,6 m✓C
1MA using scale
1A answer
1C conversion to m
(3) MP
L2

4.2.3
Space /Area for learners/ parents/ visitors to wait before they can be assisted. ✓✓O
2O opinion
(2) MP
L4
[17]

TOTAL : 75

MARKS/ PUNTE : 100

| Symbol /Kode | Explanation /Verduideliking |
|---|---|
| M | Method /Metode |
| MA | Method with accuracy /Metode met akkuraatheid |
| MCA | Method with consistent accuracy /Metode met volgehoue akkuraatheid |
| CA | Consistent accuracy /Volgehoue akkuraatheid |
| A | Accuracy /Akkuraatheid |
| C | Conversion /Herleiding |
| S | Simplification /Vereenvoudiging |
| RT | Reading from a table/a graph/document/ diagram /Lees vanaf tabel/grafiek/document/diagram |
| SF | Correct substitution in a formula Korrekte vervanging in formule |
| O | Opinion/Explanation/Reasoning /Opinie/Verduideliking/Redenasie |
| P | Penalty, e.g. for no units, incorrect rounding off, etc ./Penalisasie, bv. vir geen eenhede/verkeerde afronding, ens. |
| R | Rounding off //Afronding |
| NPR | No penalty for correct rounding /Geen penalisering vir korrekte afronding nie |
| AO | Answer only /Slegs antwoord |

These marking guidelines consists of 8 pages/ Hierdie nasienriglyne bestaan uit 8 bladsye .

MATHEMATICAL LITERACY P2
WISKUNDIGE GELETTERDHEID V2
NOVEMBER 2024
MARKING GUIDELINES/ NASIENRIGLYNE

PROVINCIAL ASSESSMENT/
PROVINSIALE ASSESSERING

GRADE /GRAAD 11

Mathematical Literacy/P2 /Wiskundige Geletterdheid/V 2 2 NW/November 2024
Grade/ Graad 11 – Marking Guideline s/Nasienriglyne

Copyright reserved /Kopiereg voorbehou Please turn over/ Blaai om asseblief NOTE:
■ If a candidate answers a question TWICE, only mark the FIRST attempt.
■ If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution,
mark the crossed out (cancelled) version.
■ Consistent accuracy (CA) applies in ALL aspects of the marking guidelines; however, it stops
at the second calculation error.
■ NOTE: consistent accuracy (CA) does not apply in cases of a breakdown.
■ If the candidate presents any extra solution when reading from a graph, table , layout plan and
map, then penalize for every extra item presented.
■ As a general marking principle, if a candidate has incurred one mistake and there is evidence
of sound mathematics thereafter, then that candidate should lose one mark only.
■ Rounding is an independent mark.
■ In opinion type questions marks will only be awarded i f relevant calc ulations are shown.

LET WEL:
■ As ‘n kandidaat ‘n vraag TWEE KEER beantwoord, sien slegs die EERSTE poging na.
■ As ‘n kandidaat ‘n antwoord van ‘n vraag doodtrek (kanselleer) en nie oordoen nie, sien die doodgetrekte
(gekanselleerde) poging na.
■ Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyne toegepas, dit hou op by die
tweede berekeningsfout.
■ LET WEL: volgehoue akkuraatheid (CA) geld nie in die geval van ‘n afbreuk nie.
■ Wanneer ‘n kandidaat aflesings vanaf ‘n grafiek, t abel, uitlegplan en kaart geneem en ekstra antwoorde
gee, penaliseer vir elke ekstra item.
■ ‘n Algemene nasienbeginsel is dat indien ‘n kandidaat een fout maak en daarna voortgaan met korrekte
wiskunde, dat die kandidaat slegs een punt verloor.
■ Afronding te l as ‘n afsonderlike punt.
■ Ten einde die verifikasie/gevolgtrekking punt toe te ken moes die kandidaat ten minste een punt gekry
het in die berekeninge wat lei tot die finale gevolgtrekking.

QUESTION 1/VRAAG [19 MARKS/ PUNTE ] Answer only AO – full marks

Q/V Solution /Oplossing Explanation/ Verduideliking T/L
1.1.1
(a) Thermometer/ Termometer ■■A 2A answer/ antwoord
(2) M
L1
E
(b)
The degree of hotness or coldness of a substance/ Die
grade van hoe warm of koud ‘n voorwerp is ■■A 2A answer/ antwoord
(2) M

L1
E
(c)
43 oC■■A 2A answer/ antwoord
(2) M
L1
E
1.1.2
400 ml■RT
0,4 litres/ liters■A 1RT correct reading/ korrek
gelees
1CA conversion/ herleiding
(2) M
L1
E
1.1.3 Distance/ Afstand = 9,7 cm – 2,9 cm■RT
= 6,8 cm■CA
= 68 mm■C
1RT reading from the diagram
/lees vanaf die diagram
1CA answer/ antwoord
1C conversion/ herleiding M
L1
M

OR/OF
Distance /Afstand = 97 mm – 29 mm■RT■C
= 68 mm■CA OR/OF
1RT reading values/ lees
waardes
1C conversion/ herleiding
1CA answer// antwoord
(3)
1.2.1 Strip Chart/map/ Strookkaart ■■A 2A answer/ antwoord
(2) MP
L1
E
1.2.2 Distance /Afstand = 117 km ■■RT

OR/OF
Distance/ Afstand = 557 km – 440 km■RT
= 117 km ■A 2RT reading from the map/ lees
vanaf kaart
OR/OF
1RT read and subtraction/ lees
en aftrek
1A answer/ antwoord
(2) MP
L1
M
1.2.3 Escourt■■RT 2RT reading from the map/ lees
Vanaf kaart
(2) MP
L1
E
1.2.4 5 towns/ dorpe■■RT
Accept 6 towns /aanvaar 6 dorpe 2RT reading from the map/ lees
Vanaf kaart
(2) MP
L1

E
[19]

QUESTION/ VRAAG 2 [24 MARKS/ PUNTE ]

2.1.1 Food/Kitchen /Kos/kombuis ■■RT 2RT answer /antwoord

(2) MP
L1
E

2.1.2 South/ Suid (S) ■A and/en East/ Oos (E)■A
OR/OF
Southern/ Suidelik and/en Eastern/ Oostelik
1A South/ Suid
1A East/ Oos

(2) MP
L2
E

2.1.3
(a) 37■RT and/en 38■RT 1RT room/ kamer 37
1RT room/ kamer 38
(2) MP
L2
E

(b) The rooms are close to the corner ■A of Mimosa ■RT and Flower Street ■RT/
Die klaskamers is naby die hoek ■A van Mimosa ■RT en Flower Straat ■RT
OR/OF
Adjacent/next to corner of Mimosa and Flower Street
Op/langs die hoek van Mimosa en Flower straat
1A corner/adjacent/next to /hoek van/langs aan
1RT Mimosa
1RT Flower

(3) MP
L3
M

2.1.4 Classroom/ Klaskamer 14■■RT 2RT number/ nommer 14

(2) MP
L2
M

2.1.5 Traffic congestion can be avoided when learners are picked up or dropped off/ Verkeersopeenhoping kan verhoed word as leerders op gelaai of afgelaai word.■■O
OR
Learners might be crossing the road, and a two-way traffic can cause accidents/ Leerders mag dalk die pad oorsteek, en tweerigting-verkeer kan ongelukke veroorsaak 2O reason/ rede

(2) MP
L4
M
2.1.6 P = 4■A
38■A ×100

= 10,52631579% ■CA

= 10,526%■R 1A numerator/ teller
1A denominator/ noemer
1CA simplification/
vireenvoudiging
1R rounding/ afronding
(4) P
L3
D
2.2.1 12 km = average speed × 60 min ■SF
12 000 m ■C = average speed × 60 min
Average speed/ gemiddelde spoed
= 12 000 m ÷ 60 min ■M
= 200 metres/ minute■CA

OR/OF

Average speed = distance ÷ time ■M
12 km ÷ 60 min ■SF
= 12 000 m ÷ 60 min ■C
= 200 m/min ■CA
1SF substitution/ vervanging
1C conversion/ herleiding
1M changing the subject of the
formula/ verander die
onderwerp van die formule
1CA simplification/
vereenvoudiging
OR/OF

1M changing the subject of the
formula/ verander die
onderwerp van di e formule
1SF substitution/ vervan gding
1C conversion/ herleiding
1CA simplification/
vereenvoudiging
(4) MP
L3
M
2.2.2 200 m/min is too fast for walking and too slow for travelling by car or taxi./ 200 m/min is te vining om te loop en te stadig om met ‘n motor of minibus te ry.■■O
Thus, the learner was cycling/running/ Dus, moet die leerder fiets ry of hardloop ■O
OR/OF
Any other sensible answer/ Enige sinvole antwoord
2O reason/ rede

1O opinion/ opinie

(3) MP

L4

D

[24]

3.2.2

Number of pallets/ hoeveelheid palette
= 8 386 ÷ 594 ■MCA
= 14,11784512 ■CA
= 15 pallets/ palette■R
OR/OF
Number of pallets/ hoeveelheid palette
= 8 385,048 ÷ 594 ■MCA
= 14,11624242 ■CA
= 15 pallets/ palette ■R

CA number of bricks from / hoeveelheid stene van 3.2.1
1MCA dividing by/ deël met 594
1CA number of pallets/ hoeveelheid palette
1R rounding up/ rond op

(3)

M
L2
M

QUESTION /VRAAG 3 [ 26 MARKS/ PUNTE ]

3.1.1 Area of 1 bicycle/ Oppv. van 1 fiets
= 180 cm × 45 cm ■SF
= 1,8 m × 0,45 m ■C
= 0,81 m2■CA
OR/OF

Area of 1 bicycle / Oppv. van 1 fiets
= 180 cm × 45 cm ■SF
= 81 cm $^2$■CA
= 0,81 m $^2$■C
1SF substitution/ vervanging
1C conversion/ herleiding
1CA simplification/
vereenvoudiging
OR/OF

1SF substitution/ vervanging
1CA simplification/
vereenvoudiging
1C conversion/ herleiding
(3) M
L2
M

3.1.2
Total floor area/ Totale vloer oppervlakte
= 0,81 m2 + 0,5 m2■MCA
= 1,31 m2 ■CA

127 bicycles/ fietse × 1,31 m2■MCA
= 166,37 m2

The statement is correct/ Die stelling is korrek. ■O CA from /vanaf 3.1.1

1MCA adding additional space
tel adisionele spapsie by
1CA simplification/
vereenvoudiging
1MCA multiplying by/ maal
met 127

1O opinion/ opinie
(4) M
L4
M
3.2.1 1 m2 need/ benodig 48 bricks/ stene
166,37 m2 needs/ benodig ?
Number of bricks/ Hoeveelheid stene
= 48 × 166,37 ■MA
= 7 985,76■A
Increase by 5% / verhoog met 5%
= 7 985,76 × 1,05 ■MCA OR/OF 105%
= 8 385,048■CA
= 8 386■R 1MA multiplying/
vermenigvuldig
1A number of bricks/
hoeveelheid stene
1MCA increase percentage/
verhoogde per sentasie
1CA simplification/
vereenvoudiging
1R rounding/ afronding
(5)
M
L3
D 3.2.3 Mass/ gewig = 8 386 × 2,230 kg ■MCA
= 18 700,78 kg ■CA
= 18 700,78 ÷ 1 000
= 18,70078 ton ■C
= 18,7 tons ■R CA number of bricks from
/hoeveelheid stene van 3.2.1
1MCA multiplying /
vermenigvuldig
1CA simplification /
vereenvoudiging
1C conversion /herleiding
1R rounding /afronding
(4)

M
L2
M
3.3.1 Indirect/inverse proportion/ Indirekte/Omgekeerde eweredigheid ■A
As one quantity increases, the other one decreases/ As een eenheid verhoog, neem die ander een af ■■O
1A correct proportion / korrekte
proporsie
2O explanation/ verduidelikking

(3) M
L1
E
3.3.2 Number of days = 30■RT ÷ number of people ■A
/ Hoeveelheid dae = 30 ÷ hoeveelheid mense

OR/OF

Number of days × number of people ■ = 30■
Hoeveelheid dae × hoeveelhed mense = 30
1RT 30
1A divided by number of
people/ deël met
hoeveelheid mense
OR/OF
1RT 30
1A number of days multiplied
by number of people/
hoeveelheid dae maal met
hoeveelhed mense
(2) M
L2
E

3.3.3
Number of days/ hoeveelheid dae = 30 ÷ 8■MCA

= 3,75 days/ dae■CA
Accept 4 days CA formula fro m/vanaf 3.3.2
1MCA substituting/ trek af 8

1CA answer/ antwoord
(2) M
L1
E
[26]
QUESTION /VRAAG 4 [31 MARKS/ PUNTE ]
Q/V Solution /Oplossing Explanation/ Verduideliking T/L
4.1.1 Anti clockwise/ Anti-kloksgewys
OR/OF
Left/ Links■■A 2A direction/ rigting

(2) MP
L1
E
4.1.2 Diagram 3 ■■A 2A correct/ korrekte diagram

(2) MP
L2
E
4.1.3 1,55 m = diagram height/ hoogte x 25
Diagram height/ hoogte = 1,55 m ÷ 25 ■MA
= 0,062 m ■A
= 6,2 cm■C
OR/OF
1,55 m = 155 cm ■C
155 cm ÷ 25■MA
= 6,2 cm■A
1MA divide by scale /deël met
skaal
1A length/ lengte in m
1C conversion/ herleiding
OR/OF
1C conversion/ herleiding
1MA dividing by scale/ deël
met skaal
1A length/ lengte (3) MP
L2
M

4.2.1
Space occupied by a 3D object/ Spasie wat deur ‘n 3D
voorwerp opgeneem word ■■A 2A answer/ antwoord
(2) M
L1
E
4.2.2  Volume = 340 mm × 325 mm × 180 mm ■SF
= 0,34 m × 0,325 m ×   0,18 m ■C
= 0,01989 m3■CA
OR/OF

Volume of the box/ volume van die boks
= 340 mm × 325 mm × 180 mm ■SF
= 19 890 000 mm3 ÷ 1 000 000 000■CA
= 0,01989 m3■C 1C conversion/ herleiding
1SF substitution/ vervanging
1CA simplification/
vereenvoudiging

NPR (3) M
L2
M
4.2.3  Number of boxes/ Hoeveelheid bokse :
Length wise/ Lengte  = 1 700 mm ÷ 340 mm ■MA
= 5■A
Width wise/ Breedte  = 1 490 mm ÷ 325 mm
= 4,584615385
= 4■A
Height wise/ Hoogte  = 1 200 mm ÷ 180 mm
= 6,6666667
= 6■A
Total/ Totaal  = 5 × 4 × 6 ■MCA
= 120 boxes/ bokse■CA
OR/OF
Length wise/ lengte : 1,7 m ÷ 0,34 m ■MA = 5■A
Width wise/ breedte :  1,49 m ÷ 0,325 m = 4,584615…
= 4■A
Height wise/  hoogte : 1,2 m ÷ 0,18 m = 6,666…
= 6■A
Total/ totaal  = 5 × 4 × 6 ■MCA
= 120 boxes ■CA 1MA dividing/ deë/
1A answer/ antwoord
1A number width wise/
hoeveelheid in breedte
1A number height wise/
hoeveelheid in hoogte
1MCA multiplying/
vermenigvuldig
1CA number of boxes/
hoeveelheid bokse

(6) MP
L3

D
4.2.4  30 June from 14:50 to 24:00 = 9 hours 10 min ■A
30 Jun ie van 14:50 to t 24:00 = 9 uur 10 min
1 July = 24 hours
1 Julie = 24 uur
2 July from 00:00 to 8:15 = 8 h 15 min ■A
2 Jul ie van 00:00 to t 8:15 = 8 h 15 min
Total elapsed time = 9h10 + 24h00 + 8h15
= 41 hours 25 min ■CA
Totale tyd verloop = 41 uur 25 min
This is within the 48 -hour service ■O
Dit is binne die 48 -uur diens
OR/OF
1A time 30 June/ tyd 30 Junie

1A time 1 and 2 July/ tyd 1 en
2 Julie
1CA adding time/ tel tyd
bymekaar
1O opinion/ opinie

OR/OF

M
L4
D

30 June from 14:50
To 1 July 14:50 = 24 hours/ first day ■ A
/30 Junie van 14:50
Tot 1 Julie 14:50 = 24 uur/ dag 1
To 2 July 14:50 = 48 hours/second day ■ A
Tot 2 Julie 14:50 = 48 ure / dag 2
But 2 July 8:15 is before 48 hours ■ A
Maar 2 Julie 8:15 is voor 48 uur
It is within 48 hours ■O
Dit is binne 48 uur 1A 1st day/dag 1

1A 2nd day/dag 2

1A conclusion/ gevolgtrekking

1O conclusion /gevolgtrekking
(4)
4.3.1  Body Mass Index/ Liggaamsmassa -indeks■■A 2A answer/ antwoord
(2) M
L1
E
4.3.2  Underweight/ Ondergewig ■■RT 2RT answer/ antwoord
(2) M
L2
E
4.3.3  18,2 kg/m2 = mass/ gewig ÷ (1,56 m)2 ■SF
Mass/ gewig = 18,2 × (1,56)2 ■M
= 44,29 kg ■A 1SF substitution/ vervanging
1M changing the subject of

the formula/ verander die
onderwerp van die f ormule
1A simplification/
vereenvoudiging
NPR (3) M
L2
M

4.3.4 Eat a balanced diet/ Eet ‘n gebalenseerde dieët OR/OF eat food with lots of fibre/fats/ eet kos met baie vessel/ vette OR/OF eat food that will assist her to gain weight/ eet kos wat sal help om gewig op te tel ■■O
2O opinion/ opinie

(2) M
L4
E
[31]
TOTAL/ TOTAAL : 100

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NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2024

MATHEMATICAL LITERACY P1
MARKING GUIDELINE

MARKS: 100

| Symbol | Explanation |
|---|---|
| M | Method |
| MA | Method with accuracy |
| CA | Consistent accuracy |
| A | Accuracy |
| C | Conversion |
| S | Simplification |
| RT | Reading from a table/a graph/document/diagram |
| SF | Correct substitution in a formula |
| O | Opinion/Explanation |
| P | Penalty, e.g. for no units, incorrect rounding off, etc. |
| R | Rounding off |
| NPR | No penalty for correct rounding minimum two decimal places |
| AO | Answer only |
| MCA | Method with constant accuracy |
| J | Justification |

This marking guideline consist of 6 pages.

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- If a candidate has crossed -out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled version)
- Consistent accuracy (CA) applies in ALL aspects of the marking guidelines; however, it stops at the second calculation error.

- If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra incorrect item presented.

QUESTION 1 [ 26 MARKS]
Ques. Solution Explanation T&L
1.1.1 Canva ✓✓A
2 RT correct service
provider
Accept Samira Hadid (2) F
L1
1.1.2 A = $127 x 4 ✓MA

= $508 ✓A 1 MA multiplying correct
price by 4
1 A answer (2) F
L1
1.1.3 Invoice is a list of goods or services rendered as
well as a statement of the amount. ✓✓O 2O explanation
(2) F
L1
1.1.4 B = $369 ÷ $123 ✓MA

= 3 ✓A 1MA dividing the correct
values
1A answer (2) F
L1
1.1.5 5 March 2024 ✓✓A
2A correct date
(2) F
L1
1.1.6 Total : R1,00 = $0,053
? = $1000
? = $1000 × R1,00 ✓MA
$0,053
? = R18 867,92 ✓CA

1MA correct exchange rate

1CA answer
(2) F
L1
1.2.1 Constant/fixed relationship ✓✓A 2A correct answer
(2) F
L1
1.2.2 R1 200,00 ✓✓RT 2RT
(2) F
L1
1.2.3 (a) C = 2 ✓✓A 2A correct number
Accept 3 (2) F
L1
(b) D = R950 ✓✓RT 2RT correct amount
(2) F
L1
1.3.1 29 + 23 + 3 + 4 = 59 learners ✓✓A 2A correct total
(2) D
L1
1.3.2 % fail = 29 × 100 ✓MA
59

= 49,15 ✓CA 1 MA correct percentage

1CA answer (2) D

L1
1.3.3 0 – 29 ✓✓A 2A correct level
(2) D
L1
[26]

QUESTION 2 [ 26 MARKS]
Ques . Solution Explanation T&L
2.1.1 Budget is a financial plan outlining income and expenses
over a period of time . ✓✓O
OR
It is a tool to help manage finances, achieve financial goals,
and to make informed decisions about spending and
saving. ✓✓O 2O correct explanation

(2) F
L1
2.1.2 31 029 965 :
Thirty -one million and twenty -nine thousand, nine
hundred and sixty -five. ✓✓A 2A correct

(2) F
L1
2.1.3 % increase

= 44 104 248 - 42 441 422 ✓RT ×100 ✓MA
42 441 422 ✓MA
= 3,92
= 4% ✓CA 1RT correct values
1MA multiplying by 100
1MA correct
denominator

1CA answer PR (4) F
L3
2.2.1 Cost = R50 + R12,00 × (No of km travelled – 3) ✓✓A 2A correct formula
(2) F
L2
2.2.2 Daniel: R50,00 + R12,00 × (No of km used – 3)

= R50,00 + R12,00 × (80 km ■3) ✓SF

= R50,00 + R12,00 × (77 km) ÷ 1 km

= R50,00 + R924,00 ✓A

= R974,00 ✓CA

Jerry: R14,00 × R80,00 ✓A

= R1 160,00 ✓CA

His statement is incorrect. ✓J

1SF substitution to a
formula

1A simplification

1CA answer for Daniel

1A multiplying correct
amounts
1CA answer for Jerry

1J justification (6) F
L4
2.2.3 Distance = R50,00 + R12,00 × (No of km travelled ■3)

R1 214,00 = R50,00 + R12,00 × (D ■3) ✓SF

R1 214,00 – R50,00 = R12,00 × (D – 3)

R1 164,00 = R12,00 × (D ■ 3) ✓S

R1 164,00 = D – 3
R12,00

97 = D – 3 ✓S

D = 100 ✓CA

1 SF substitution to a
formula

1S simplification

1S simplification

1 CA answer (4) F
L3

2.3 Discount on school fees = R48 800,00 × 10 ✓MA
100
= R4 880,00 ✓A

Discount on hostel fees = R66 800 × 5 ✓MA
100
= R3 340,00 ✓A

Total Discount = R4 880,00 + R3 340,00 ✓A

= R8 220,00 ✓CA 1MA multiplying by a

correct percentage
1A simplification

1MA multiplying by
correct percentage
1A simplification

1A adding correct amounts

1CA answer (6) F
L4
[26]

QUESTION 3 [ 17 MARKS]
Ques. Solution Explanation T&L
3.1 India ✓✓A 2A correct country
(2) D
L2
3.2 103 317 638 ; 15 400 547 ; 15 400 547 ; 12 862 740 ;
12 615 365 ; 8 485 749 ; 6 692 796 ; 4 684 727 ;
3 740 280 ; 2 713 117 ; 2 544 102 ; 2 505 605 ;
1 854 162 ✓✓M 2M arranging in correct
order

(2) D
L2
3.3 Difference = 103 317 638 – 1 854 162 ✓RT
= 101 463 478 ✓CA 1RT correct values
1CA answer
(2) D
L2
3.4 5 ✓✓RT 2RT
(2) D
L1
3.5 Thailand ✓✓A CA 3.2
2 A answer
(2) D
L2
3.6 Bar Graph /Line graph ✓✓A 2A Type of graph
(2) D
L2
3.7 Probability = 9 ✓M ✓M
13
= 0,69 ✓CA 1M fraction
1M division
1 CA answer (3) P
L2
3.8 3 countries ✓✓A 2A answer
(2) D
L2
[17]

Question 4 [31marks]
Que. Solutions Explanation T&L
4.1.1 Inflation ✓✓A 2A answer (2) F
L1
4.1.2 Cost = R2 400,00 + R50,00 × number of learners ✓SF
= R2 400,00 + R50,00 × 100 ✓MA
= R2 400,00 + R5 000,00 ✓M
= R7 400,00 1 SF formula
1MA multiplying by 100
1M addition

(3) F
L3
4.1.3

1A starting point (0 ;0) ✓
1A end point (100 ;15 000) ✓
1A straight line ✓

(3) F
L2
4.1.4 24 learners buying 24 t-shirt indicated as B on the graph
✓✓A 2A correct number of
learners indicated on the
graph (2) F
L2
010002000300040005000600070008000900010000110001200013000140001500016000
0 10 20 30 40 50 60 70 80 90 100 110Cost in Rands
Number of LearnersB

4.1.5 End of Year One:
10,57 500 7 500100
7 500 787,5
8 287,50RR
RR
R= + ■
=+
=

End of Year Two:
10,58 287,50 8 287,50100
8 287,50 870,1875
9 157,6875RR
RR
R= + ■
=+
=

End of Year Three:
10,59 157,6875 9 157,6875100
9 157,6875 961,5571875
10 119,24RR
RR
R= + ■
=+
=

Statement is valid

✓M

✓CA

✓CA

✓CA

✓O

1 M calculating interest

1CA balance for 1st year

1 CA balance for 2nd year

1 CA balance for 3rd year

1O opinion

3 M multiplication

1 A answer
1 O opinion (5)
F
L3
OR

Balance 7 500 1 ,105 1 ,105 1 ,105 10 119,24 RR= ■ ■ ■ =

Statement is valid ■O
4.2.1 Discreet ✓✓A 2A answer (2) D
L1
4.2.2 ✓A
37 = 1588 + F ✓M
44 ✓A

1588 + F = 1628 ✓A

F = 1628 – 1588

F = 40 ✓
1M adding
1A substituting 37

1A dividing by 44

1A multiplying 37 by 44

1CA answer (5)

D
L3
4.2.3 12 13 14 15 15 17 18 21 21 22 23 24 24 26 26 26 30 31
32 33 37 38 39 39 40 40 40 41 41 41 43 43 49 49 50 51
53 54 55 60 62 69 64 86 ✓M
Range = 86 – 12 ✓M
= 74 ✓CA

Statement is not correct the range is big, so the data is far
apart from each other. ✓J 1M arranging in order of
descending or ascending /
OR RT correct values
1M calculating range
1CA answer

1J correct justification
(4) D
L3
4.2.4 28 learners ✓✓A 2A correct answer
(2) D
L1
4.2.5 P = 1 ✓A
44 ✓A
= 0,023 ✓CA 1A numerator
1A denominator
1CA answer
(3) P
L2
[31]

TOTAL: 100
■M ■M ■M ■A

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NATIONAL
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GRADE 1 1

NOVEMBER 2024

# MATHEMATICAL LITERACY P2
# MARKING GUID ELINES

MARKS: 100

| Symbol | Explanation |
|---|---|
| M | Method |
| MA | Method with accuracy |
| CA | Consistent accuracy |
| A | Accuracy |
| C | Conversion |
| S | Simplification |
| RT | Reading from a table/graph/ diagram |
| SF | Correct s ubstitution in a formula |
| O | Opinion/Explanation/Reasoning |
| P | Penalty, e.g. for no units, incorrect rounding off etc. |
| R | Rounding Off/Reason |
| NPR | No penalty for correct rounding minimum two decimal places |
| AO | Answer only |
| MCA | Method with consistent accuracy |
| RCA | Rounding with consistent accuracy |

MARKING GUIDELINES

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.

- If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled) version.

- Consistent Accuracy (CA) applies in ALL aspects of the marking guidelines; however, it stops at the second calculation error.

- If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalize for every extra incorrect item presented.

KEY TO TOPIC SYMBOL:
F = Finance; M = Measurement; MP = Maps, plans and other representations; P = Probability

QUESTION 1 [20 MARKS]
ANSWER ONLY FULL MARKS

Ques. Solution Explanation Level
1.1.1 Analo gue clock 2A correct type clock
(2) M
L1

1.1.2 Ten minutes past ten o’clock in the morning.

OR
Ten past ten in the morning.

OR
Ten past ten before noon. 2A correct time

(2) M
L1

1.1.3 Time taken = 10:27
-10:10
= 00:17
■ 17 minutes 1M subtracting time
1A correct time

(2) M
L1

1.2.1 Perimeter is the distance around an object or shape . 2A definition
(2) M
L1

1.2.2 Perimeter = 230 cm + 200 cm + 9 5 cm + 88 cm + 135 cm +
112 cm
= 860 cm

Accept if calculated as follows:
Perimeter = (230 × 2) + (200 × 2)
= 860 cm 1MA adding all the
correct values
1A perimeter

1MA multiplying length
and width by 2
1A perimeter
(2) M
L1

1.3.1 Side or length = 43
100
= 0,43 m 1C conversion
1A answer
(2) M
L1

1.3.2 Area of a square = side × side
= 0,43 × 0,43
= 0,1849
≈ 0,18 m2
(Accept 0,185 m2) CA from 1.3.1
1SF substitution
1MCA area in m2
NPR
(2) M
L1

1.4.1 5 casters (wheels) 2A number of casters
(2) MP
L1

1.4.2 Tool = Allen key 2A correct tool
(2) MP
L1 ■✓A
■M
■A
■✓A
■MA
■A
■C
■A
■SF
■MCA
■✓A
■✓A ■✓A
■✓A
■✓A
■MA
■A

1.4.3 M8 × 15 mm
M5 × 18 mm 1A M8 × 15 mm
1A M5 × 18 mm
(Accept any order)
(2) MP
L1
[20]

■A
■A

QUESTION 2 [2 0 MARKS]

Ques. Solution Explanation Level
2.1.1 N2 2A correct national road
(2) MP
L1

2.1.2 Southeast OR SE 2A correct direction
(2) MP
L1

2.1.3 Map distance = 7 km × 100 000
= 700 000 cm
■ 700 000
56 000
= 12, 5 cm
≈ 13 cm

OR

Map distance = 56 000
100  000
= 0,56 m
■ 7
0,56
= 12, 5 cm
≈ 13 cm 1C conversion
1M dividing by scale
1CA map distance
1R rounding

OR

1C conversion
1M dividing by scale
1CA map distance
1R rounding (4) MP
L2

2.2.1  Speed = ■■■■■■■■
■■■■
100 km/h = 11,2 ■■
■■■■
■ Time = 11,2 ■■
100  ■■/■
= 0,112 hours × 60
= 6,72 minutes
(Accept 7 minutes)  1SF substitution
1M changing subject of
formula
1C converting hours  to
minutes
1CA time
NPR
(4) MP
L3

2.2.2
Total Distance = (11,2 km × 2) + 156 km
= 178,4 km  1M multiplying by 2
1M adding 156 km
1A total distance (3) MP
L2

■✓A
■✓A
■C
■M
■CA
■R
■SF
■M
■C
■CA
■M
■A ■M ■CA
■R ■C
■M
Accept if calculated
the time  as follows:

= 0,112 × 60
= 6 minutes, 43,2 sec
(full marks)

2.2.3 Number of litres of petrol = 178 ,4
100 × 5,7
= 10,1688 litres
■ Cost = R24,45 × 10,1688
= R248,63
■ Her statement is valid

OR
CA from 2.2.2
1M dividing by 100
and multiplying
with 5,7
1A number of litres
1M multiply with
R24,45
1MCA petrol cost
1O opinion

OR
F
L4
Number of litres of petrol = 5,7
100 × 178 ,4
= 10,1688 litres
■ Cost = R24,45 × 10,1688
= R248,63
■ Her statement is valid 1M dividing by 100
and multiplying
with 178,4
1A no. of litres
1M multiply with
R24,45
1MCA petrol cost
1O opinion
(5)
[20]

■A
■M
■MCA
■O ■M ■M
■A
■M
■MCA
■O

QUESTION 3 [3 4 MARKS]

Ques. Solution Explanation Level
3.1.1 Volume is the amount of space inside the fish tank. 2A definition
(2) M
L1

3.1.2 Volume of a cylindrical prism = π × radius2 × height

38,8 ft3 = 3,142 × 2,12 × height
38,8 ft3 = 13,85622 × height
■ Height = 38,8
13,85622
= 2,8 foot 1SF substitution
1M divide correct values
1CA height

(3) M
L2

3.1.3 ■ = (■ - 320) × 5
9
= (72 ■ - 320) × 5
9
= 22,222…
≈ 22,2 ■
■ Hein’s claim is valid 1SF substitution
1A temperature
1O opinion

(3) M
L4

3.1.4 Volume of stones = 87% – 75%
= 12%
■ 12
100 × 38,8
= 4,656 ft3
≈ 4,7 ft3

OR

Volume of water and stones in tank = 87
100 × 38,8
= 33,756 ft3

Volume of water before stones added = 75
100 × 38,8
= 29,1 ft3
■ Volume of stones = 33,756 ft3 – 29,1 ft3
= 4,656 ft3
≈ 4,7 ft3 1M subtracting
percentages
1A correct %
1M multiply with 38,8
1CA volume of stones
1R rounding

OR

1M calculating 87%

1M calculating 75%

1M subtracting volumes
1CA volume of stones
1R rounding (5) M
L3

3.1.5 Probability = 1
7 × 100 %
= 14,285 7…
≈ 14,29%
(Accept 14,3% OR 14,286%) 1A correct fraction
1M multiply with 100%
1CA probability as %
NPR
(3) P
L2

■✓A
■SF
■M
■CA
■SF
■A
■O
■M
■A
■M
■CA
■R
■A ■M
■CA ■R ■CA ■M
■M
■M

3.2.1 Total SA = [ 2 (l × w) + 2 (l × h) + 2 (w × h)]

= [ 2 (240 × 112) + 2 (240 × 70) + 2 (112 × 70)]
= 53 760 + 33 600 + 15 680
= 103 040 mm2 1C conversion
1SF substitution
1S simplification
1CA total SA
(4) M
L2

3.2.2 Number of complete pallets = 2 500
500
= 5 pallets 1MA dividing correct values
1A number of pallets
(2) M
L1
3.2.3 Weight of one pallet = 9 187
5
= 1 837,5 kg
■ = 1 837 ,5
1 000
= 1,8375 ton

OR

Weight of one pallet = 500 × 3,675
= 1 837,5 kg
■ = 1 837 ,5
1 000

= 1,8375 ton  1M dividing weight by 5
1C conversion
1CA weight of one pallet

OR

1M multiplying 500 with
3,675
1C conversion
1CA weight of on e pallet
(3) M
L2

3.2.4  Volume of a rectangular prism = Length × Width × Height
= 240 mm × 112 mm × 70 mm
= 1 881 600 mm3
■ 1 881 600 × 2  500
= 4 704 000 000 mm3
■ 4 704  000
000
1 000  000  000
= 4,704 m3
≈ 4,7 m3
■ Hein's claim is correct

OR

Volume of a rectangular prism = Length × Width × Height

= 0,24 m × 0,112 m × 0,070 m
= 0,0018816 m3
■ 0,0018816 × 2  500
= 4,704 m3
≈ 4,7 m3

■ Hein's claim is correct.
1SF substitution

1M multiply volume with
2 500 bricks

1C convert  volume to m3
1CA volume of all bricks
1O opinion

OR

1C conversion
1SF substitution

1M multiply volume with
2 500 bricks
1CA volume of all bricks
1O opinion
(5) M
L4

■C
■SF
■S

■CA
■MA
■A
■M
■C
■CA
■CA ■M
■C
■O ■CA ■C ■M ■SF
■C ■SF
■M
■CA
■O

3.2.5 Total cost = R2,60 × 2 500
= R6 500 + R650 (delivery)
= R7 150
≈ R7 200 1M multiply cost with
number of bricks
1M adding delivery costs
1CA total cost
1R rounding
(4) F
L2
[34]

■M
■M
■CA
■R

QUESTION 4 [26 MARKS]

Ques. Solution Explanation Level
4.1.1 Graphic scale OR Bar scale OR Linear scale
Ratio scale OR Num eric scale 1A bar scale
1A number scale
(2) MP
L1

4.1.2 Scale = 6,7 cm
6,7 cm = 12 miles
6,7 ■■
6,7 = 12 ■■■■■
6,7
1 cm : 1,791044776 miles
■ Actual distance = 10,5 cm × 1,7910…
= 18,80597015 miles
≈ 18,81 miles
(Accept 18,806 miles OR 19 miles) 1A measuring scale
in cm
1M divide by 6,7
1CA scale
1M multiply 10,5
cm with the scale
1CA actual distance
NPR
(5) MP
L3

4.1.3 Time = ■■■■■■■■■
■■■■■
= 121 ,4 ■■■■■
85 ■■■■■ /■
= 1,4282… hours
■ 0,4282… × 60
= 25,694… minutes
■ Time = 1 hour 25 minutes 42 seconds

Arrival time = 13:30:00 (departure)
01:25:42 (travelling)
00:25:00 (stop at petrol station)
= 15:20:42
■ His claim is invalid.

1SF substitution
1C converting
hours to minutes
1CA travel time

1M adding time
1CA arrival time
1O opinion
(6) MP
L4

4.1.4 Probability = 2
8
(Accept simplified form = 1
4) 1A numerator
1A denominator
(2) P
L2

4.2.1 Radius = 9,5
2 = 4,75 cm
Area of a circle = π × radius2
= 3,142 × 4,752
= 70,891375
≈ 70,89 cm2 1M finding radius
1SF substitution
1CA area of white
cloth

(3) M
L2

4.2.2 Length of wooden frame = 22 inches × 2,54 ■C
= 55,88 cm
Area of rectangle = length × width
2 682 = 55,88 × width
■ Width = 2 682
55,88
= 47,995 7 …
≈ 48 cm
(Aanvaar 47,996 cm) 1C converting
length
1A answer

1SF substitution
1M dividing area
by length

1CA width
(5) M
L3 ■A ■A ■A
■A
■M
■CA
■M
If calculated
as follows,
DO NOT
penalize:

Travel time
= 1h26 min
Arrival time
= 13:30
01:26
00:25
= 15:21

■SF
■C
■CA
■M
■CA
■O
■A
■A
■SF
■CA
■SF
■M
■CA ■A ■A
■CA
■A ■A ■A
■M

4.2.3 Width of one rectangular bar = 48
6
= 8 cm
■ Lwandile’s claim is valid.

OR

Width of one rectangular bar = 47,996
6
= 7,9993… cm
≈ 8 cm
■ Lwandile’s claim is valid.

CA from 4.2. 2
1MCA dividing width by 6
1CA width of rectangular bar
1O opinion
OR
1MCA dividing width by 6
1CA width of rectangular bar
1O opinion
(3) M
L4
[26]
TOTAL : 100
■MCA
■CA
■O
■MCA
■CA
■O

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NATIONAL
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GRADE 1 1

NOVEMBER 2023

MATHEMATICAL LITERACY P 1

MARKING GUIDELINE

MARKS: 100

Symbol Explanation
M Method
MA Method with accuracy
CA Consistent accuracy
A Accuracy
C Conversion
S Simplification
RT/RG/RM Reading from a table/Reading from a graph/Reading from a map
F Choosing the correct formula
SF Substitution in a formula
J Justification
P Penalty, e.g., for no units, incorrect rounding off etc.
R Rounding Off/Reason
AO Answer only
NPR No penalty for correct rounding

This marking guideline consists of 6 pages.

QUESTION 1 [ 23 MARKS ]

Ques . Solution Explanation
AO: FULL MARKS T&L
1.1.1 A = R33 150 + R1 625
= R34 775 1MA for adding
correct values
1A answer
(2) F
L1
1.1.2 R33 150 ÷ 26
= R1 275,00
1MA for dividing
correct values
1A answer
(2) F
L1
1.1.3 B = R7 927,86 – (R375 + R850,25 + R1 500)
= R7 927,86 – R2 725,25
= R5 202,61 1MA for subtracting
correct values
1A answer
(2) F
L1

1.1.4
Percentage = R850,25
R7 927,86 ×100

= 10,72%
1RT correct values
1M multiplying by 100
1CA answer
(3) F
L1
1.2.1 R22,50
2RT reading correct
value
(2) F

L1
1.2.2 Vaal Plaza Main line 2RT correct tollgate

(2) F
L1

1.2.3 140 : 225
28 : 45 1RT correct values
1A simplifying
correctly
(2) F
L1
1.2.4 R271,00 × 2
= R542,00 1MA multiplying
correct rate by 2
1A answer
(2) F
L1

1.3.1 Pie Chart
2RT correct graph
(2) D
L1
1.3.2
Percentage comedies = 100% ■ (21% + 6% + 24% + 11%)
= 38%
1MA subtracting
1A percentage
(2) D
L1

1.3.3 Number of romance = 24% × 220
= 52,8
= 53 students
1MA multiplying
correct
percentage
1A correct answer
Accept 52 (2) D
L1
[23]

■MA
■A
■■RT ■MA
■A
■A ■MA
■RT
■M
■CA
■■RT
■RT
■A
■MA
■A
■■RT
■MA
■A
■MA
■A

## QUESTION 2 [25 MARKS]

Ques Solution Explanation/Marks
AO: FULL MARKS T/L
2.1.1
Jan 2023: R12,47 × 1,069 = R13,33

Feb 2023: R13,33 × 1,07 = R14,26 1RT correct rate
1M multiplying
1A answer
1MA multiplying with
correct rate
1CA answer
(5) F
L3

2.1.2 Prices increased
Prices still went up, but by a lower percentages/ the percentages are not negative. 1A increased
1O reason
(2) F
L4
2.2.1
7 May 2023 1A 7th
1A May
(2) F
L1
2.2.2 Total = R349 × 3
= R1 047
≈ R1 000 1MA multiplying by 3
1A answer
1R rounding to nearest
1 000
(3) F
L2
2.2.3 Labour per hour = R6 500 ÷ 7,5 hrs
= R866,67 1MA dividing by 7,5
1A correct answer
(2) F
L2
2.2.4 Total = R5 499 + R1 047 + R1 699 + R6 500
= R14 745 × 1,15
= R16 956,75 CA from 2.2.2
1M adding values
1MA multiplying by
1,15
1CA answer
(3) F
L2
2.2.5

% change = 5 795 – 5 499
5 499 × 100%

= 5,38%

∴ Valid

1RT correct values
1MA correct %
calculation
1CA simplification

1O statement
(4) F

L4

2.3 Block 1: 600 kWh × 229,00c = 137 400c
Block 2: 112 kWh × 278,46c = 31 187,52c
Total = 168 587,52c ÷ 100
= R1 685,88

1MA multiplying Block 1
1MA answer Block 2
1C divide by 100
1CA answer
(4) F
L3
[25]

■M ■RT
■CA
■A
■O
■A ■A
■MA
■A
■A ■MA
■R
■MA
■A
■MA ■M
■CA
■RT
■MA
■CA
■O
■MA ■MA
■C
■CA

QUESTION 3 [19 MARKS ]

Ques Solution Explanation T&L

3.1 Forty -five million eight hundred thousand dollars
2RT correct value
(2) D
L1

3.2 Median:

5; 25; 47; 50; 52; 55; 55; 80; 90

= $52 000 000 / $52 million
1RT correct values
1MA method median
1A correct answer
(3) D
L3

3.3 Federer total = $121,2 – $30,5
= $90,7 million

Total = $130 + $121,2 + $115 + $95 + $92,8 + $92,1
+ $90,7 + $90 + $83,9
= $910,7 million / $910 000 000 1M subtracting
1A simplification

1M adding
1CA simplification

(4) D
L3

3.4
Mean = $451,7
9

= $50,2 million / $ 50 200 000
1M adding values
1MA dividing by 9
1CA answer
(3) D
L2

3.5 Percentage = 55
115 × 100

= 47,83% 1MA correct values
multiply by 100
1A simplification
(2) D
L2
3.6 70 : 90
7 : 9 1RT correct value s
1CA simplification of
ratio
(2) D
L2
3.7 Probability = 4
9

= 0,444
1RT numerator
1RT denominator
1CA rounding to 3
decimal places
(3) P
L2
[19]
■CA ■A ■M ■■RT
■RT
■A ■MA
■M
■CA ■M
■MA
■MA
■A
■RT
■CA
■RT
■RT
■CA

QUESTION 4 [33 MARKS]

Ques Solution Explanation/Marks T&L
4.1.1 Dependent: Amount in Rands
Independent: Number of curry bunnies
1RT dependent
1RT independent
(2) F
L1
4.1.2 R250 ÷ 10

= R25,00
1MA dividing by 10
1A simplification
(2) F
L1

4.1.3
Total Expenses = 450 + (10 × number of curry bunnies)
1RT fixed cost
1RT cost per curry bunny
(2) F
L2
4.1.4

1A starting point
1A 2 correct points
1A end point
(3) F
L2
4.1.5 Profit ONE month = R5 000 – R2 450
= R2 550

Profit FOUR months = R2 550 × 4
= R10 200

∴ Valid
1A profit one month

1MA profit 4 months
1CA answer

1O statement
(4) F
L4
4.2.1 A = 20 + 13
= 33
1RT adding correct
values
1A answer
(2) D
L2
4.2.2 15:00 – 17:59 2A correct answer
(2) D
L2
4.2.3 18:00 – 20:59 2RT correct time interval
(2) D
L2
045090013501800225027003150360040504500049505400
0 50 100 150 200 250Amount in Rands
Number of Curry Bunnies Income and Expenses of Curry

Bunny Sales
■■RT ■RT
■RT
■MA
■A
■RT ■RT
■MA ■A
■CA
■O
■RT
■A
■■A

Ques Solution Explanation/Marks T&L

4.2.4 6,5 ■ × 77 = 500,5 ■

Cost = 500,5 ■ × R21,92
= R10 970,96 1MA calculating litres
1M multiplying
1CA answer
(3) D
L3

4.2.5 Probabilty = 32
77 1RT numerator
1RT denominator
(2) P
L2

4.3.1 BANK B
Interest amount is the same for each year.
1RT correct bank
1O statement/opinion
(2) F
L2

4.3.2
BANK A: R1 277,29 × 1,085
= R1 385,86

Difference = R1 385,86 – R1 340
= R45,86

∴ Valid 1MA multiplying by rate
1A simplification
1M difference

1O statement
(4) F
L4

4.3.3
68,49 × 19,83
= R1 358,16

∴ Yes, she will have enough. 1MA multiplying by exchange rate
1A simplification
1O statement
(3) F
L4
[33]

TOTAL: 100

■M ■MA
■CA
■RT ■RT
■RT
■O
■MA
■A
■M
■O
■MA
■A
■O

NATIONAL
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GRADE 11

NOVEMBER 2023

# MATHEMATICAL LITERACY P2
# MARKING GUIDELINE

MARKS: 100

This marking guideline consists of 9 pages.

| Symbol | Explanation |
|---|---|
| M | Method |
| MA | Method with accuracy |
| CA | Consistent accuracy |
| RCA | Rounding consistent accuracy |
| A | Accuracy |
| C | Conversion |
| S | Simplification |
| SF | Correct substitution in a formula |
| J | Justification |
| O | Opinion/Example/Definition/Explanation/Justification/Verification |
| RT/RG/RM | Reading from a table/graph/map |
| P | Penalty, e.g. for no units, incorrect rounding off etc. |
| R | Rounding off |
| NPR | No penalty rounding or omitting units |
| AO | Answer only, full marks |

## MARKING GUIDELINES

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.

- If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled version) .

- Consistent Accuracy (CA) applies in ALL aspects of the marking guidelines; however, it stops at the second calculation error.

- If the candidate presents any extra solution when reading from a graph, table, layout

plan and
map, then penalise for every extra incorrect item presented.

LET WEL:

- As ■ kandidaat ■ vraag TWEE keer beantwoord merk slegs die EERSTE poging.

- As ■ kandidaat ■ antwoord van ■ vraag doodtrek (kanselleer) en nie oordoen nie, merk die doodgetrekte (gekanselleerde) poging.

- Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyn toegepas, maar dit hou by die tweede berekeningsfout op.

- Wanneer ■ kandidaat aflees van ■ grafiek, tabel, uitlegplan en kaart en ekstra antwoorde gee, penaliseer vir elke ekstra item.

KEY TO TOPIC SYMBOL:
F = Finance; M = Measurement; MP = Maps, plans and other representations ; P= Probability

QUESTION 1 [23]

Quest Solution Explanation Level
1.1.1
3,142 × 450 ✓ M
= 1 413,9 mm ✓ A

1M multiply by 450
1A correct answer

(2) M
L1
1.1.2
Option B:

290 ÷ 200 m ✓ M
= R1,45 per metre ✓ MA

1M divide by 200 m
1MA answer

(2) M
L1
1.1.3
R290 : R390
29 : 39 ✓ A

1M correct order
1A correct simplified values

(2) M
L1
1.1.4
Radius = 450 mm ÷ 10 ✓ C
= 45 cm ÷ 2 ✓ M

= 22,5 cm ✓ A
1C divide by 10
1M finding the radius
1A correct answer

(3) M
L1
1.2.1
N2; N3; N6; N10 ✓✓A

(Any TWO )

2A correct national roads

(2) MP
L1
1.2.2
Strip Chart / Map ✓ ✓ A

2A name

(2) MP
L1

1.2.3 ✓ RT ✓ C
Distance = 964 km × 1000

= 964 000 m ✓ A
1RT correct values
1C conversion

1A 964 000 m
(3) MP
L1
✓M
=

1.2.4 (a) Port St. Johns OR Port Edward ✓✓A 2 A town
(2) MP
L1
1.2.4 (b) ✓A
Distance = 201 + 36 ✓ M
= 237 km ✓ A
1A correct distances
1M Addition
1A 237 km
(3) MP
L1
1.2.5 R617; R101; R56 ; R61 ✓✓RM

(Accept any two answer s)
2RM correct provincial
roads

(2) MP
L1

[23]

QUESTION 2 [18]

Quest . Solution Explanation Level
2.1.1
Bar Scale ✓✓ RM
2RM correct answer

(2) MP
L1
2.1.2
South West ✓✓ RM
2RM correct
direction

(2) MP
L2
2.1.3 ✓M
2 cm : 250 km
✓M
9,6 cm : 9,6 × 250
2

≈ 1 200 km ✓ CA

Mr. Antonie is incorrect ✓ J
1M measured value

1M multiply by 250
and divide by 2

1CA correct answer

1J correct
justification
(4) MP
L3
2.2.1

Average Speed = 1 635
17,25 ✓ MA

= 94,782 ✓A

= 94,78 km / hr ✓R
1MA divide by 17,25

1A correct answer

1R correct rounding
(3) M
L2
2.2.2 (a) 1 litre = 12,5 km
No. of litres = 1 635
12,5 × 1 ✓MA
= 130,8 ✓A
0,80 litre = 10 km
No of litres = 1 635

10 × 0,80 ✓MA
= 130,8 ✓A
Mr Amos statement is incorrect . ✓J
1MA dividing by
12,5
1A correct answer

1MA dividing by 10

1A correct answer
1J correct deduction
(5) MP
L4
2.2.2 (b) Cost = 130,8 × R24,75 ✓M

= R3 237,30 ✓A 1M multiply by 130,8

1A correct answer
(2) M
L2
[18]

QUESTION 3 [32]
Quest Solution Explanation Level
3.1.1
Mass of cake = 900 ÷ 1 000 ✓M

= 0,9 kg ✓A
1M divide by 1000

1A correct answer

(2) M
L1
3.1.2
Mass of one slice = 900
12 ✓ M

= 75 g ✓ A

1M divide by 12

1A correct answer

(2) M
L2
3.1.3
Calories of one slice of cake = 75
100 × 400 ✓✓ M

= 300 calories ✓A
1M multiply by 75
1M divide by 100

1A correct answer
(3) M
L2
3.1.4 Convert min to hrs

✓M
75 min ÷ 60 = 1,25 hrs ✓ A

1M conversion ratio

1A correct answer

(2) M
L1
3.1.5 90 guests = 90 slices ✓M

Number of Cakes = 90 ÷ 12 ✓M

= 7,5 cakes ✓CA

≈ 8 cakes ✓R

1M number of slices

1M divide by 12

1CA number of cakes

1A correct rounding

(4) M
L1
3.2.1
Cups = 8 × 3
4 ✓M

= 6 cups ✓A
1M multiplication

1A correct answer
(2) M
L2
3.2.2 Cocoa = 8 × 90 g ✓M

Required Amount = 720 g ÷ 240 g
= 3 ✓M
✓M
Cost = 3 × R 62,75

= R188,25 ✓A 1M total grams
1M amount

1M multiplication
1A correct cost

(4) M
L3

3.2.3 ■ = (■ – 32) ÷ 1,8
✓SF
= (320 – 32) ÷ 1,8
✓M
= 288 ÷ 1,8

= 160 ■ ✓ CA

1SF substitution

1M 268 ÷ 1,8

1CA correct rounding

(3) M
L2
3.2.4
Starting time 09h 03 + 55 min ✓M
Finishing tim e = 09h 58 ✓ A

1M adding time
1A correct answer

(2) M
L2

3.3.1 Radius = 86 mm ÷ 2 = 43 mm ✓ A
Convert: 43 mm ÷ 10 = 4,3 cm ✓ C

Volume of one can = 3,142 × 4,3 × 4,3 × height ✓SF
546,10 cm3 = 58,09558 cm2 × height
Height = 546,10 cm3 ÷ 58,09558 cm2
= 9,4 cm ✓CA 1A radius value
1C conversion

1SF correct values

1CA height value
(4) M
L3

3.3.2 ✓M
Height of label = 80% × 9,4 cm
= 7,52 cm ✓A

Difference = 9,4 – 7,52

= 1,88 cm ✓M

His statement is invalid ✓J
CA value from Q 3.3.1
1M calculating 80 % of
9,4
1A for 7,52 cm

1M difference value of
1,88 cm

1J justification

(4) M
L3
[32]

QUESTION 4 [ 27]

Quest  Solution  Explanation  Level

4.1.1

Shows a building's plan as seen from above. It is a 2 - dimensional view of the building . ✓✓ A

2A correct explanation

(2) MP
L1

4.1.2   One window  ✓✓ A

2A correct  answer

(2) MP
L2

4.1.3        1 mm represents 100 mm

Length of wall
✓ M
114 mm = 114 × 100 ✓ M
= 11 400 mm ÷ 1 000 ✓ M
= 11,4 m  ✓ CA

Width of Wall
✓ M
77 mm = 77 × 100
= 7 700 ÷ 1 000 ✓ M
= 7,7 m   ✓ CA

1M for measurement
1MA using conversion factor
1M conversion to m
1CA for 11,4 m

1M for measurement
1M conversion to m

1CA 7,7 m
(7) MP
L4

4.1.4                          ✓ MA
Width = 19,38 m2 ÷ 10,2 m
= 1,9 m ✓ A

Times lesser = 10,2 ÷ 1,9 m ✓ MA
= 5,37 times

Mrs Smith's statement is invalid. ✓ J

1MA 19,38 ÷ 10,2
1A correct answer

1MA divide by correct values

1J correct justification
(4) M
L4

4.2.1
Area Circle  = ■× r2
= 3,142 × 1,65 × 1,65 ✓ SF
= 8,55 cm2  ✓ A

Area square = side × side
= 0,9 × 0,9
= 0,81 cm2 ✓ A

Area of coin = 8,55 – 0,81 ✓ M
= 7,74 cm2
= 7,7 cm2 ✓ A

1SF correct radius
1A correct answer

1A correct answer

1M subtraction

1A correct rounded off
(5) M
L3
4.2.2 Mass of coin = 1,47 × 19,30 ✓ M

= 28,371 grams
≈ 28,4 g ✓ A 1M correct values

1A correct answer
(2) M
L2

4.3.1
Probability gold coin = 12
16

= 0,75 ✓ A

1A numerator
1A denominator

1A correct answer

(3) P
L2
4.3.2 The probability of selecting a gold coin has a much greater chance of happening than not happening. ✓✓ O

OR

Very likely to select a gold coin. ✓✓ O

(Accept any relevant answer .)
2O correct reasoning

2O correct reasoning

(2) P
L4

[27]

TOTAL: 100

✓ A
✓ A

MARKS: 100

This marking guideline consists of 6 pages.

SYMBOL EXPLANATION

M Method
MA Method with accuracy
CA Consistent accuracy
A Accuracy (Answer)
C Conversion
S Simplification
RT/RG /RD Reading from a table/ graph / diagram
NPR No penalty for units/rounding
SF Correct substitution in a formula
O Opinion / reason/deduction/example
J Justification
R Rounding off /
F deriving a formula
E Explanation
U Units
AO Answer only full marks MATHEMATICAL LITERACY

COMMON TEST

MARCH 2022

MARKING GUIDELINE
NATIONAL
SENIOR CERTIFICATE

GRADE 11

QUESTION 1 [20 MARKS]
SOLUTION EXPLANATION T/L
1.1.1 6hours +0,5×60 ■MA
6hrs and 30minutes ■A
1MA, Multiplying by 60
1A, Time in hours and minutes
(2) M
L1
1.1.2 It means that the 30 minutes will be charged at R110. ■■E 2E, Explanation
(2) M
L1
1.1.3
■M
Income = R110 ×6 +R110 ■MA
= R770■CA 1M, Multiplying by 6
1MA, Adding R110
1CA, Answer
(3) F
L1
1.1.4 No of hours = R440
R110 ■MA
=4 ■A

1MA, Dividing by the rate
1A, Answer
(2) M
L1
1.1.5 B ■■A 2A, Correct graph
(2) B
L1
1.2.1 Radius =16cm
2 ■MA
=8cm ■A
1MA, Dividing by 2

1A. Answer
AO (2) M
L1
1.2.2 Total length = 50,3cm × 30■ M

=1 509cm ■CA 1M, Multiplying by 30
1CA, Answer
AO (2) M
L1
1.2.3 0,5 inches ■■RT
OR
1 inch■■RT
2 2RT, Correct reading
(2) M
L1
1.2.4 ■RT ■M
Difference = 80,5 – 72,8
=7,7 oF ■CA
1RT, Reading correct values
1M, Subtracting correct values
1CA, Difference
(3) M
L1
[20]

QUESTION 2 [25 MARKS]
SOLUTION EXPLANATION T/L
2.1.1 Opening balance is a balance brought forward from the previous account period. ■■E
2E, Explanation
(2) F
L1
2.1.2 Balance = R5 719,47 ■■A
2A, Answer
(2) F
L1
2.1.3 ■RT ■M
Total deposits = R272,45 +R250+R2100+R250
=R2 872,45 ■CA
1RT, Reading correct values
1M, Adding correct values
1CA, Answer
(3) F
L2
2.1.4 ■MA
Days = (31 – 15) +1+18
= 17+18 ■M
=35 days ■CA 1MA, Days in March
1M, addition
1CA, Answer.
(3) M
L2
2.1.5 ■RT ■ MA ■ MA
Closing balance = R5 234,09- R2387,07 +R2872,45
= R5 717,47
CA from question2.1.3

1RT, Opening balance.
1MA, Subtracting withdrawals
1MA, Adding deposits
(3) F
L2
2.2.1 ■M
Income = R25 000+ R17500
=R42 500 ■A 1M, Adding correct values
1A, Answer
AO (2) F
L1
2.2.2 Water and electricity ■■A
2A, Answer
(2) F
L1
2.2.3 Markerting exp%= R2000 × 100% ■M
R17990 ■M

=11,12% ■MA 1M, Dividing Correct values
1M, Percentage concept
1CA, Answer
NPR (3) F
L2
2.2.4 a) ■MA
Total cost = R160 +R0,80×20 ■M
=R176 ■CA
1MA, Multiplying by 20minutes
1M, Addition
1CA, Answer
(3) F
L3
2.2.4 b) Combination relation ■■A
2A, Answer
(2) B
L1
[25]

QUESTION 3 [33 MARKS]
SOLUTION EXPLANATION T/L
3.1.1 Overweight. ■■RT
2RT, Correct answer
(2) M
L2
3.1.2 BMI = 83,5kg
(1,7m)2■■SF
=28,8927 ■CA
=28,9kg/m2■R
2SF, Substitution correct weight
and height
1CA, Answer
1R, Rounding
(4) M
L3
3.1.3 Exercise regularly ■■O
OR
Eat healthy food ■■O
OR
Follow the diet programme ■■O

2O, Opinion
2O, Opinion

(4) M
L4
3.1.4 Weight in pounds = 2,2 ×83,5 ■MA
=183,7 pounds ■A 1MA, Multiplication by 2,2
1A, Answer
AO
(2) M
L2
3.2.1 ■M ■M
Width of the baseline = 8,23m+1,37m+1,37m
= 10,97m 1M, Addition
1M, Adding 1,37m both sides
(2) M
L2
3.2.2 ■SF ■A
P = 2×23,77m +2×10,97m
= 69,48m ■CA 1SF, Substitution
1A, Correct values
1CA, Answer
(3) M
L2
3.2.3 ■M
23,77m – 6,4m-6,4m
10,97m ■MCA
2 ■MA
5,485m ■CA
1M, Subtracting both sides 6,4m
1MCA, 10,97m
1MA, Dividing by 2
1CA, Answer
(4) M
L4
3.2.4 ■SF
Area =23,77m×1,37m ■M
=32,5649m2■CA

1SF, Substitution
1Multiplying
1CA, Answer.
(3) M
L2
3.3.1 No of calories = 83,5kg × 583 ■M
70kg ■S
= 695,44calories ■A 1M, Multiplying by 583
1S, Dividing by 70 kg
1A, Answer.
(3) M
L3
3.3.2 Mass = 695,44 calories
9 ■MA
= 77,27711g
1000 ■C
= 0,077kg ■A
The statement is invalid. ■J CA from Question 3.3.1
1MA, Dividing by 9
1C, Dividing by 1000
1A, Answer
1J, Justification
(4) M
L4
3.3.3 No of hours = 1600

583 ■M
=2,744 ■A 1M, Dividing by 583
1A, Answer
NPR (2) M
L3

[33]

QUESTION 4 [22]
SOLUTION EXPLANATION T/L
4.1.1 Amount in rands ■■RT 2RT, Answer
(2) F
L1
4.1.2 a)

1A, Starting point
1CA, Joining straight line point
1CA, Label
(3) F
L3
4.1.2 b) 50 breads ■■RT
2RT, Answer
(2) F
L2
4.1.2 c) R975■■RT
2RT, Answer
(2) F
L2
4.1.3 Income for one bread = R125
10
=R12,50 ■M
Total income = R12,50 ×No of bread sold ■CA

1M, Rate of R12,50
1CA, Equation (2) F
L3

4.2.1 Volume = 40cm ×10cm×10cm ×4 ■■SF
=16 000cm3■A 2SF, Correct Substitution
1A, Answer
(3) M
L3

Amount in rand
Number of breads

4.2.2 Rect. =16 000g ■C
Cylindrical = 3501,19×4 ■MA
=14004,76g
A rectangular compartment will have more mass ■CA CA from question 4.2.1
1C, Conversion of cm3to grams

1MA, Multiplying by 4
1CA, Opinion (3)
M
L4
4.2.3 5,08cm :40cm■MA
5,08cm 5,08cm
1:7,874… ■CA 1MA, Correct ratio order and dividing
1CA, Answer
AO
(2) M
L2
4.2.4 = (450-32) ÷1,8 ■SF
=418÷1,8 ■S
=232 ■CA 1SF, Substitution
1S, Simplification
1CA, Answer (3) M
L2
[22]
TOTAL MARKS: 100

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2022

MATHEMATICAL LITERACY P2
MARKING GUIDELINE

MARKS: 100

This marking guideline consists of 8 pages.

Symbol Explanation
M Method
M/A Method with accuracy
MCA Method with consistent accuracy
CA Consistent accuracy
A Accuracy
C Conversion
S Simplification
RT/RG/RM Reading from a table OR Reading from a graph OR Read from a map
F Choosing the correct formula
SF Substitution in a formula
J Justification
P Penalty, e.g. for no units, incorrect rounding off etc.
R Rounding off OR Reason
AO Answer only
NPR No penalty for rounding

QUESTION 1 [20]
Ques . Solution Explanation Topic
&

Level
1.1 1.1.1 R40,00
2RT correct value

(2) F
L1
1.1.2 Hours: 8 ✓ C
Convert 15 min = 15 ÷ 60
= 0,25 + 8
= 8,25 hrs. ✓ A 1C divide by 60
1A adding correct
values
(2) M
L1
1.2 1.2.1 Departure 14h20
Arrive – 11h30 ✓ MA
Lapse time = 2h50 min ✓ A 1MA subtraction time
1A simplified time
(2) M
L1
1.2.2 Cost = R50,00 + (R7,00 x R 3,00)
= R50,00 + R21,00
R71,00 ✓ A 1RT correct values
1A answer
(2) F
L1
1.3 1.3.1 It is the total length of the sides in a shape . ✓✓
OR
It is the distance around the outside of the
shape . ✓✓ R
Accept any other relevant reason . 2R Explanation

(2) M
L1
1.3.2 ✓ MA
Perimeter rectangular diagram = 210 + 210 +
100 + 100
= 620 cm ✓ CA 1MA adding sides

1CA answer
(2) M
L1
1.3.3 Numerical scale or Ratio scale ✓✓ A 2A concept scale

(2) M&P
L1
1.3.4 Every 1 unit on the map represents 100 units in
reality . ✓✓ A 2A Explanation
(2) M&P
L1
1.4 1.4.1 R44 ✓✓ A 2A correct road
(2) M&P
L1
1.4.2 Provincial roads serve as feeders to the national
roads. ✓✓ A
OR
Provincial roads also serve as trunk roads in
areas where there is no national roads. ✓✓ A
Accept any relevant explanation . 2A Explanation

(2) M&P
L1
[20]
✓✓ RT

✓RT

QUESTION 2 [30]
Ques . Solution Explanation Topic
&
Level
2.1
2.1.1 Bar scale ✓✓ 2A 2A correct scale
(2) M&P
L1
2.1.2 Width of bar (measured) = 2,1 cm ✓ A
Length of animal = 15,1 cm ✓ A

2,1 cm = 1,9 m
✓ M
15,1
2,1 × 1,9 m = 13,6619 m ✓ CA
= 13,7 m ✓ R

OR

✓A
2,1 cm = 1,9 m
✓ A ✓ M
15,1 cm = 15,1 cm ×1,9 m
2,1 cm

= 13,6619 m ✓ CA

= 13,7 m ✓ R
1A measured value for bar
width
2,1 cm
Accept 2 – 2,3 cm
1A measured value
diagram for
15,1 cm
1M multiply by 1,9 m
1CA correct value
1R one decimal number

1A measured value for bar
width
2,1 cm
1A measured value
diagram for 15,1 cm
1M multiply by 1,9 m
1CA correct value
1R one decimal number
Mark according to your
measured length.
(5) M&P
L2
2.1.3 1 ton = 1 000 kilogram
20 ton = 20 × 1 000 ✓ A
= 20 000 kilogram ✓ A 1A × 1 000
1A correct value

(2) M
L1
2.2 48 000 – 30 000 = 18 000 ✓ MA
✓ A
Percentage illegal fishing = 18 000
30 000 × 100

= 60% ✓ A 1MA Subtracting correct
number of tuna
1A × 100
1A correct percentage

(3) F
L3

2.3 2.3.1 M9 ✓ RT
M18 ✓ RT
2RT correct roads

(2) M&P
L1
2.3.2 N2 ✓✓ RT
2 RT
(2) M&P
L1
2.3.3 R75 ✓✓ RT 2 RT
(2) M&P
L1
2.3.4 6 provincial roads ✓✓ 2A
OR
R75; R102; R334; R335; R367; R368 ✓✓ 2A 2A correct
number
2A correct list of
roads
(2) M&P
L2
2.3.5 Measured distance : 5 cm ✓ A
5 cm : 13 km ✓ M
5 cm : 1300 000 cm ✓ C
5 ■■
5 ■■ : 1 300 000 cm
5 cm ✓ S

1 : 260 000

Yes, his statement is valid . ✓✓ 2J 1A measured
value
1M ratio concept
1C conversion
1S simplify
values

2J conclusion
(6) M&P
L4
2.3.6 Time = 13 km
80 km /h ✓ SF

= 0,1625 hrs × 60
= 9,75 min ✓ C

= 0,75 × 60 = 45 seconds ✓ C
Total time = 9 min 45 seconds ✓ S 1SF correct
substitution
1C converting hrs
to min
1C converting
min to seconds
1S simplifying
total time
(4) M
L2
[30]

QUESTION 3 [23]
Ques . Solution
Explanation Topic
&
Level
3.1 3.1.1 Perimeter = 2 (length + breadth)

= 2 (8 500 mm + 3 500 mm)
= 2 (12 0 00 mm) ✓
= 24 000 mm ✓ 1SF correct values
1S simplification
1CA answer

(3) M
L2
3.1.2 Area of floor = Length × Width ✓ C
= 8,5 m × 3,5 m ✓
= 29,75 m2 ✓ 1C convert to m
1M multiply 5,8 m with
3,5 m
1CA correct value
(3) M
V2
3.1.3 Volume = length × breadth × height (thickness)

= 8,5 m × 3,5 m × 0,1 m ✓

= 2,975 m3 ✓ 2A Reason

1SF substitute correct
values
1CA answer for 2,975 m3

NPR
(2) M
L3
3.1.4 Facing west ✓ and east. ✓ 2RT directions
(2) M&P
L3 ✓SF

3.2 3.2.1 ✓ C
Area of southern wall = 8,5 m × 5 m ✓ SF

= 42,5 m2 ✓ CA

Dimensions of 1 brick (with mortar)
Length = 222 + 10 ✓ M
= 232 mm
Height = 73 + 10
= 83 mm

Area of 1 brick = length × height
= 232 mm × 83 mm ✓ SF
= 19 256 $mm^2$ ÷ 1 000 000 ✓ C
= 0,019 256 $m^2$ ✓ CA

No. of bricks = Area of the wall in $m^2$
Area of one brick in $m^2$

= 42,5 $m^2$
0,019256 $m^2$ ✓ SF

= 2 207 ,10
= 2 208 bricks ✓ CA 1C Convert cm to m
1SF correct values
1CA correct value for
42,5 $m^2$

1M adding 10 mm

1SF correct values
1C divide by conversion
factor
1CA for 0,019256 $m^2$

1SF correct values

1CA number of bricks
(9) M
L3
3.2.2
10% × 2 208 bricks = 220,8 + 2 208 ✓ M
Total amount of bricks ≈ 2 428,8
≈ 2 429 bricks ✓ CA

OR

110% × 2 208 = 2 428,8 ✓ S
≈ 2 429 number of bricks ✓ CA CA from 3.2.1
1M adding correct no of
bricks
1CA total no of bricks

1S simplifying correct
number of bricks
1CA total number of
bricks
(2) M

L2
3.2.3 2 429 total number of bricks (answer to QUESTION 3.2.2)
R4,75 price of one brick

2 429 x R4,75 ✓ M
= R11 537,75 ✓ CA
1CA total number of bricks
1M price of one brick
1CA total amount

(2) M
L4
[23]

QUESTION 4 [ 27]
Ques . Solution Explanation Topic& Level
4.1
4.1.1 Scale 1 : 30
✓ C
A = 45 ×1
30 = 1,5 cm ✓ CA
✓ M

✓ M
B = 105 × 1
30 = 3,5 cm ✓ CA 1C conversion to cm
1M divide by 30
1CA value of 1,5 cm

1M using 105 cm
1CA value of 3,5 cm

(5) M&P
L4
4.1.2 Volume = l × b × h
✓ C
= 105 cm × 45 cm × 30 cm ✓ SF
= 141 750 cm3 ✓ CA
1C conversion
1SF substitute correct values
1CA Simplification
(3) M
L2
4.1.3 1 cm3 = 1 m■
141 750 cm3 = 141 750 m■ ✓ A

1 litre = 1 000 m■
✓ M
141 750 ml
1 000 ml = 141,75 litres

0,9 × 141,75 = 127,575 litres ✓ CA

1A ratio concept

1M divide by 1 000
1CA value of 127 ,575
litres
(3) M
L2
4.2 4.2.1 Actual length = 50,4 mm × 150 ✓ M
= 7 560 mm ÷ 1 000 ✓ M
= 7,56 m ✓ CA

Actual width = 25,6 mm × 150 ✓ M
= 3 840 mm ÷ 1 000 ✓ M
= 3,84 m ✓ CA
1M × 150
1M ÷ 1 000
1CA correct value

1M × 150
1M ÷ 1 000
1CA correct value
(6) M
L2
4.2.2 Area of rectangle = length × breadth
= 7,56 m × 3,84 m ✓✓SF
M
= 29,03 m2 ✓ CA 1SF correct values
1M multiply values
1CA answer
(3) M
L3

4.3 Probability not winning: 10 000
10 000 – 1000
10 000
✓A

= 9 000
10 000 × 100
✓A

= 90% ✓ A
1A numerator
1A denominator

1A for 90%

(3) P
L2
4.3.1 Unlikely ✓✓ A 2A Explanation

(2) P
L1

4.3.2 10% : 90 % ✓ A

1 : 9 ✓ A
1A ratio concept

1A simplified
values

(2) P
L1
[27]

TOTAL : 100

NATIONAL
SENIOR CERTIFICATE

GRADE 1 1

NOVEMBER 2022

MATHEMATICAL LITERACY P 1
MARKING GUIDELINE

MARKS: 100

| Symbol | Explanation |
|---|---|
| M | Method |
| MA | Method with accuracy |
| CA | Consistent accuracy |
| MCA | Method with consistent accuracy |
| A | Accuracy |
| C | Conversion |
| S | Simplification |
| RT/RG/RM | Reading from a table/Reading from a graph/Read from a map |
| F | Choosing the correct formula |
| SF | Substitution in a formula |
| J | Justification |
| P | Penalty, e.g. for no units, incorrect rounding off etc. |
| R | Rounding Off/Reason |
| AO | Answer only |
| NPR | No penalty for rounding |

This marking guideline consists of 7 pages .

QUESTION 1 : FINANCE AND DATA HANDLING [19]

Ques Solution Explanation T&L
1.1.1 Clerk/Cashier 2RT correct occupation
(2) F
L1
1.1.2 Total deductions = R650 + R774 + R61,25
= R1 485,25

OR

Total deductions = R6 125 –R4 639,75
= R1 485,25 1M adding the
deductions
1A answer

1M subtracting
1A answer (2) F
L1

1.1.3 Overtime = R25 × 45
= R1 125 1RG values R25 and 45
1M multiplication (2) F
L1
1.1.4 Net pay is the amount received by employee after subtracting total deductions ✓ from gross income 2J explanation
(2) F
L1
1.1.5 Increase = 7,5
100× 5 000 ✓ MA ✓M

= R375 ✓A 1MA percentage of 7,5%
1M multiplication with
R5 000
1A answer (3) F
L1
1.2.1 Histogram ✓✓ RT 2RT correct answer
(2) D
L1

1.2.2
Total = 30 + 25 + 40 + 35 + 25 + 20 + 15 + 10 + 10 + 5
= 215
1M addition
1CA answer (2) D
L1
1.2.3 40 ✓✓ RG 2RG answer (2)
1.2.4 68
100× 50 = 34 marks 1M multiplication with
% with 50
1A answer (2) D
L1
[19]
✓ M
✓ A

✓J
✓M✓ A ✓M

✓✓ RT

✓ A
✓ M
✓ A
✓ RG ✓ M

✓CA

QUESTION 2: FINANCE [33]

Ques . Solution Explanation T&L
2.1.1 R2 470 783 2RT correct lump sum
payable to beneficiaries
(2) F
L1

2.1.2
R2 836 836 – R166 417
= R2 670 419 1RT correct value for
2022
1M subtraction R166 417
from the value of 202 2

1CA correct answer (3) F
L1
2.1.3 2022
Annual amount = R26 383 × 12

= R3 16 596

= R3 16 600
1M multiplying the
correc t value by 12
1S simplification

1R correct rounding (3) F
L2
2.1.4 Difference in l ump sums for 202 2 and 202 1
= R919 363 – R884 198
= R35 165
R35 165 is not more than R 35 615 statement is invalid . 1RT and subtracting
correct values R919 363
and R884 198
1CA simplification
1J justification. (3) F
L4
2.1.5 Advantage :
It is a saving for the future
One gets income after retiring
Beneficiaries get some income when the bread winner has
passed on .
Choose ONE or ANY relevant answer . 2J any correct reason
given

(2) F
L4
2.2.1 Selling price for Samsung Galaxy A72
= R7 499,25 + R2 497,75
= R9 999,00
1M addition
1A answer (2) F
L1
2.2.2 Total profit = R1 099,75 + R2 499,75 + R3 699,75 +
R7 249,75
= R14 549,00

Total income = (3 299,25 + 1099,75 + 7499,25 + 2499,75 +
11099,25 + 3699,75 + 21749,25+7249,75)
= R58 196,00

Ratio of total profit : total income
= R14 549,00 : R58 196,00
= 1 : 4 1M adding the total costs

1M adding the profits

1M forming ratio
1CA reduced to simplest
ratio form
(4) F
L3
2.2.3 A31 cellphones ordered = 2

3 × 15 = 10
A72 cellphones ordered = 1
3 ×15 = 5
Cost price of A3 1: 10 × R3 299,25 = R32 992,50
Cost price of A7 2: 5 × R7 499,25 = R37 496,25

Total cost price of the order = R32 992,50 + R37 496,25
= R70 488,75 1A for number of A31
phone ordered
1A for number of A72
phone ordered .
1MCA cost price for A31

1MCA cost price for A72
1MCA addition
(5) F
L4
✓✓ RT
✓ M
✓ S
✓ R
✓ RT
✓ CA
✓ J
✓✓ J ✓ M
✓M
✓A
✓M
✓M
✓CA
✓A
✓A
✓MCA
✓MCA
✓MCA ✓CA ✓RT
✓M

Ques Solution Explanation T&L
2.2.4 Percentage decrease = R3 299,25 – R3 399,75
R3 399,75×100% ✓SF ✓SF

= -2,96% ✓S

= 3,0% ✓R
1SF substituting the
numerator values
1SF substituting
denominator value
1S simplification
1R rounding
(NPR for negative
answer) (4) F
L3

2.3
Amount charged : 6,110 × R16,992900 = 103,826619
: 4,073 × R17,326130 = 70,56932749
: 2,817 × R24,06370 = 6 7,7874429

Total: R24 2,18

Amount charged including VAT: R24 2,18 × 1,15
= R2 78,51

INVALID statement
1M multiplying by the
rate

1S simplification

1M VAT calculation
1CA simplification
1J justification (5) F
L4
[33]

QUESTION 3: DATA HANDLING AND PROBABILITY [ 19]

Ques . Solution Explanation T&L
3.1 April 2018 1RT for the month
1RT for the year
correct (2) D
L1

3.2
Range = R59 960 – R52 361
= R7 599
1RT identifying the
values
1M subtraction
1CA answer (3) D
L2

3.3
Six hundred and sixty -six billion, four hundred and twenty -
four million

1A value of billions
1A value of millions
(2) D
L1

3.4
Mean sales in millions = 666 424 000
12

= R55 535 333,33

1RT value of the
numerator
1M dividing by 12
1A correct answer
(3) D
L2

3.5 Descending order :
59 960; 59 270; 58 435; 57 915; 56 846 ; 55 646 ; 54 981 ;
54 467; 54 249; 53 844; 52 718; 52 361

Median in millions = 55 646 + 54 981

2
= R55 313,50 1M arranging in
descending /ascending
order

1M concept of median.
1CA answer
(3) D
L3

✓✓ RT
✓ RT
million ✓CA ✓ M
✓ A
✓ A
✓M ✓ RT
✓A
✓M
✓M
✓CA ✓M
✓ J ✓ S
✓CA }
✓ M

3.6 Annual % increase rate on monthly p remium

= 60 406
666 424 × 100%

= 9,06 % 1RT correct values
1M calculation of
percentage.
1CA answer

(NPR) (3) D
L2

3.7
P (month with decline from 2018 to 2019) = 5
12 2A identifying the number
of 5 months
1A answer denominator
(3) P
L3
[19]

QUESTION 4 : DATA HANDLING [29 MARKS]

Ques . Solution Explanation T&L
4.1.1 R650

OR
Fixed cost = R250 + 400

= R650 2RT value from expenses
formula
OR
1RT for adding both
values
1A answer (2)

F
L1
4.1.2
(a) A = R650 + R25 × 20
= R1 150 1SF substitution in expenses formula
1A simplification and answer (2) F
L2
4.1.2
(b) Total expenses = R775 + R900 + R1 150
= R2 825

Total Income = R375 + R750 + R1 500
= R2 625

Expenses – Income = R2 825 – R2 625
= R200 1MCA adding the expenses with CA from
4.1.2
1S simplification
1M adding income values
1S simplification

IMCA subtracting the two values that give the loss of R200
(5) F
L4
✓ CA ✓ M ✓RT
✓✓A
✓✓ RT
✓ RT
✓ A
✓ SF ✓ A
✓ MCA
✓ S
✓M
✓ S

✓ MCA ✓A

4.1.3

Starting point (0;650)
Any other correctly plotted point
Joining points with a straight line
(3) F
L2

4.1.4 Number of pair of shoes = 13 at break -even point 2RT value of 13 from graph (2) F
L2

4.2 1 Japanese yen (¥) = R0,1352364
(¥) 25 000 = R?
Amount received = 25 000 × R0,1352364
= R3 380,91 2MA multiplication of the two values
1A answer
(Accept R3 380,90)
(3) F
L2

Income and Expenses (R)
Number of pairs of shoesGRAPH FOR INCOME AND EXPENSES SHOE REPAIR BUSINESS
INCOME
EXPENSE
✓A
✓ A
✓A

✓
A ✓✓RT
✓✓MA
✓A ✓A ✓A
✓A

4.3.1

| Weight Category | Female Tally | Female Frequency | Male Tally | Male Frequency |
|---|---|---|---|---|
| 50–59 | // | 2 | / | 1 |
| 60–69 | //// | 4 | // | 2 ✓A |
| 70–79 | //// | 4 | | 5 ✓A |
| 80–89 | / | 1 | /// | 3 ✓A |
| 90–99 | / | 1 | / | 1 ✓A |
| 100–109 | / | 1 | 0 | 0 ✓A |

1A tally and frequency for each row of male s

(5) D

L2
4.3.2  60–69
70–70 1RT for correct category
1RT for correct category
 (2) D
L2
4.3.3  5 + 3 = 8 2RT for the correct
answer of 8
(2)
D
L3
4.3.4
P(more than 70  kg) = 16
25×100%

= 64%
1RT for the correct value
of numerator and
denominator
1M percentage
calculation
1A answer (3) P
L3
[29]

TOTAL:   100

✓RT
✓✓RT
✓RT
✓M
✓A
✓RT

Please turn over MARKS:  100
This marking guideline consist of 7 pages. Symbol Explanation
M Method
M/A Method with Accuracy
CA Consistent Accuracy
A Accuracy (Answer)
AO  Answer only full marks
C Conversion
S Simplification
RT / RG / RM/RP Reading from table / Reading from graph / Reading from map/Reading from plan
F Choosing the correct formula
E Explanation/Comment
D Correct definition
SF Substitution in formula
O Opinion
J Justification
P Penalty e.g. for no units, incorrect rounding, etc
R Rounding off / Reason MATHEMATICAL LITERACY
MARCH COMMON TEST 2020
MARKING GUIDLINES
GRADE 11

QUESTION 1
Q Solution  Explanation  T&L
1.1.1

1M multiplying 20 by 1/4
1A correct answer
(2) M
L1
1.1.2
1M for dividing 25 by 2
1A correct answer
(2) M
L1
1.1.3 1M multiplying 3,5 by 25
1M multiplying by 453,59
1M for dividing by 1000
1C correct conversion
(4) M
L2
1.1.4
= 232 1SF correct substitution
1CA correct answer
1R Correct rounding
(3) M
L2
1.1.5
= 611 min
= 17 :26 OR 5:26 pm 1M dividing by 4
1R rounding off
1M multi plying 13 by 4 7
1A correct answer
1C correct conversion
1M addition
1CA correct time
(7) M
L4
2
1.2.1 2D correct definition
(2) F
L1
1.2.2 C = 008 RT 2RT correct employment
number
(2) F
L1

Q Solution Explanation T&L
1.2.3 1M multiply by 7,5%
1A the correct value of A
1A correct value of B
1M subtracting 425 from
5000
1A correct value of D
(5) F
L2
1.3.1 2A correct answer
(2) F
L1
1.3.2 1M for dividing by R1,85
1R for rounding off 32,5
1M multiplying by 33
1CA correct answer/accuracy
(4) F
L2
1.3.3
= 66,60
=
= R3 600 1M subtracting R8, 50 from

R75,10
1M Diving by 1 ,85
1A correct answer
1M Multiplying by 100
1CA correct answer
(5) F
L3
[38]
QUESTION 2
2.1.1 The amount of money left over when all payment have been made. 2D Correct definition
(2) F
L1
2.1.2 1M mul tiplying by 110 %
1A for the new fees
1M dividing by 1 650
1CA correct
answer/accuracy
(4) F
L3
2.1.3 2D correct definition
(2) F
L1
2.1.4 (R2 500 – R1 650 = R 850) 1M subtracting 1650 from 2500
1M multiplying 850 by 750
1A answer
1O correct verification
(4) F
L4

Q Solution Explanation T&L
2.1.5 1M addition
1A total expenditure 1A correct answer
(3) F L2
2.1.6
OR 2O any good suggestion
2O any good suggestion
(2) F
L4
2.2.1 2A for the correct formula.
(2)F L2
2.2.2 1M multiplying 50 by 200 by 12 1A total cost 1M multiplying 100 by 200 by 12 1A for total income 1M for the difference 1CA answer/accuracy 1O correct conclusion
(7) F L4
2.3.3 1M for R10 000
1M dividing 364 972 by 10 000 1A correct answer
(3) F
L2
[29]
QUESTION 3 [21 MARKS]
3.1 2D correct definition
(2) F
L1
3.2 1M multiplying by 1,15
1A answer
(2) F L2
F
S2
3.3 1M for multiplying 50 by 0,7298 1M for multiplying 228 by 0,9373 1CA Correct

answer
(3)F L2

TOTAL: 100 Q Solution Explanation T&L
3.4
1M for multiplying by 300
1M for multiplying by 250 1A Correct answer
1M dividing 32,85 by
1,6055 1CA answer/accuracy 1M adding correct values 1A correct answer
(7) F L3
3.5 To discourage people from reckless or careless use of
electricity.  The greater the usage, the more the consumer
pays.     O 1O correct reason
(2)F
L4
3.6 1A for each correct line segment

(5) F L2
[21]
QUESTION 4[12 MARKS]
Q Solution Explanation T&L
4.2.1 1A for correct answer 1J correct reason
(2) P&R L4
4.2.2 2A for correct answer
(2) P&R L1
4.2.3
= 1 451.6 % 1SF correct substitution 1M dividing by 291 1A correct answer     1 R
correct rounding
(4) P&R L2
4.2.4 2J correct reason/justification
(2) P&R L4
4.2.5
2RG correct reading from graph
(2) P&R L4
[12]

MEMO – QUESTION 3.6
0.8393 0.9413 1.0433 1.1453 1.2473 1.3493 1.4513 1.5533 1.6553 1.7573
0 50 100 150 200 250 300 350 400 450 500 550 600 650 700 750 800 850 900 950 1000 1050
1100 1150 1200 1250 1300 1350 1400 1450 1500 1550 CHARGE/KILOWATT HOUR (R)
NUMBER OF KILO WATT HOURS  UMHLATHUZE RESIDENTIAL ELECTRICITY TARIFFS
A
A  A  A A

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NATIONAL
SENIOR CERTIFICATE

GRADE 1 1

NOVEMBER  2020

MATHEMATICAL LITERACY P 1
MARKING GUIDELINE
EXEMPLAR

MARKS:  100

Symbol  Explanation
M  Method
MA  Method with accuracy

CA  Consistent accuracy
A  Accuracy
C  Conversion
S  Simplification
RT/RG/RM  Reading from a table/Reading from a graph/Read from map
F  Choosing the correct formula
SF  Substitution in a formula
J  Justification
P  Penalty, e.g. for no units, incorrect rounding off etc.
R  Rounding Off/Reason
AO Answer only
NPR  No penalty for rounding

This marking guideline consists of 8 pages .

MARKING GUIDELINES

NOTE:

■ If a candidate answers a question TWICE, only mark the FIRST attempt.
■ If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled version)
■ Consistent accuracy (CA) applies in ALL aspects of th e marking guidelines, however it stops at the second calculation error.
■ If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra incorrect item presented.

LET WEL:

■ As ■ kandidaat ■ vraag TWEE keer beantwoord, merk slegs die EERSTE poging.
■ As ■ kandidaat ■ antwoord van ■ vraag doodtrek (kanselleer) en nie oordoen nie, merk die doodgetrekte (gekanselleerde) poging.
■ Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyn toegepas, maar dit hou by die tweede berekeningsfout op.
■ Wanneer ■ kandidaat aflesings vanaf ■ grafiek, tabel, uitlegplan en kaart geneem en ekstra antwoorde gee, penaliseer vir elke ekstra verkeerde item.

QUESTION 1 [2 3 marks ]

Ques . Solution  Explanation  T&L
1.1.1  Annual gross salary = R10 500 × 12
= R126 000
1M Multiply by 12
1A Gross per annum  (2) F
L1

1.1.2  Monthly food expense = R10 500 × 36%
= R3 780  1M % Calculation
1CA Amount  (2) F
L1

1.1.3  Housing % : Food %
= 21% : 36%
= 7 : 12 ■CA
1M Correct values and order
1CA Simplest form  (2) F
L1

1.1.4 Savings % = 100% – (21% +36% +10% + 1,9%)

= 100% – 68,9%
= 31,1%   1M Adding  correct values
1M Subtracting from 100
1CA  Percentage                (3) F
L1

1.2.1 Primary data
2A Correct data type
(2) D
L1

1.2.2  41 ■■ RT
2RT Highest  mark
(2) D
L1

1.2.3 Median is the middle value of a set of data which is arranged from small to big . 2A Explanation
(2) D
L1

1.2.4  35
2A Correct mark
(2) D
L1

1.2.5 3 2RT No. of learners failed
(2) D
L1

1.3.1  Loss is when the cost is more than the income.  ■■ A
OR
Loss incurred when selling price is less than cost price of an item.  ■■ A 2A Correct explanation

(2) F
L1

1.3.2  % loss =  50
750× 100%  ■M
= 6,67% ■ CA 1M Fraction multiplied by
100%
1CA Percentage
NPR                          (2) F
L1
[23]
■ M
■ A
■M
■M
■■A
■■ RT ■M
■CA
■■ A ■M
■CA
■■A
A

## QUESTION 2 :  FINANCE [ 30 marks ]

Ques . Solution Explanation Topic
/Level
2.1.1 SmartMAXFocussed Education Plan 1 2A Correct investment plan
(2) F
L1

2.1.2 Number of units =
100470, 8266
= R82,6647
=
6647,8207, 8038
= 97,23703104 1C Converted to Rands
1CA Value

1M Division

1CA No. of units
(4) F
L2

2.1.3 % loss = 12 924,75 – 6 995,25
= R5 929,50
=
10075, 1292450, 5929■
= 45,88%
OR
Percentage loss =
10012924,7525, 6995■
= 54,12%
= 100% – 54,12%
= 45,88% 1M Subtraction of values
1S Simplification
1M Dividing correct values
1M Multiply by 100%

OR

1M Dividing correct values
1M M ultiply by 100%
1S Simplification
1M Subtraction of %
(4) F
L3

2.1.4 B = R8 038,07 – R6 995,25
= R1 042,82 1MA Subtraction
1CA Correct answer (2) F
L2

2.1.5 R765,57
2RT Correct value
(2) F
L2

2.1.6 % increase =
10075,33275,33202,366■■
= 9,998%
= 10% 1RT Correct values
1SF Substitution
1S Simplification
1R Nearest % (4) F
L2

2.2.1 Number of plates ■■ RT
2RT Number of plates
(2) F
L2

2.2.2 Fixed expenses = R500 ■■ RT 2RT Fixed expenses
(2) F
L2

2.2.3 Income = R50 × Number of plates sold ■ M ■ A 1M Multiplication with R50
1A Correct formula (2) F
L2

2.2.4 R0 OR (No Profit) ■■ RT 2RT No profit
(2) F
L2 ■ R ■ S ■■ RT
■ M
■ S
■M ■ CA
■ M ■ C
■ CA
■■RT ■C A ■ MA
■RT ■ M
■ M
■ SF ■ S

■ M
■ M

2.2.5 Loss for 8 plates = Expenses – Income
■RT ■ RT
= 740 – 400 ■M
= R340 ■A
OR

Expenses = 500 + 8 × 30 = R740 ■M
Income = 50 × 8 = R400 ■M
Loss = 740 – 400 = R340 ■A 1RT R7 40
1RT R400
1M Subtraction
1A Loss
(From graph allow 3 40±10)
OR
IM for R 740
1M for R400
1M subtraction
1A for R3 40 exact answer (4) F

L3

[30]

QUESTION 3 : DATA HANDLING (18 marks) AND PROBABILITY (3 marks)
Ques . Solution Explanation T/L
3.1 Total number of spas
= 30 394 + 13 856 + 3 984 + 6 057 + 46 282 + 48 679
=149 252 ■M ■A
1M Adding correct values
1A Total (2) D
L1

3.2 Mean = 149 252
6 ■M
= 24 875,33
= 24 875 ■ R
CA from 3.1
1M Divisio n
1R Whole number
(2) D
L2

3.3 Europe an spas as a % = 46 282
149 252×100 ■M
= 31% ■CA 1M Fraction with correct
values and
multiplication by 100
1CA Percentage (2) D
L2

3.4 Range = 48 679 – 3 984
= 44 695 ■S
Number of regions above range = 2 CA■
1S Calculate range
1CA Number of regions (2) D
L3

3.5 30 394 : 48 679 = 1 : 48 679
30 394 ■M ■M

=1 : 1,60 ■ CA 1M Ratio
1M Fraction
1CA Unit ratio
NPR (3) D
L3

3.6 Revenue in sub-saharan Africa = 6,6 – 5,0 ■M

= 1,6 ■S
Total revenue for spas
= 22,9 + 6,6 + 1,6 + 2,8 + 33,3 + 26,5 ■M
= $93,7 billion ■CA 1M Subtraction
1S Simplification
1M Addition
1CA Total revenue
Penalise 1 mark if not in
billions (4) D
L3

3.7 1,6; 2,8; 6,6; 22,9; 26,5; 33,3 ■M

Median revenue = 6,6 + 22,9
2 ■M
= $14,75 billion ■CA CA the value $1,6 from 3.6
included in the data
1M Arranging in order of
descending or ascending
1M Concept of median
1CA Answer in billions (3) D
L3

3.8 P (Region s with more than 40 000 spas)

= 2
6 ×100 ■M

= 33,33% ■CA 1RT Correct numerator and
denominator
1M Multiplication by 100

1CA Percentage
NPR (3) P
L2

[21]
■RT

QUESTION 4 : FINANCE (12 marks), DATA HANDLING (11 marks) AND
PROBABILITY (3 marks)
Ques . Solution Explanation T/L
4.1 Cost =
Kilolitre Cost
6 6 × 0 = 0 ■M
9 9 × 9,35 = R84,15
10 10 × 11,16 = R111,6
Total = 25 litres 84,15 + 111,60 = R195,75

R195 ×115% = R225,11
OR

Cost = ( 6 × 0 ) + (9 × 9,35 ) + (10 × 11,16 )
= R84,15 + R111,60
= R195,75
Including VAT = R195,75 × 15%
= R29,3625
= R195,75 + R29,3625
= R225,11

Increased block rate tariffs to encourage saving of water ■A
OR
Also assist small businesses or families with free water ■A
Accept any other sound reason .

1M Cost of first 6 k■

1M Cost for both 9 and 10
kilolitres
1CA Total cost

1M Multiply by 15%
1CA Cost including VAT

1A Reason

(6) F
L4

4.2 R0,019 = 1 RWF
R? = 745 614,04 RWF ■M
R? = 0,019 × 745 614,04 ■M
= R14 166,66676 ■S
Bank charges = 14166,66676 ×10
100 ■M
= R1 416,666676 ■A

Andile received = R14 166,66676 – 1 416,66676 ■ M
= R12 750

Statement is v alid. ■A

OR
Bank charges = 10
100 × 745 614,04 RWF ■M
= 74 561,404 ■A
Andile received in RWF = 745 614,04 – 74 561,404 ■M
= 671 052,636 ■S
In Rands: R0,019 = 1 RWF
R? = 671 052,636 RWF ■M
Andile received = R0,019 × 671 052,636 ■M
= R12 750
Statement is valid ■CA 1M Concept of ratio

1M Multiplication
1S Simplification value in R
1M Multiplication of 10%
1A Value of 10%
1M Subtraction

1A Valid

OR
1M Multiplication of 10%
1A Value of 10%
1M Subtraction
1S Simplification value

1M Concept of ratio
1M Multiplication

1CA Valid (6) F
L4

■CA ■M
■M ■M ■M ■M
■M
■CA ■CA

4.3.1 Total absentee s = 67 ■ M
Absentees on Wednesday =16 ■A
P(absent on Wed) = 16
67 ■CA 1M Addition to 67
1A Absentees on Wed
1CA Fraction
(3) P
L2

4.3.2

1A for Monday
boys at 5
1A for Tuesday girls at 6
1A for Thu rs for girls at 7
1A for Thu rs for boys at 9
1A for Fri for boys at 7

(5) D
L2

4.4.1 Value of C:
64,2 = ■ + 1 853
30 ■M
64,2 × 30 = C + 1 853

C = 1 926 – 1 853 ■M
= 73 ■CA
Answer invalid
1M Addition (1 853) and
division by 30

1M Subtraction
1CA Value of C

1A Invalid (4) D
L4

4.4.2 D = 0 ■A
No learners scored 30 – 39 marks ■A 1A Value of D
1A Explanation
(CA value of D from 4 .4.1
includ ed in the data ) (2) D
L4

[26]

TOTAL: 100
024681012
Monday Tuesday Wednesday Thursday FridayNumber of absent learners
Days of the weekNumber of absent learners during the week
Girls Boys■A
■A
■A
■O ■A ■A

Mathematical Literacy (Marking Guideline) 1 September 2019 Common Test
NSC - Grade 11
Please Turn Over MARKS: 100

| Symbol | Explanation |
|---|---|
| M | Method |
| M/A | Method with Accuracy |
| CA | Consistent Accuracy |
| A | Accuracy (Answer) |
| AO | Answer only full marks |
| C | Conversion |
| S | Simplification |
| RT / RG / RM/RP | Reading from table/ graph /map/plan |
| F | Choosing the correct formula |
| E | Explanation/Comment |
| D | Correct definition |
| SF | Substitution in formula |
| O | Opinion |
| J | Justification |
| P | Penalty e.g. for no units, incorrect rounding, etc |
| R | Rounding off / Reason |

This marking guideline consists of 6 pages. MATHEMATICAL LITERACY
SEPTEMBER 2019
COMMON TEST
MARKING GUIDELINE GRADE 11
NATIONAL

Copyright Reserved  Please Turn Over  QUESTION 1     [31 marks]
Question  Solution  Explanation  T/L
1.1.1
2D correct definition  (2) F
L1
1.1.2
 1M adding earnings
1A correct answer  (2) F
L1
1.1.3
 1M for multiplying by 7 ,5%
1A correct answer
(2) F
L1
1.1.4
 1M subtracting deductions
 from gross salary
1A correct answer       (2) F
L1
1.1.5
■
■
2J for justification
(2) F
L4
1.2
 1A correct match
1A correct match
1A correct match  (3) M
L3
1.3.1
 1M for subtracting 0,5 cm
1A correct answer
(2) M
L2
1.3.2
PN = 3 cm – 1 cm = 2 cm ■A
■CA1M for subtracting correct values
1A correct value for LN
1M adding all values
1CA correct perimeter
1M for multiplying by 100
1CA correct answer/accuracy
(6)           M
L3
1.3.3 Width = 0,5m and depth = 0,4 m ■M
Volume  = 30 x 0,5 x 0,4 ■SF
OR
 ■M
 ■A 1C converting cm to m
1SF substitution
1A c orrect answer
OR
1SF correct substitution
1M dividing by 1  000 000
1A correct answer       (3) M
L2

Copyright Reserved  Please Turn Over  Question  Solution  Explanation  T/L
1.3.4
Ratio = 1:  4 : 5
Number of bags of cement   =
50918,4431 ■M
=   28,9 ■A
= 29 ■R
1A for 14 439,18kg
1M for multiplying by 1/10
1A a nswer
1M for dividing by 50
1A correct answer
1R correct rounding
1O correct opinion
(7)                              M
L4
[31]
QUESTION 2    [22 marks]
Question  Solution  Explanation  T/L
2.1.1
2E correct explanation
(2) F
L1
2.1.2
1M subtracting 109,1 from
111,7
1M dividing by 109,1
1A correct answer
(3) F
L2
2.2.1
2RT correct amount
(2) F
1
2.2.2
1SF correct substitution
1A correct answer
(2) F
L2
2.2.3
1A for correct value
1A for correct  value
1A for correct  value
1A for correct  value
(4) F
L2

Copyright Reserved  Please Turn Over  Question  Solution  Explanation  T/
L
2.2.4  Money that went out of the account reducing the
available balance. ■■E 2E correct opinion
(2) F
L1
2.2.5
CA from  2.2.3
1M for adding all the
credit val ues
1A correct answer
(2) F
L2
2.3.1
1M subtraction

1A correct answer
(2) F
L2
2.3.2 The average salary per annum increased ■ A from 2002
to 2017.
and distribu te electricity
due to the increase in salary. ■■O
OR
(any valid reason) 1A for increase
2O correct opinion
(3) F
L4
[22]

Copyright Reserved Please Turn Over QUESTION 3 [27 mar ks]
Question Solution Explanation T/L
3.1.1 D = 1,8 m x 100 ■M = 180cm ■A
H = 2,34 m x 100 = 234 cm ■A 1M multiplying by 100
1A correct diameter
1A correct height
AO (3) M
L1
3.1.2 R = 180 ÷ 2 ■M = 90cm ■A 1M dividing by 2
1A correct radius
AO (2) M
L1
3.1.3
■SF
1SF correct substitution
1A correct answer
1A for unit
(3) M
L2
3.1.4
= 5955,35 ■■R 1M dividing by 1 000
1A correct answer
1R rounding
(3) M
L2
3.1.5
■M
= 58,5 ÷ 100 ■C = 0,585
= 1 m ■R 1M dividing by 4
1A correct answer
1M for changing the subject of
the formula
1M dividing volume by 25450,2
1CA correct answer
1C for conversion
1R rounding
(7) M
L3
3.2.1
1A for 4 tins
1A for 5 tins
1A number of layers
1M for multipl ying the number
of tins by 4 layers
1A number of tins
1O correct opinion
(6)
3.2.2

OR
■C
■M

■A 1M multiplying 410 by 80
1C dividing 32 800 by 1 000
1A correct answer
1C conversion
1M multiplying by 80
1A correct ans wer
(3) M
L2
[27]

TOTAL : 100 QUESTION 4 [20 marks] T/L
4.1.1 56km ■■RD 2RD correct reading
(2) MP
L1
4.1.2 University of Cape Town ■■A 2A correct answer
(2) MP
L1
4.1.3 19■■A 2A correct answer
(2) MP
L1
4.1.4
1A number of medical stations
1A total number of st ations
(2) MP
L2
4.1.5
■M
■A
■SF
1M for subtraction
1C conversion to seconds
1A for correct seconds
1M mult iplying by 1000
1A correct metres
1SF substitution
1CA correct answer
(7) MP
L3
4.1.6
=
1SF correct substi tution
1A correct hours
1M for multiplying by 60
1A correct hours
1 CA correct minutes
(5) MP
L3
[20]

NATIONAL
SENIOR CERTIFICATE

GRADE 1 1

NOVEMBER 2018

MATHEMATICAL LITERACY P 1
MARKING GUIDELINE

MARKS:  100

| Symbol | Explanation |
|---|---|
| M | Method |
| MA | Method with accuracy |
| CA | Consistent accuracy |
| A | Accuracy |
| C | Conversion |
| S | Simplification |
| RT/RG/RM | Reading from a table/Reading from a graph/Reading from map |
| F | Choosing the correct formula |
| SF | Substitution in a formula |
| J | Justification |
| P | Penalty, e.g. for no units, incorrect rounding off etc. |
| R | Rounding Off/Reason |
| AO | Answer only |
| NPR | No penalty for rounding |

This marking guideline consists of 6 pages .

QUESTION 1  [21]
Ques  Solutions  Explanation  T&L
1.1.1  R2 578 799
Two million, five hundred and seventy eight thousand seven
hundred and ninety nine rand

2A Write in words (2) F
L1

1.1.2
% Deposit =
100799 578 285,819 386■

= 15%
1M Correct values
1M Multiply by 100

1A Answer in % (3) F
L1

1.1.3
R386 819,85
Transaction Cost  = R5,75 + R1,10 × 386  819 ,85
100
= R5,75 + R4 255,02
= R4 260,77
1A Correct value

1M Dividing by 100
1CA Transaction cost   ( 3)
NPR  F
L1

1.2.1
Distance = 82,3 – 26,9 ■M■RT
= 55,4 km ■ CA

1RT Correct distanc es
1M Subtraction
1CA Distance (3) M
L1

1.2.2
Time taken = 04:54:45 – 03:05:14 ■M
= 01:49:31 ■CA

1MA Subtracting correct
times
1CA Time (2) M
L1

1.2.3
Distance in metres = 68,9 × 1 000 ■C
= 68 900 m ■A
1C Multiply by 1 000
1A Distance in metres (2) M
L1

1.3.1
12, 8, 7, 5, 2, ■RG ■M

1RG Correct values
1M Descending order (2) D
L1

1.3.2
Bar graph OR Column graph ■■ A
2A Correct graph (2) D
L1

1.3.3
Total number of houses = 12 + 8 + 7 + 5 + 2 ■RG
= 34 ■A
CA from 1.3.1
1RG Values from the
graph
1A Number of houses (2) D
L1

[21]
■■A
■M
■A
■A
■CA ■M ■M

QUESTION 2 [29]
Ques Solution Explanation T&L

2.1.1
R2 250 ■■ RT
2RT Break -even
amount (2) F

L1

2.1.2
Cost of A = R1 500 + R5,00(50) ■SF
= R1 500 + R250 ■ S
= R1 750 ■CA
1SF Substitution
1S Simplification
1CA Answer
(3) F
L1

2.1.3
Income = R15,00 × 2 50 ■■
= R3 750,00
1RT Correct values
1M Multiplication
(2) F
L1

2.1.4
Profit = Income – Expenses
= R5 250 – R3 250 ■RT ■M
= R2 000 ■A
1RT Correct values
1M Subtraction
1M Profit (3) F
L1

2.2.1
1st year = R60 000 × 8,5 % ■M
= R5 100 ■A
1M Multiplication
1CA Interest (2) F
L2

2.2.2
1st year total amount = R5 100 +R60 000 ■M
= R65 100 ■ CA
2nd year total amou nt = R65 100 × 8,5 %
= R5 533,50 ■CA
Total at the end of 2 years = R65 100 + R5 533,50 ■M
= R70 633,50 ■CA CA from 2.2.1
1M Adding interest
1CA Amount
1CA % calculation
1M Adding interest
1CA Total amount
(5) F
L2

2.3.1 Water used = 587-561
= 26 k■ ■M
Cost = ( 0 × 6 k■) ■A + (20 × R10,02) ■ M
= R200,40 ■CA
Total cost = 200,40 + 80,70
= R281,10 ■CA
1M Water used
1RT Free k■
1M Multiplying by
R10,02
1CA Water cost
1CA Cost including
additional charge (5) F
L2

2.3.2
VAT amount = R80,70 × 15% ■M
= R12,105 ■S
= R12,10 ■R

1M Multiplying
1S Simplification
1R Rounding
(Accept R 12,11 )(3) F
L1

2.4.1
Inflation rate is a measure of inflation expressed in % showing the increase in price of goods and services. ■■A
OR
The rate at which price increases over time if there is a decline in the purchasing value of money ■■A
2A Explanation

(2) F
L1

2.4.2
Price of brown bread in 2017
= (1 + 6,59 %) × R9,99
= R10,65
1M Multiply
correct values
1A Cost (2) F
L1

[29]

## QUESTION 3 [17]

Ques Solution Explanation T&L

3.1.1
12 miles
Distance = 12 ×1,609
= 19,308 km ■A
1RT Correct value
1A Answer in km
NPR (2) M
L1

3.1.2
Humidity of Cape Town =
10068
=
2517

1RT Correct value

1A Simplified
fraction (2) M
L1

3.1.3
Time of sunset in Cape Town = 17:27
12-hour format = 05:27 pm

2A Correct time (2) M
L1

3.1.4
■=(■ ×1,8) + 32
= (17 × 1,8 ) + 32
= 62,6
= 63

1SF Correct value
1S Simplification
1R Rounding (3) M
L2

3.2.1
Volume = Length × Width × Height
= 30 in × 12 in × 7,1 in
= 2 556 in3
1SF Substitution
1S Simplification
1A Correct unit (3) M

L2

■ RT ■ A ■ M
■ SF ■ A
■■ A
■ S
■ ■ R

■ SF
■S ■A ■ RT

3.2.2
Volume of the tank = 2 556 in3 × 85%
= 2 172,6 in3

Increased volume after stones added = 2 556 × 97%
= 2479,32 in3
Volume of stones = 2479,32 – 2 172,6
= 306,72 in3 ■CA
OR
Volume of stone s = 97% - 85% ■M ■M
= 12% × 2556 ■M ■M
= 306,72 in3 ■CA CA from 3.2.1
1M Multiply by
85%
1CA Volume

1CA Volume
1M subtraction
1CA Volume

1M Using correct
values
1M Subtraction
2M Multiplication
by 12% and 2556
1CA Volume of
stones (5) M
L3

[17]

QUESTION 4 [13]
Ques Solutions Explanation T&L

4.1
Hartford ■■RM
2RM Correct city
(2) M&P
L1

4.2
Distance on the map = 7,5 cm
2,5 cm = 100 miles ■M

5,25,7 = 3 cm ■S
3 × 100 = 300 miles ■CA

1M Scale measure
(use the scale as
from actual map)
1S Division

1CA Multiply by
100 (3) M&P
L2

4.3
84■and 87 ■ RG
OR
81,■88 and 90 ■RG
2RG Combination
of roads
(2) M&P
L1

4.4
North East ■■A

2A Correct direction
(2) M&P
L1

4.5
Road 80

2RG Correct road
(2) M&P
L1

4.6
Probability = ■
■■
1A Numerator
1A Denominator (2) P
L2

[13]
■M
■CA
■M ■CA

■■ RG
■ RG ■ RG
6 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2018)

Copyright reserved Please turn over QUESTION 5 [20]
Ques Solutions Explanation T&L

5.1
Poults not hatched in December = 28 795 – 25 422 ■M
= 3 373 ■CA 1M Subt racting
correct
values
1CA Not hatched (2) D
L1

5.2
Mean
= 28 927 + 28 409 + 27 179 + 28 795 + 29 961 + 29 906 + 30 030 +
28 597 + 28 825 + 29 441 + 29 271 + 29 725
= 349 066
12 ■M
= 29 088,93 ■CA

1M Adding
1M Dividing by 12
1CA Mean
NPR (3) D
L2

5.3
No modal value ■■A
2A Modal value (2) D
L1

5.4
Range = highest value – lowest value
= 25 719 – 22 782 ■MA
= 2 937 ■CA
1MA Subtracti ng
correct values
1CA Range (2) D
L2

5.5
25 719, 25 422, 25 332, 25 075, 24 786, 24 616, 24 067, 23 645,
23 598, 23 572, 23 179, 22 782 ■A

Median =
2067 24 616 24■ ■MA
= 24 341 ■ CA
1M Arranged

1MA Median concept
with correct values
1CA Median (3) D
L2

5.6 Total hatched during 2016-2107= 291 7893 ■M
Hatched in March= 25 719 ■ RT
Ratio. 25 719 : 29 1793 ■ CA 1M Addition
1RT
1CA Express a ratio.
(3) D

L2

5.7
P(July eggs) = 29 271
349 066 × 100 ■M ■M
= 8,3 9 % ■CA CA from 5.1.2
1M Fraction
1M Multiplying by 100
1CA % (3)
NPR P
L2

5.8
Compound bar graph ■■A
OR
Bar graph ■■A
OR
Line graph ■■A
2A Type of graph

(2) D
L1

[20]

TOTAL : 100

■ M

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2017

MATHEMATICAL LITERACY P1
MARKING GUIDELINE

MARKS: 100

This marking guideline consists of 7 pages.

Symbol Explanation
M Method
M/A Method with Accuracy
A Accuracy
CA Consistent accuracy
RT/RG/RM Reading from a table/Reading from a graph/Read from map
SF Substitution in a formula
P Penalty, e.g. for no units, incorrect rounding off etc.
NP No Penalty
S Simplification
R Rounding/Reason

QUESTION 1

Quest. Solution
AWARD FULL MARKS FOR ANSWER ONLY Explanation Marks

1.1.1 January 2017 ■■ 2RT Correct
month L1
(2)

1.1.2 R258,20 + R4 956,38 + R2 582 + R1 956,20 ■
= R9 753,08 ■ 1M Count
correct values
1CA Total
deductions L1

(2)

1.1.3 Unemployment Insurance Fund ■■ 2A Write in
full L1
(2)

1.1.4 3
5 ■ 1 290 ■
= R774,00 ■ 1M Correct
ratio

1 CA Amount L1

(2)

1.2 Perimeter
4 cm x 2 + 2 cm + 3 cm + 6 cm + 7 cm ■
=26 cm ■ 1M/A correct
values
1A Perimeter L1

(2)

1.3.1 North East ■■ 2A Direction L1
(2)

1.3.2 1 cm on the map represents 250 000 cm on the ground /in
reality. ■■ 2A Explanation L1

(2)

1.4.1 17, 19, 21, 23, 25, 26, 26, 27, 28, 29 ■■ 2A
Arrangement L1
(12)

1.4.2 7 ■■ 2A Minimum
temper ature L1
(2)

1.4.3 Number appearing most frequently .■■ 2A explanation L1
(2)
[20]

QUESTION 2

Quest. Solution Explanation Marks
2.1.1 R1 450,00 ■■
2RT amount
L1
(2)

2.1.2 1 450 ■
1 495 x 100 ■
=96,98 ■
= 97,0% ■ 1M Correct values
1M (x100) L1

(4)

2.1.3 Total amount to be paid per day.
R1 400 x 3 = R4 200,00 ■
R1 035 x 2 = R2 070,00 ■
R1 495 x 1 = R1 495,00 ■
Total = R7 765,00 ■

1A 6 sleeper
1A 4 sleeper
1A 8 sleeper
1A Total L1

(4)

2.1.4 Total cost for 4 days = R7 765 x 4 ■
= R31 060,00 ■
31 060 x 14%
= R4 348,40 ■
R31 060 + R4 348,40 ■
= R35 408,40 ■
OR
R7 765 x4 ■
= R31 060 ■
R31 060 x 114% ■
R35 408,40 ■■ CA from 2.1.2
1M multiply ing by 4
1M R7 765
1CA Cost without
VAT

1M x 14%
1CA
L3

(5)

2.2.1 Deposit = 12,5% ■ x R31 060 ■
= R3 882,50 ■ CA from 2.1.2 2M
Multiplying by 12,5%
and 31 060
1CA L1

(3)

2.2.2 Balance = R35 408 – R3 882,50 ■
= R31 525,9 0 ■
OR
VAT on Deposit = 3 882,50 x 0,14
= R543,55
Balance including VAT
R27 177,50 x 1,14
= R30 982,35 + 543,55
= R31 525,90 CA from 2.1. 2 and
2.2.1
1M Subtraction
1CA Balance L1

(2)

Quest. Solution Explanation Marks
2.2.3 Thirty one thousand five hundred twenty five rands
and ninety cents ■■ (2)

2.2.4 Donation = 20 x R14,2058 ■
= R284, 116 ■
= R284,12 ■
2M Multiplying
1S
1A Donation
(Accept R284,10) L3

(3)

2.3. Cost of parking = 4 x R12,00 ■
= R48 ,00 ■

1M/A
(Multiply 4 days by
12)
1CA
Parking fees L1

(2)
[27]

QUESTION 3

Quest. Solution Explanation Marks
3.1.1 Minimum daily food = 250g + 500 g ■
cups = 750 g ■
125 g

= 6 cups ■ 1M Addition
1M Division

1CA Cups L1

(3)

3.1.2 Number of days =10 000 ■
750 ■ ■■

= 13,3 ■
= 13 days ■ 1SF,
1 C ( kg to g)
1S Division
1CA Days - L2

(4)

3.1.3 Tommy 's weight = 42
1 000 ■
= 0,042 tons ■
1C
1A Ton L1

(2)

3.2.1 43
100 ■=0,43 ■
1MA
1A L1

(2)

3.2.2 Area of a main bedroom = 3 m x 3,5 m ■

= 10,5 m2 ■ 1M

1CA Area L2

(2)

3.2.3 Number of boxes of tiles = 10,5 m2
(0,43 x 0,43 ) m 2x 13 ■■
= 10,5 m2
0,1849 m2 x 13■
= 56,79
13
= 4,37 ■
= 5 boxes ■ 1SF and 1C

1S
1CA

1CA
(rounding upward for
boxes ) L2

(5)
[18]

QUESTION 5

Quest. Solution
ANSWER ONLY (FULL MARKS) Explanation Marks

5.1.1 445 million cubic metres ■
Heyshope dam ■ 1A Maximum
1A units L2
(2)

5.1.2 Difference = 445 – 180,9 ■■
= 246,1 ■ 1A C orrect values
1M subtraction
1CA Difference L1

(3)

5.1.3 57,9+36,3+71,5+20,8+57+18,9+13,5+18,1+180, 9 9 ■

Mean = 474,9 ■
9
= 52,77 million cubic metres ■
1M division by 9
1S

1CA L2

(3)

5.1.4  (i)    Klipfontein  ■
(ii)   Ohrigstad  ■
(iii)  Glen Alpine  ■ 2 RT
any two dams
identified  L1

(2)

5.1.5  Range = 445 – 10 ■
= 435  ■ 1M
1CA (using
values from a
wrong column )  L2

(2)

Quest.  Solution  Explanation  Marks
5.2.1 Absentees Week1 = 47
Absentees Week 2 = 42
Therefore Week 1 had most absentees ■■ M/A
A
(2)
5.2.2

Each bar correctly plotted -allocate 1 mark [ 5x 1= 5]        L2
(5)

5.3.1 P(Green ball)  = 5
18■■
(Allow answer in % or decimal and no penalty for
rounding)  Accept 0,278 OR 27,8% . 1M numerator
1M denominator  L2

(2)

5.3.2 P( Blue ball)= 10
18 ■×100 ■

= 55,56%  ■
1M numerator

1M multiplication
by 100
1CA  Percentage to
2 decimal places  L2

(3)
[24]

TOTAL:  100

NASIONALE
SENIOR SERTIFIKAAT

GRAAD 11

NOVEMBER 2016

WISKUNDIGE GELETTERDHEID V1
MEMORANDUM

PUNTE: 100

Hierdie memorandum bestaan uit 4 bladsye .

SLEUTEL

Simbool Verduideliking
M Metode
A Akkuraatheid
CA Deurlopende Akkuraatheid
RT/G/M/P Lees vanaf die tabel/grafiek/kaart/plan
SF Vervanging in ■ formule
S Vereenvoudiging
P Penalisering (geen eenhede, verkeerde ronding ens.)

VRAAG 1 [23]
Vr Oplossing Verduideliking Punt
1.1.1 Bennie 's Service Station Buffalo City ■■ 2M L1(2)
1.1.2 A= R3 390,96 –R2 093,46 ■
= R1 297,50 ■ 1RT
1A L1(2)
1.1.3 ■ 1 234 ,96
■2 469 ,92 ■× 100 ■
= 50% 1M teller en noemer
1M Vermenigvuldig met
100 L2(2)
1.1.4 R212 ,82 + R4,99 + R75 ,87 + R697 ,13 +
R2 469 ,92 + R726 ,16 ■ = R4 186 ,89 ■ 1M
1CA indien 1 waarde
weggelaat is of
R159,50 bygetel is L1(2)
1.1.5 B = BTW = R3 390,96 × 14%■
= R474,73 ■
OF
B = R3 865,96+R0,09 – R3 390,96 ■
=R 474,73 ■ 1M
1A
L1(2)
1.1.6 R2 469,92 – R4,99 ■■
= R2 464,93 ■ 1 Korrekte waardes
1 M
1CA L1(3)

1.2.1 Beginsaldo is die balans wat gereflekteer of
vertoon word op ■ staat op die eerste dag voor
enige ander transaksies . 2 A
Verduideliking
L1(2)
1.2.2 9,50

500 × 100■ = 0,019 × 100■ = 1,9%■ 1M deel deur 500
1M
1CA wanneer die
waarde wat gebruik
word verskillend van
die staat is L1(3)
1.2.3 R110,00 + R55,00+R1,10 × 2■
R165,00 + R2,20 ■
= R167,20 ■ 1M
1S
1CA L2(3)
1.2.4 Agtienduisend Vyfhonderd en Twee en tagtig
Rand ■en sewe sent ■ 2A Uitbreiding
L1(2)

VRAAG 2 [19]
Vr Oplossing Verduideliking Punt
2.1.1 Lengte = 30 vt■
Breedte = 25 vt■ 2 RD
L1(2)
2.1.2 30 vt × 25 vt ■
= 750 ft2 ■ OF
=750 ■■
10,764■
= 69,68 m2 ■
(Aanvaar 69,677 ■2 ) 30 vt = 9,1435
25 vt = 7,6196
■A = 9,1435 × 7,6196
= 69,6698 m2
= 69,67 m2 1M
1S
1 M Deel deur 10,764
1CA
L2(4)
2.1.3 Aantal kunsmis =20 × 15■
100 ■
= 300 vt2
100 vt2 ■
= 3 × 2 ■ pond
= 6 pond ■
OF
2 pond × 3■■ = 100 vt2 × 3■
6 pond ■ = 300 vt2■ 1M teller
1M noemer
1S
1M Vermenigvuldig met
2
1CA
L2(5)
2.1.4 0,15 × 2 ■ = 0,3 ■
Ongeveer = 1
3 koppie ■
OF
1
0,15■= 6,66666667
Dus 2
0,666666667 ■
= 0,3 koppies = ■ koppie ■ 1M Vermenigvuldig met
2
1 A
1A

L1(3)

2.2.1  Oggend + Aand
(10■■+15■■+10■■ ) ■ × 2■
=35■■ × 2
=  70 ■■■
2 M

1CA L1(3)
2.2.2  10 ■■+10 ■■■ =20 ■■■ 1M
1A L1(2)

VRAAG 3 [19]
Vr Oplossing  Verduideliking  Punt
3.1.1  4■■ 2 RT  L1(2)
3.1.2  Water moniteringstasie ■■ 2 RT  L1 (2)
3.1.3  MA-1 ■■ 2 RT  L1(2)
3.1.4  103 mm ■■ (10,3 cm)  2 RT  L1(2)
3.1.5  10,3 × 250  000■
100  000■
= 25,75 km ■ 1C
1 Deling
1CA L3(3)

3.2.1  Tafel B■■(Aanvaar H/I)  2A RP  L1(2)
3.2.2  Suidoos ■■ 2A RP  L1(2)
3.2.3  Onmoontlik/Zero/0% ■■ 2A L1(2)
3.2.4  Tafel C ■■ 2RP L1(2)

VRAAG 4 [27]
Vr Oplossing  Verduideliking  Punt
4.1 32,49 ; 29,63 ; 23,62 ; 17,89 ; 14,59 ; 12,03 ; 10,31 ;
9,89; 9,57■■
2 M L1(2)
4.2 Mediaan =R14,59 ■■ 2 M L2(2)
4.3 Omvang  = R33,73 – R9,68■■ =R24,05 ■ 2RT L2(3)
4.4 R-3,32 ■■ 2M L1(3)
4.5 Margarien 500g■■ 2 RT  L1(2)
4.6 Geen modus ■■ 2RT L2 (2)
4.7 Gebied  Margarien  Rys Olie Tee Witsuiker
Stedelik  21,68  23,45  17,25  9,68  26,31

1punt vir elke voedselprys korrek afgesteek × 5 = 5 punte  + 1 = 6 punte
1punt vir korrekte grafiek  L2(6)
4.8 Mense in landelike gebiede betaal meer  of minder vir
sekere items ■■ 2A Verduideliking
L3(2)
4.9 0,56 +  0,72 + 1,11 + 1,24 + 3,79 + (-0,17)  + 2,66 +
(-0,21)  + (-3,32)  ■
R6,38
9■ =■0,71■ 1M
1deel deur
1CA
L2(3)
4.10
2■
9■ × 100 = 22,2% =22% ■ 1teller
1noemer
1CA L2(3)

VRAAG 5 [12]

Vr Oplossing Verduideliking Punt
5.1 R36,99 ■■ 2RT L1(2)
5.2 R200,00 ■■ 2 RT L1(2)
5.3 ■278 ,78
13 ■ = R21,44 ■ = R20,00 ■ 1M deel deur 13
1S
1A L1(3)
5.4 6 × R17,99 OF
= ■107 ,94 ■
1,14 ■ = R94,68 ■ 17,99
1,14 ■ = 15,78 ■ × 6
= R94,68 ■ 1A vir 6 1M ÷ 1,14
1CA L2(3)
5.5 R99 ,98
2■ = R49,99 ■ 1M
1A L1(2)
5.6 Totaal = 22,49 + 29,26 + 25,59 + 99,98 + 22,00 +
17,99 + 9,99 + 13,49 + 1,00 ■ = R241,79 ■
OF Totaal = 478,78 – 200 – 278,78 = R241,79
1M
1CA L2(2)

TOTAAL : 100

051015202530
Margarien Rys Olie Tee Wit SuikerPryse in Rand
Geselekteerde voedselpryseStedelike voedselpryse
Stedelike voedselpryse
■ ■
■
■ ■
■

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NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2016

MATHEMATICAL LITERACY P1
MEMORANDUM

MARKS: 100

This memorandum consists of 4 pages .

KEY

| Symbol | Explanation |
|---|---|
| M | Method |
| A | Accuracy |
| CA | Consistent accuracy |
| RT/G/M/ | Reading from the table/graph/map/plan |
| SF | Substitution in a formula |
| S | Simplification |

P Penalty (no units, incorrect rounding etc.)

QUESTION 1 [23]
Q Solution Explanation Mark
1.1.1 Bennie's Service Station Motors Buffalo City ■■ 2M L1(2)
1.1.2 A= R3 390,96 –R2 093,46 ■
= R1 297,50 ■ 1RT
1A L1(2)
1.1.3 ■ 1 234 ,96
■2 469 ,92 ■× 100 ■
= 50% 1M numerator and
denominator
1M multiply by 100 L2(2)
1.1.4 R212 ,82 + R4,99 + R75 ,87 + R697 ,13 +
R2 469 ,92 + R726 ,16 ■ = R4 186 ,89 ■ 1M
1CA if one value is
omitted or R159,50 is
added L1(2)
1.1.5 B = VAT = R3 390,96 × 14%■
= R474,73 ■
OR
B = R3 865,96+R0,09 – R3 390,96 ■
= R 474,73 ■ 1M
1A
L1(2)
1.1.6 R2 469,92 – R4,99 ■■
= R2 464,93 ■ 1 Correct values
1 M
1CA L1(3)

1.2.1 Opening balance is the balance that is
reflected or displayed on the first day listed in
the statement before any other transactions. 2 A
Explanation
L1(2)
1.2.2 9,50
500 × 100■ = 0,019 × 100■ = 1,9%■ 1M dividing by 500
1M
1CA when the different
value from the
statement is used L1(3)
1.2.3 R110,00 + R55,00+R1,10 × 2■
R165,00 + R2,20 ■
= R167,20 ■ 1M
1S
1CA L2 (3)
1.2.4 Eighteen Thousand, Five Hundred ■
and Eighty Two Rand, and Seven Cents ■ 2A expanding
L1(2)

QUESTION 2 [19]
Q Solution Explanation Mark
2.1.1 Length= 30 ft■
Width = 25 ft ■ 2 RD
L1(2)
2.1.2 30 ft × 25 ft ■
= 750 ft2 ■ OR
=750 ft2
10,764■
= 69,68 m2 ■
(Accept 69,677 ■2 ) 30 ft = 9,1435

25 ft = 7,6196
■A = 9,1435 × 7,6196
= 69,6698 m2
= 69,67 m2 1M
1S
1 M Dividing by 10,764
1CA
L2 (4)
2.1.3 Amount of fertiliser =20 × 15■
100 ■
= 300 ft2
100 ft2 ■
= 3 × 2 ■ pounds
= 6 pounds ■
OR
2 pounds × 3 ■■ = 100 ft2 × 3■
6 pounds ■ = 300 ft2■ 1M numerator
1M denominator
1S
1M Multiply by 2
1CA
L2 (5)
2.1.4 0,15 × 2 ■
= 0,3 ■
Approximately = 1
3 cup ■
OR
1
0,15■= 6,66666667
Therefore 2
0,666666667 ■
= 0,3 cups = ■ cup ■ 1M Multiply by 2
1 A
1A
L1(3)

2.2.1 Morning + Evening
(10■■+15■■+10■■ ) ■ × 2■
=35■■ × 2
= 70 ■■■
2 M

1CA L1(3)
2.2.2 10 ■■+10 ■■■ =20 ■■■ 1M
1A L1(2)

QUESTION 3 [19]
Q Solution Explanation Mark
3.1.1 4■■ 2 RT L1(2)
3.1.2 Water monitoring station ■■ 2 RT L1 (2)
3.1.3 MA-1 ■■ 2 RT L1(2)
3.1.4 103 mm ■■ (10,3 cm) 2 RT L1(2)
3.1.5 10,3 × 250 000■
100 000■
= 25,75 km ■ 1C
1 Division
1CA L3(3)

3.2.1 Table B■■ (Accept H/I) 2A RP L1(2)
3.2.2 South East ■■ 2A RP L1(2)
3.2.3 Impossible/Zero/0% ■■ 2A L1(2)
3.2.4 Table C ■■ 2RP L1(2)

QUESTION 4 [27]
Q Solution Explanation Mark
4.1 32,49 ; 29,63 ; 23,62 ; 17,89 ; 14,59 ; 12,03 ; 10,31 ;
9,89; 9,57■■
2 M L1(2)
4.2 Median=R14,59 ■■ 2 M L2(2)
4.3 Range=R 33,73 – R9,68■■ =R24,05 ■ 2RT L2(3)
4.4 R-3,32 ■■ 2M L1(3)
4.5 Margarine 500g ■■ 2 RT L1(2)
4.6 No Mode ■■ 2RT L2 (2)
4.7 Area Margarine Rice Sunflower Ceylon Tea White Sugar
Urban 21,68 23,45 17,25 9,68 26,31

1mark per food item correctly plotted × 5 = 5 marks + 1 = 6 marks
1mark for correct graph L2(6)
4.8 Rural area people pay more or less on certain
items ■■ 2A Explanation
L3(2)
4.9 0,56 + 0,72 + 1,11 + 1,24 + 3,79 + ( -0,17) + 2,66 +
(-0,21) + ( -3,32) ■
R6,38
9■ =■0,71■ 1M
1Dividing
1CA
L2(3)
4.10 2■
9■ × 100 = 22, 2% = 22% ■ 1numerator
1denominator
1CA L2(3)

QUESTION 5 [12]
Q Solution Explanation Mark
5.1 R36,99 ■■ 2RT L1(2)
5.2 R200,00 ■■ 2 RT L1(2)
5.3 ■278 ,78
13 ■ = R21,44 ■ = R20,00 ■ 1M dividing by 13
1S
1A L1(3)
5.4 6 × R17,99 OR
= ■107 ,94 ■
1,14 ■ = R94,68■ 17,99
1,14 ■ = 15,78 ■ × 6
= R94,6 8 ■ 1A for 6 1M ÷ 1,14
1CA L2(3)
5.5 R99 ,98
2■ = R49,99 ■ 1M
1A L1(2)
5.6 Total = 22,49 + 29,26 + 25,59 + 99,98 + 22,00 +
17,99 + 9,99 + 13,49 + 1,00 ■ = R241,79 ■
OR Total = 478,78 – 200 – 278,78 = R241,79 1M
1CA
L2(2)

TOTAL : 100

Prices in Rands Urban food prices
■ ■
■
■ ■
Selected food items ■
■

MARKS:  100

SYMBOL  EXPLANATION
M Method
MA Method with accuracy
CA Consistent a ccuracy
A Accuracy
C Conversion
S Simplification
RT /RG  Reading  from a table /Reading from a graph
F Choosing the correct formula
SF Substitution in a formula
O Opinion
P Penalty , e.g. for no units, incorrect rounding off etc.
R Rounding off/Reason

This memorandum consists of 7 pages .

GRADE 11

MATHEMATICAL LITERACY P1

EXEMPLAR 2013

MEMORANDUM  NATIONAL
SENIOR CERTIFICATE

QUESTION 1 [ 19 marks]
Ques  Solution  Explanation  Topic

1.1.1
2,35 ■ = 2,35 × 1 000 m■
= 2 350 m ■
1A answer
(1) M
L1

1.1.2
Cost per kg = kg399,19R
= R6,66
1M dividing by 3 kg
1A answer
(2) F
L1

1.2.1
Percentage mark -up = %10000,43R00,43R 75,53R×–
= %10043R75,10R×
= 25%
1SF substitution
1S simplification
1CA answer
(3) F
L2

1.2.2
Number of kilograms = R54000 R2
= 37,04
1M dividing
1A answer
(2) F
L1

1.2.3
A = R76,00 + 30% of R76,00 = R76,00 + R22,80 = R98,80

OR

A = 1,3 × R76,00
= R98,80
1M adding 30%
1A mark up
1CA answer

(3) F
L2

1.2.4
Cost = 1,2 kg × R53,75 + 0,5 kg × R85,00
= R64,50 + R42,50 = R107,00

2M using correct selling prices
1S simplification
1CA answer
(4) F
L1(2)
L2(2) ■A
■M
■SF
■S
■CA
■M
■A
■CA
■M ■A
■CA ■A
■M
■A
■M ■M
■S
■CA

Ques Solution Explanation Topic

1.2.5
Cost price = 50 × R43,00
= R2 150
Selling price = 50
× R53,75
= R2 687,50 Profit = R2 687,50 – R2 150
= R537,50
OR

Profit per kilogram = R53,75 – R43,00 = R10,75
Profit on 50 kg = 50

× R10,75
= R537,50
1M/A multiplying
1S simplifying
1S selling price
1CA answer

1M/A multiplying
1S simplifying
1M/A multiplying
1CA answer
(4) F
L1(2)
L2(1 )
L3(2)
[19]
■M
■S
■CA ■S
■M
■S
■M
■CA

QUESTION 2 [1 5 marks ]
Ques Solution Explanation Topic

2.1
Deposit = 10% of R6 599,99
= ×10010 R6 599,99
= R659,999
≈R660

1M finding 10%
1CA answer
(2) F
L1(2)

2.2
Amount to be financed (P) = R6 599,99 – R660,00
= R5 939,99 Interest = R5 939,99
× 0,115 × 2
= R 1 366,20 for two years
Total due = R5 939,99 + R1 366,20 = R7 306,19
1A subtracting deposit
1A value of i .
1A two years
1CA a dding i to
outstanding amount
(4) F
L1(2)
L2(2)

2.3
Amount = 2419,3067R
= R304,4245.....
≈ R304,42
1M dividing by 24
1A answer
(2) F

L2

2.4
Height = 12
× 2,54 cm = 30,48 cm

1M multiplying by 2,54 1A height
(2) M
L1

2.5.1
Amount = 12 × R300
= R3 600
1M multiplying by 12
1A answer
(2) F
L1

2.5.2
Amount = R300 – (R120 + R150)
= R30
1M subtracting from R300 1M adding costs
1A answer
(3) F
L1(2)
L2(1)
[15]

■M
■CA
■A
■A ■A
■CA
■M
■A
■A
■M
■A
■M ■M
■A ■M

QUESTION 3 [ 20 marks ]
Ques Solution Explanation Topic

3.1.1
Height of tin = 430 mm = 43 cm
1A length
1C conversion
(2) M
L1

3.1.2
Distance = 4 570 mm – 1 780 m m
= 2 790 m m

1M subtracting correct
values
1A correct answer

(2) M
L1

3.1.3
Length = 4 260 – 2 610
= 1 650 mm
Area = 1 600 × 1 650
= 2 640 000 m m2
1 A value of length
1SF substitution
1CA answer
(3) M
L1(1)
L2(2)

3.1.4
A = 4, 830 m (2 × 9,75mm + 6,4 m ) – 6,4 m × 0,43 m
= 4, 83 m (25,9 m ) – 2,752 m2
= 122,345 m2
1C conversion
1SF value of k
1SF value of b
1SF value of t and p.
1S simplification
1CA area
(6) M
L1(2)
L2(2)
L3(2)

3.2.1
Amount of paint = ■8345,122
= 15,293.. ■
≈ 16 ■
1M dividing by 8
1CA amount of paint
1R rounding up
(3) M
L1(2)
L2(1)

3.2.2
Number of 5 ■ = ■■
516
= 3 ,2 = 4
Cost = 4
× R215,85
= R 863,40
1M dividing by 5 ■
1 CA number of containers
1M multiplying by cost
1CA cost of paint
(4) M/F
L1(2)
L2(2)
[20]
■M
■A
■SF ■A
■M ■SF ■SF
■CA ■CA
■S ■CA
■M
■CA ■M

■CA
■R ■SF ■C

QUESTION 4 [14 marks]
Ques Solution Explanation Topic

4.1.1
R63
2A correct regional road
(2) M
L1

4.1.2
Pearston; Somerset E ast; Cookhouse; Bedford
3A any three correct
(3) M
L1

4.1.3
25 km + 49 km + 10 km
= 84 km
1A correct reading from map
1M adding 1CA total distance
(3) M
L1

4.1.4
North -easterly OR NE

2A correct direction
(2) M/P
L1

4.2
3 cm : 15 km
= 3 cm : 1 500 000 cm
= 1 : 500 000
1A correct measurement
1M writing as a ratio
1C converting to cm
1CA simplified ratio
(4) S/M/P
L2
[14]
■■A
■A ■A ■A
■M
■CA
■■A
■M ■A
■C
■CA ■A

QUESTION 5 [ 32 marks ]
Ques Solution Explanation Topic

5.1.1
Total = 20 + 5 + 5 + 30 + 40 + 15 + 5 + 10
= 130
1M adding 1A sum
(2) D
L1
5.1.2 Percentage preferring casual wear
= %10013040×
= 30,77%
1A correct values
1M calculating %
1CA solution
(3) D
L1

5.1.3
Fancy dress
2A correct answer
(2) D
L2

5.1.4 DRESS CODE FOR THE DANCE
051015202530354045
FORMAL
TRADITIONAL
DRESS
FANCY DRESS
CASUAL
WEAR
Dress CodeNumber of Learners

7A each bar

(7) D
L1

5.1.5 (a)
P= 13060
1A numerator
1A denominator
(2) P
L2

5.1.5 (b)
P = 13010 130–

= 130120
1A numerator

1A solution
(2) P
L3
■A
■A ■A ■A
■A ■A
■A ■A ■M
■M ■A
■CA
■■A
■A
■A ■A
■A

Ques Solution Explanation Topic

5.2.1
39 49 56 58 59 67 75 75 75 79 89 98 99
1A solution
(2) D
L1

5.2.2
75
2A solution
(2) D
L1

5.2.3
Mean = 14976
= 69,7…
≈70
1M adding 1A dividing by 14
1A simplification

(3) D
L1
L2

5.2.4
Median = 268 66+
= 67
1M concept
1A solution (2) D
L2

5.2.5
Range = 99 – 39
= 60
1M subtracting correct values
1A solution (2) D
L1

5.2.6
P= 133
1A numerator
1A denominator
(2) P
L2

5.2.7
P = 145
1A numerator
1A denominator
(2) P
L2
[32]

TOTAL: 100 ■M
■A
■A
■A
■A
■A ■A ■M
■A ■M

■A ■A ■A ■A

MARKS:

SYMBO L
A
CA
C
J
M
MA
P
R
RT/RG
S
SF
O

L
MAT H
100

EXPLA
Accura c
Consist e
Conver s
Justific a
Method
Method
Penalty
Roundi n
Readin g
Simplifi c
Correct
Own o p
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pinion
his memo r

NATIO
OR CE R

G

VEMB E

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MORA

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son/Opini o
racy
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e/graph
on in a for m
randum c o
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RTIFIC A
RAD E
ER 20 1
LITE R
ANDU M
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ATE
E 11
13
RACY
M
g off, etc.
7 pages.
P1

QUESTION 1 (30 MARKS)

Question Solution Explanation Level +
Topic
1.1 Fixed Expenses
= 6000 + 1500 + 465 + 250 + 1 750 + 500 √M√SF
= R 10 465, 00 √A 1M Method
used 1SF Correct values substituted 1A Correct answer (3) Finance
L1
1.2 Cost of ingredients
= 15,50 + 0,25 + 11,50 + 9,50 + 12,75 + 8,00 + 12,50 √M√SF
= R70,00 √A 1M Method
used 1SF Correct values substituted 1A Correct answer (3) Finance
L1
1.3 Cost for Ingredients for one Giant Muffin

= 70, 00 ■ 50 √SF
= R1,40 √MA 1SF Correct
substitution 1MA Method and accuracy (2)
Finance L1

1.4 Cost of Electricity for one Muffin
= (6 x 1,09) ■50√SF
= 0,1308
= R0,13 √MA 1SF Correct
substitution 1MA Method and accuracy
(2) Finance
L2

1.5 Cost of one Giant Muffin
= 1,40 + 0,13 √SF
= R1,53 √MA
1SF Correct
substitution 1MA Method and accuracy
(2) Finance
L1

1.6.1 Profit = 5,00 – 1,53 √SF√SF
= R3,47 √MA
2SF Correct
substitutions 1MA Method and accuracy
(3) Finance
L1

1.6.2 % Profit = 3,47
1,53 x 100
√SF
= 226,7973856 √A
= 226,80% √C 1SF Correct
values used 1A Accuracy 1C Conversion to 2 decimal places (3) Finance
L1

.
Question Solution Explanation Level +
Topic

1.7 Number of muffins = ■■ ■■■,■■
■,■■ √SF
= 3015,850144 √A
= 3016 Muffins √C 1SF Correct
values used 1A Accuracy 1C Conversion up to whole muffin (3) Finance
L2

1.8.1 A = 12 500 + 12 500 x 12,5% x 5 √SF
= 12 500 + 7 812,50 √CA
= R20 312,50 √A
1SF Correct
substitution 1CA Accurate calculation 1A Correct answer (3) Finance
L2

1.8.2 Monthly Payment = 20 312, 50 ■ 60 √M √C
= R 338,5416667
= R 338,54 √C 1M Correct
method used ( ■■
1C using 60 1C Correct answer to 2 decimal places
(3) Finance
L2

1.8.3 Extra muffins needed = 338, 54 ■ 3,47 √M√SF
= 97,56195965
= 98 muffins more √C 1M Correct
method used 1SF Correct
values used
1C Correct answer round up ( 3 ) Finance
L2
[30]

QUESTION 2 (22 MARKS)

Question Solution Explanation Level +
Topic
2.1.1 Surface Area = 2 x 15 x 1,5 + 2 x 8 x 1,5 √SF
= 45 + 24 √MA
= 69 ■■ √CA 1SF Correct
values substituted 1MA Correct method and accuracy 1CA Correct answer (3) Measure
L1
2.1.2 20 cm ■ 100 = 0,2 m √C
Tile area = 0,2 x 0,2
= 0,04 ■■each √CA

No. of Tiles needed = 69 ■ 0,04 √M
= 1725 tiles √CA 1C Correct
conversion 1CA Accurate answer 1M Correct method 1CA Correct answer (4) Measure
L2
2.2 Volume = 15 x 8 x 1,5 √SF
= 180 ■
■ √A
= 180 kilolitres √C
1Sf Correct
substitution 1A Accurate answer 1C Correct conversion (3) Measure
L2
2.3 Cost = 180 x 5, 55 √SF √SF
= R 999,00 √CA
2SF Correct
values used 1CA Consistent accuracy (3) Finance
L1
2.4 Water evaporated = 180 x 1,5% √SF √M
= 2,7 kilolitres √CA
1SF Correct
values used 1M Correct method (x 1,5%) 1CA Consistent accuracy (3) Measure
L2

Question Solution Explanation Level +
Topic

2.5.1 Width of paving = 17 – 15 or 10 – 8
2 2 √M
= 1m √CA
1M Correct
method used 1CA Correct answer (2) Measure
L1
2.5.2 50 ■ 100 = 0,5 ■ 2 pavers per metre √C

No. pavers for length = 17 x 2 x 2 = 68 pavers √M
No. pavers for width = 8 x 2 x 2 = 32 pavers √M
Total No. of Pavers = 68 + 32 = 100 pavers √CA
Width and length with 2 m short can be swapped around 1C Correct
concept 1M correct method 1M Correct method 1CA Correct Answer (4) Measure
L3 L2
[22]

QUESTION 3 (20 MARKS)

Question Solution Explanation Level +
Topic
3.1.1 North √A 1A Correct
answer (1) Maps
etc. L1

3.1.2 Go through the passage from the front door
onto the veranda. Turn right and walk on the veranda until you get to a classroom Turn
left along the veranda and enter the third door on your right. √√CI 2 CI correct
directions given  Give 1 mark for every 2 correct instructions (2) Maps
etc.  L1
3.1.3 Double door √A
Reason = 2 arcs and a line – allows more children to go through in one go √J 1A correct
answer 1J Correct justification (2) Maps
etc.  L1
3.2.1 Length = 3 cm √A
Breadth = 2,2 cm √A 1A Correct
answer 1A correct answer (2) Maps
etc.  L1

Question Solution Explanation Level +
Topic
3.2.2 Real life length = 3 x 400 √M
= 1 200 cm / 12 m √CA
Real life breadth = 2,2 x 400 √M
= 880 cm / 8,8 m √CA 1M correct
method used 1CA Consistent accuracy 1M correct method used 1CA Consistent accuracy
(4) Maps etc.
L1
3.3

1 mark for correct 1 door 1 mark for correct number of windows (can
vary in size) 1 mark for roof (roof may have angled sides) As indicated
with drawing

.
(3) Maps etc.
L1
3.4 No, they are not all the same size. √A
Not all classes need the same amount of space. Some classes less learners etc. Accept
any
reasonable answer here. √J 1A Correct
answer 1J Justification/Opinion (2) Maps etc.
L1
3.5.1 Annual School  fees = 10 x 950 √M
= R 9 500 √CA 1M Correct
method 1CA Correct answer
(2) Finance
L1
3.5.2 Discount = 9 500 x 15% √M
= R1 425 √CA
Accept  x  by 0,15 instead of 15% 1M Correct
method 1CA Correct answer (2) Finance
L2
[20]

QUESTION 4 (28 MARKS)
Question Solution Explanation Level +
Topic
4.1.1 16 √RG 1RG Correct
reading from graph  (1) Data
L1
4.1.2 5 √RG 1RG Correct
reading from graph  (1) Data
L1

4.1.3 4 Overs √RG√RG 2RG Correct
adding and reading from graph (2) Data etc.
L1
4.1.4 6 Overs √RG√RG 2RG Correct
adding and reading from
graph (2) Data etc.
L1
4.1.5 P(9 runs SA) = ■
■■ √SF = 20% √C

o r
1
5 1SF Correct
values Substituted 1C Correct conversion
(2) Probability
L1
4.1.6 Total number of runs SA
= 5 + 10 + 15 + 9 + 0 + 6 + 15 + 12 + 8+ 9 + 4 + 11 + 16 + 1 + 9 + 24 + 12 + 18 + 13 +
9 √MA
= 206 runs √CA 1MA Correct
method and accuracy 1CA Correct answer (2) Data etc.
L1
4.1.7 Average = 194 ■ 20 √M
= 10,3 runs per over √CA 1M Correct
method 1CA Consistent accuracy (2) Data etc.
L2
4.1.8 Total number of runs New Zealand
= 6 +12 + 12 + 7 + 2 + 8 + 12 + 10 + 10 +
12 + 2 + 5 + 12 + 2 + 7 + 18 + 6 + 12 + 15 + 2 √MA
= 172 runs √CA
SA won because they made more runs √J 1MA Correct
method and
accuracy 1CA Correct answer 1J Correct justification (3) Data etc.

L1

Question Solution Explanation Level +
Topic
4.2.1 Social Networking √RG 1Rg Correct
reading from graph (1)
Data etc.
L1
4.2.2 % usage = 72 x 100 √√SF
400
= 18% √CA
2SF Correct
values substituted 1CA Correct answer (3) Data etc.
L1
4.2.3 400 x 8% √M
= 32 learners √A
■ 8% = SMSs √CA

1M Correct
method used 1A Accuracy 1CA Correct answer (3) Data etc.
L3
4.2.4 No. Degrees So cial Networking
= 80 x 360
400 √M√SF
= 72° √CA
1M Correct
method used 1SF correct values substituted 1CA Correct answer (3) Data etc.
L2

4.2.5 P( games) = 45 √SF = 11,25 % √CA
400 √SF 1SF Correct
numerator 1SF correct denominator 1CA Correct percentage (3) Probability
L1
[28]

TOTAL: 100

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Province of the
EASTERN CAPE
EDUCATION

NATIONAL
SENIOR CERTIFICATE

GRADE 1 1

NOVEMBER 2012

MATHEMATICAL LITERACY P1
MEMORANDUM

MARKS: 100

SYMBOL EXPLANATION
A Accuracy
CA Consistent Accuracy
C Conversion
J Justification (Reason/Opinion)
M Method
MA Method with accuracy
P Penalty for no units, incorrect rounding off, etc.
R Rounding off
RT/RG Reading from table/graph
S Simplification
SF Correct substitution in a formula
O Own opinion

This memorandum consists of 6 pages.

QUESTION 1 LO+AS
1.1 1.1.1 Sale Price of CD = 125 – 15% ■M
= R106,25 ■CA

Or
Sale Price of CD = 125 – (125 x 0,15) ■M
= 125 – 18,75
= R106,25 ■CA 1M Method used
1CA Consistent
accuracy

1M Method used
1CA Consistent
accuracy (2) 11.1.3
1.1.2 Sale Price of DVD = 215 – (215 x ) ■M
= 215 – 53,75 ■CA
= R161,25 ■A

Or

Sale Price of DVD = 215 – (215 x 0,25) ■M
= 215 – 53,75 ■CA
= R161,25 ■A

Or
Sale Price of DVD = 215 – (215 x25%) ■M

= 215 – 53,75 ■CA
= R161,25 ■A

Or
Sale Price of DVD = 215 – 25%) ■M
= 215 – 53,75 ■CA
= R161,25 ■A 1M Correct method
used
1CA Consistent
accuracy
1A Accurate answer

1M Correct method
used
1CA Consistent
accuracy
1A Accurate answer

1M Correct method
used
1CA Consistent
accuracy
1A Accurate answer

1M Correct method
used
1CA Consistent
accuracy
1A Accurate answer
(3) 11.1.3
1.2 1.2.1 Mowing Lawn = ■SF = ■CA 1SF Correct
substitution
1CA Accurate
simplification (2) 11.1.1
1.2.2 Watering the flowers:
Time taken = 120 – 45 – 20 – 30 ■SF
= 25 minutes ■CA
Watering time = ■SF = ■CA 1SF Correct values
used
1CA Consistent
Accuracy
1SF Correct values
used
1CA Consistent
Accuracy (4) 11.1.1
1.3 1.3.1 Flour = x 50 ■M
= 1 000 g or 1 kg ■CA 1M Correct method used
1CA Consistent accuracy (2) 11.1.1
1.3.2 Cheese = x 50 ■M
= 750 g ■CA
1M Correct method used
1CA Consistent accuracy (2) 11.1.1

1.4 1.4.1 Using; Distance = Speed x Time

561 = Speed x 6 ■SF
Speed = ■M
= 93,5 km/h ■CA
= 94 km/h ■R 1SF Correct values
substituted
1M Correct method used
1CA Consistent accuracy
1R Correctly rounded up
(4) 11.1.1
1.4.2 Litre/km = ■M
= 17,9980…… ■CA
= 18 km/litre ■R 1M Correct method used
1CA Consistent accuracy
1R Correctly rounded up
(3) 11.1.3
1.4.3 Cost of Petrol
= 31,17 x 10,05 ■SF ■M
= R313,2585
= R313,26 ■CA
1SF Correct values used
1M Correct method used
1CA Correct answer (3) 11.1.2

[25]

QUESTION 2 LO+AS
2.1 Fixed expenses = R200 ■RG

Rental , electricity, salaries, etc.
Accept any logical answer here . ■O 1RG Correct reading
from graph
1O Correct, logical
opinion (2) 11.2.3
2.2 Expenses making coffee tables = wood, nails
glue, etc.
Accept any material needed . ■■O 2O any 2 correct
materials given (2) 11.2.1
2.3 Break -even = 6,5 ■RG = 7 tables ■CA 1RG Correct reading
from graph
1CA Correct answer (2) 11.2.1
2.4 Expenses 20 tables = R500,00 ■RG ■A 1RG Correct reading
from graph
1A Accuracy in reading
(2) 11.2.1
2.5 Income 20 tables = R 900,00 ■RG ■A 1RG Correct reading
from graph
1A Accuracy in reading
(2) 11.2.1

2.6 Profit 20 tables = 900 – 500 ■SF ■M
= R400,00 ■CA 1SF Correct values
1M Correct method
1CA Consis tent accuracy
(3) 11.2.3
2.7 Selling Price 1 table = 900 20 ■SF ■M
= R 45,00 ■CA
Or
Selling Price 1 table =
= (any correct graph
values used )
■SF ■M
= R45,00 ■CA 1SF Correct values
1M Correct method

1CA Consistent accuracy

1SF Correct values
1M Correct method
1CA Consistent accuracy
(3) 11.2.3
2.8 Income 15 tables = 45 x 15 ■M
= R675,00 ■CA 1M Correct method
1CA Consistent accuracy
(2) 11.2.3

2.9 Profit would increase . ■O
Reason : 60 x 20 tables = R1 200 ■M

1 200 – 900 = R300 more profit
■M■CA 1O Correct opinion
1M Correct method with
graph values used
1M Correct method
1CA accurate
calculations (4) 11.2.3
2.10 No of tables sold = ■M

= 13 tables ■CA 1M Correct method

1CA consistent accuracy
(2) 11.2.1

[24]

QUESTION 3 LO+AS
3.1 20 cm
7,5 cm 7,5 cm

25 cm 25 cm

7,5 cm 20 cm

20 cm ■■Correct net drawn
with 6 rectangles

■■ correct
measurements placed
in required places (not
all sides need
measurements)
(4) 11.3.1
3.2 Surface Area =
(2 x 20 x 25)+(2 x 20 x 7,5)+(2 x 25 x 7,5)
■■■ SF and M
= 1 000 + 300 + 375 ■CA
= 1 675 ■CA 3 SF M correct values used
and correct method x 3
1CA Consistent accuracy
1CA Correct answer (5) 11.3.1
3.3 Cost of cardboard = 1 675 x 0,06 ■M
= 100,5 cents ■CA

= 101 cents ■R 1M Correct method
1CA Consistent accuracy
1R Rounding up (3) 11.3.1
3.4 Cost of Printing = 1 675 x 0,04 ■SF■M

= 67 cents ■CA 1SF Correct values used
1M Correct method used
1CA Consistent accuracy (3) 11.3.1
3.5 Volume = 25 x 20 x 7,5 ■SF ■M

= 3 750 ■CA 1SF Correct values used
1M Correct method used
1CA Consistent accuracy (3) 11.3.1
3.6 Grams in box = 3 750 7,5 ■SF ■M
= 500 g cereal ■CA 1SF Correct values used
1M Correct method used
1CA Consistent accuracy (3) 11.3.2

3.7 2 m x 100 = 200 cm and 15 m x 100 = 1 500 cm ■C
Width of net = 7,5 + 25 + 7,5 = 40 cm ■M
Length of net = 20 + 7,5 + 20 + 7,5 = 55 cm ■M
200 40 = 5 widths
1500 55 = 27, = 27 lengths ■C
Number of boxes = 5 x 27 = 135 boxes ■CA

Or
2 m x 100 = 200 cm and 15 m x100 = 1 500 cm ■C
Area of roll = 200 x 1 500
= 300 000 ■■MA
Number of boxes = 300 000 1 675
= 179,1044…
= 179 boxes ■CA
(1 mark less for not allowing for waste of space) 1C conversion to cm
2 M Correct method
1C correct conversions
1CA Consistent
accuracy (5)
1C Conversions to cm.

1MA Correct method
and accuracy

1CA Consistent
accuracy (4) 11.3.2

Or
Area of roll = 2 x 15 ■M
= 30 ■A

Surface Area of box in m$^2$ = ■C
= 0,1675

Number of boxes =
= 179,104…
= 179 boxes

2MA Method and
Accuracy

1C Conversion to

1M Method

1A Accuracy [26]

QUESTION 4 LO+AS
4.1 4.1.1 Tazz ■RG 1 RG correct reading from graph (1) 11.4.4
4.1.2 Mercedes ■RG 1 RG correct reading from graph (1) 11.4.4
4.1.3 Suburb A much lower income than
Suburb B because Suburb A
purchases lower cost range of cars
in the main. ■■OR
(Accept any logical opinion and
reasoning .) 1O Opinion
1R Reason (2) 11.4.4
4.1.4 Not a realistic picture ■O as it only
deals with Johannesburg and not
the rest of the country where less
money is earned ■R
(Accept any logical opinion and
reasoning .) 1O Opinion
1R Reason (2) 11.4.4

4.2 4.2.1
(a) Mean = 23+41+42+50+50+51+54+
55+56+57+60+61+65+66+
66+67+68+69+70+70+70+
72+74+76+79+82+85+86+88 ■M
29
= 1853 ■MA
29
= 63,89655….
63,90 ■CA

1M Correct Method
1MA Method and
accuracy
1CA Consistent accuracy
(3) 11.4.3
(b) 23 41 42 50 50 51
54 55 56 57 60 61
65 66 66 67 68 69
70 70 70 72 74 76
79 82 85 86 88
Mode = 70 ■MA

1MA correct method used
and accuracy (1) 11.4.3
(c) 23 41 42 50 50 51
54 55 56 57 60 61
65 66 66 67 68 69
70 70 70 72 74 76
79 82 85 86 88
Median = 66 ■M■CA 1M Correct method used

1CA Consistent accuracy
(2) 11.4.3

4.2.2

| Interval | Frequency |
|---|---|
| 20 – 29 | 1 |
| 30 – 39 | 0 |
| 40 – 49 | 2 |
| 50 – 59 | 7 |
| 60 – 69 | 8 |
| 70 – 79 | 7 |
| 80 – 89 | 4 |

1 Mark for each correct frequency
in the following intervals
20 – 29 ■
30 – 39 ■
40 – 49 ■
50 – 59 ■
60 – 69 ■
70 – 79 ■
80 – 89 ■ (Any 4) (4) 11.4.2

4.2.3
1 Mark for correct
frequency intervals added
1 Mark for correct graph
used
3 Marks for any 3 correct
bars on graph (5) 11.4.2

4.2.4 Mean and median as both tell us
that most learners got around the
60 – 69 mark. This is also shown by
the histogram . ■O ■R (Accept all
valid choices and reasons .) 1O Correct choice made
1R Valid reason given (2) 11.4.4

4.2.5 The test was easy ■O because
most learners got above 50% . ■R 1O Correct choice
1R Correct reason for
choice (2) 11.4.4

[25]
TOTAL: 100

0510
20 -
2930 -
3940 -
4950 -
5960 -
6970 -
7980 -
89Frequency
Marks Test Results